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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Heat and Thermodynamics

JEE Main Physics · Previous Year Questions with Answer Key

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QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer) 407 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

One mole of diatomic gas having rotational modes only is kept in a cylinder with a piston system. The cross-section area of the cylinder is $4\ \mathrm{cm}^2$. The gas is heated slowly to raise the temperature by $1.2^{\circ}\mathrm{C}$ during which the piston moves by 25 mm . The amount of heat supplied to the gas is ____ J.

(Atmospheric pressure $= 100\mathrm{kPa}, R = 8.3\ \mathrm{J/mol.\ K}$) (Neglect mass of the piston)

(A) 24.8 (B) 10.96 (C) 15.04 (D) 29.98

Open solution

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Initial pressure and volume of a monoatomic ideal gas are P and V. The change in internal energy of this gas in adiabatic expansion to volume $V_{\text{final}} = 27\ \mathrm{V}$ is ____ J.

(A) $-2PV(3\sqrt{3}-1)$ (B) $\frac{4}{3}PV$ (C) $-\frac{4}{3}PV$ (D) $\frac{3}{4}PV$

Open solution

Q3. JEE Main 2026 (Online) 6th April Evening Shift

A cylinder with adiabatic walls is closed at both ends and is divided into two compartments by a frictionless adiabatic piston. Ideal gas is filled in both (left and right) the compartments at same P, V, T. Heating is started from left side until pressure changes to $27\mathrm{P}/8$. If initial volume of each compartment was 9 litres then the final volume in right-hand side compartment is ____ litres. (for this ideal gas $\mathrm{C_P}/\mathrm{C_V} = 1.5$)

(A) 3 (B) 4 (C) 14 (D) 9

Open solution

Q4. JEE Main 2026 (Online) 6th April Evening Shift

If 2 mole of an ideal monoatomic gas at temperature T, is mixed with 6 mole of another ideal monoatomic gas at temperature $2T$ then the temperature of mixture is:

(A) $\frac{5}{2}T$ (B) $\frac{5}{4}T$ (C) $\frac{7}{2}T$ (D) $\frac{7}{4}T$

Open solution

Q5. JEE Main 2026 (Online) 6th April Morning Shift

Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K . Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K . The final pressure in the vessels is ____ kPa .

(A) 100 (B) 120 (C) 90 (D) 105

Open solution

Q6. JEE Main 2026 (Online) 5th April Evening Shift

An ideal gas at pressure P and temperature T is expanding such that $PT^3 =$ constant. The coefficient of volume expansion of the gas is ____ .

(A) $\frac{2}{T}$ (B) $\frac{1}{T}$ (C) $\frac{4}{T}$ (D) $\frac{3}{T}$

Open solution

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Q7. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Statement I : Change in internal energy of a system containing n mole of ideal gas can be written as $\Delta \mathrm{U}=n \mathrm{C}_v\left(T_{\mathrm{f}}-T_i\right)=\frac{nR}{\gamma-1}\left(T_{\mathrm{f}}-T_i\right)$, where $\gamma=\frac{C_p}{C_v}, T_i=$ initial temperature, $T_{\mathrm{f}}=$ final temperature.

Statement II : Relation between degree of freedom f and $\gamma(=C_p/C_v)$ is $\left(\gamma=1+\frac{2}{f}\right)$

Choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

Q8. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Consider the following statements:

A. Zeroth law of thermodynamics gives concept of temperature

B. First law of thermodynamics gives concept of internal energy

C. In isothermal expansion of ideal gas, $\Delta Q \neq \Delta W$

D. Product of intensive and extensive variables is extensive

E. The ratio of any extensive variable to mass will be an extensive variable

Choose the correct combination of statements from the options given below:

(A) C, D and E Only (B) A, B and C Only (C) A, B and D Only (D) B, C and D Only

Q9. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : If the average kinetic energy of H_2 and O_2 molecules, kept in two different sized containers are same, then their temperatures will be same.

Reason R : The r.m.s speed of H_2 and O_2 molecules are same at same temperature.

Choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

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Q10. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

The temperature of a metal strip having coefficient of linear expansion α is increased from T_1 to T_2 resulting in increase of its length by ΔL_1. The temperature is further increased from T_2 to T_3 such that the increase in its length is ΔL_2.

Given $T_3 + T_1 = 2T_2$ and $T_2 - T_1 = \Delta T$, the value of ΔL_2 is ____ .

(A) $\Delta L_1 \left[1 + 2\alpha^2(\Delta T)^2\right]$

(B) $\Delta L_1 \left[1 + \alpha^2(\Delta T)^2\right]$

(C) $\Delta L_1[1 + 2\alpha\Delta T]$

(D) $\Delta L_1[1 + \alpha\Delta T]$

Q11. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

One gas of n_1 mole of molecules at temperature T_1, volume V_1, and pressure P_1, and another gas of n_2 mole of molecules at temperature T_2, volume V_2, and pressure P_2, are mixed resulting in pressure P and volume V of the mixture. The temperature of the mixture is ____ .

(A) $\frac{(T_1+T_2)}{2}$

(B) $\frac{T_1T_2PV}{T_2P_1V_1+T_1P_2V_2}$

(C) $\frac{(T_2P_1V_1+T_1P_2V_2)}{T_1T_2PV}$

(D) $\frac{[T_1-T_2]}{2}$

Q12. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

An ideal gas undergoes a process maintaining relation between pressure (P) and $\text{volume}(V)$ as $P = P_0\left(1 + \left(\frac{V_0}{V}\right)^2\right)^{-1}$, where P_0 and V_0 are constants. If two samples A and B (two moles each) with initial volumes V_0 and $3V_0$ respectively undergo above mentioned process and attain same pressure, then the difference at the temperatures of these samples, $T_B - T_A$ is ____ .

($R =$ gas constant)

(A) $\frac{9P_0V_0}{8R}$

(B) $\frac{11P_0V_0}{10R}$

(C) $\frac{7P_0V_0}{6R}$

(D) $\frac{13P_0V_0}{11R}$

Q13. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

A mixture of carbon dioxide and oxygen has volume 8310 cm^3, temperature 300 K, pressure 100 kPa and mass 13.2 g. The number of moles of carbon dioxide and oxygen gases in the mixture respectively are _____.

(Assume both carbon dioxide and oxygen gases behave like ideal gases) [R = 8.31 J/mol K]

(A) 0.15 and 0.18 (B) 0.25 and 0.08 (C) 0.21 and 0.12 (D) 0.13 and 0.20

Q14. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Heat is supplied to a diatomic gas at constant pressure. Then the ratio of $\Delta Q : \Delta U : \Delta W$ is _____.

(A) 2 : 3 : 5 (B) 5 : 3 : 2 (C) 2 : 5 : 7 (D) 7 : 5 : 2

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Q15. JEE Main 2026 (Online) 28th January Evening Shift

The mean free path of a molecule of diameter 5×10^{-10} m at the temperature 41°C and pressure 1.38×10^{5} Pa, is given as _____ m. (Given $k_B = 1.38 \times 10^{-23}$ J/K).

(A) $2\sqrt{2} \times 10^{-10}$ (B) $10\sqrt{2} \times 10^{-8}$ (C) $2\sqrt{2} \times 10^{-8}$ (D) 2×10^{-8}

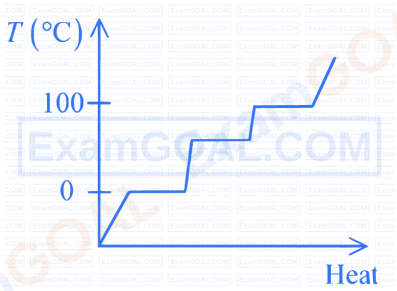
Open solution

Q16. JEE Main 2026 (Online) 28th January Morning Shift

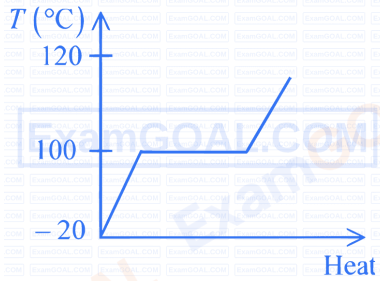
Which of the following best represents the temperature versus heat supplied graph for water, in the range of -20°C to 120°C?

Open solution

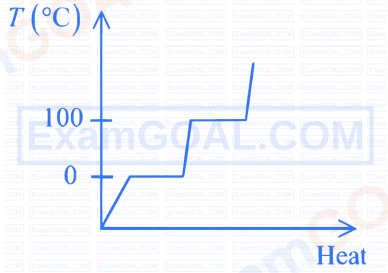
(A)



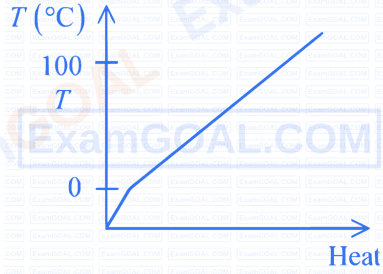
(B)



(C)



(D)



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Q17. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

10 kg of ice at $-10^{\circ}\mathrm{C}$ is added to 100 kg of water to lower its temperature from 25 $^{\circ}\mathrm{C}$. Consider no heat exchange to surroundings. The decrement to the temperature of water is ____ $^{\circ}\mathrm{C}$.

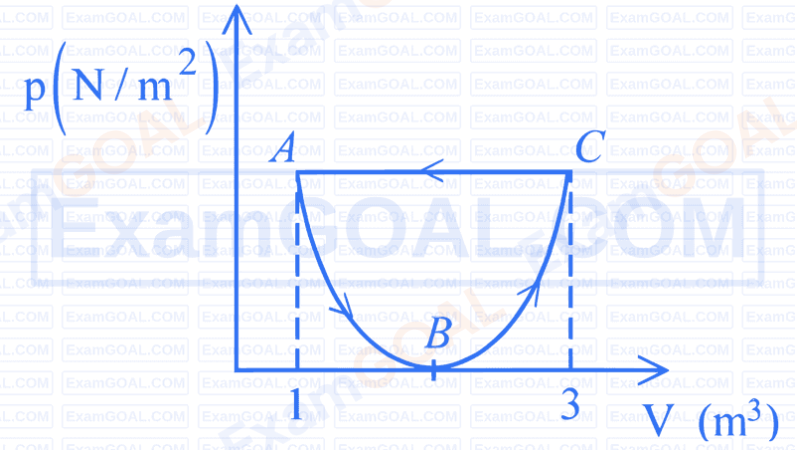
(specific heat of ice $= 2100\ \mathrm{J/Kg.^{\circ}C}$, specific heat of water $= 4200\ \mathrm{J/Kg.^{\circ}C}$, latent heat of fusion of ice $= 3.36 \times 10^5\ \mathrm{J/Kg}$)

(A) 15 (B) 10 (C) 6.67 (D) 11.6

Q18. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

In the following $p - V$ diagram the equation of state along the curved path is given by $(V-2)^2 = 4ap$ where a is a constant. The total work done in the closed path is

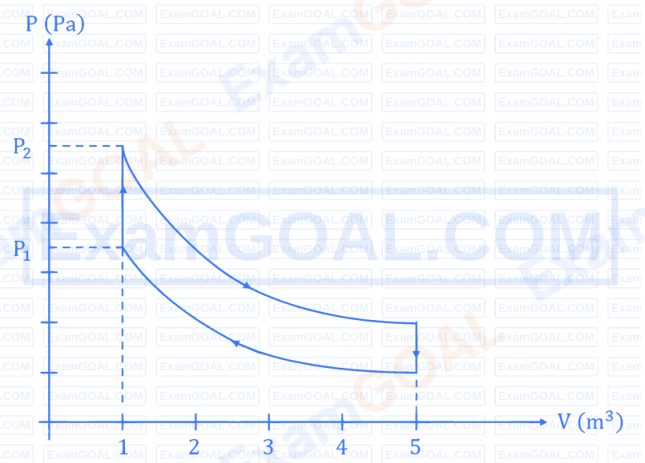


(A) $+\frac{1}{3a}$ (B) $-\frac{1}{a}$ (C) $\frac{1}{2a}$ (D) $-\frac{1}{3a}$

Q19. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule ($P_1 = 21.7\ \mathrm{Pa}$ and $P_2 = 30\ \mathrm{Pa}, \mathrm{C}_v = 21\ \mathrm{J/K.mol}, R = 8.3\ \mathrm{J/mol.K}$). The value of α is ____ .



(A) 21 (B) 28 (C) 24 (D) 15

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Q20. JEE Main 2026 (Online) 24th January Morning Shift

Density of water at $4^{\circ}\mathrm{C}$ and $20^{\circ}\mathrm{C}$ are $1000\ \mathrm{kg/m^3}$ and $998\ \mathrm{kg/m^3}$ respectively. The increase in internal energy of 4 kg of water when it is heated from $4^{\circ}\mathrm{C}$ to $20^{\circ}\mathrm{C}$ is ____ J.

(specific heat capacity of water $= 4.2\ \mathrm{kJ/kg\ K}$ and 1 atmospheric pressure $= 10^5\ \mathrm{Pa}$)

Open solution

(A) 268799.2 (B) 315826.2 (C) 234699.2 (D) 258700.8

Q21. JEE Main 2026 (Online) 23rd January Evening Shift

One mole of an ideal diatomic gas expands from volume V to $2V$ isothermally at a temperature $27^{\circ}\mathrm{C}$ and does W joule of work. If the gas undergoes same magnitude of expansion adiabatically from $27^{\circ}\mathrm{C}$ doing the same amount of work W, then its final temperature will be (close to) ____ $^{\circ}\mathrm{C}$.

$(\log_e 2 = 0.693)$

Open solution

(A) -56 (B) -117 (C) -30 (D) -189

Q22. JEE Main 2026 (Online) 23rd January Evening Shift

The internal energy of a monoatomic gas is 3nRT. One mole of helium is kept in a cylinder having internal cross section area of $17\ \mathrm{cm^2}$ and fitted with a light movable frictionless piston. The gas is heated slowly by suppling 126 J heat. If the temperature rises by $4^{\circ}\mathrm{C}$, then the piston will move ____ cm.

(atmospheric pressure $= 10^5\ \mathrm{Pa}$)

Open solution

(A) 1.55 (B) 14.5 (C) 15.5 (D) 1.45

Q23. JEE Main 2026 (Online) 23rd January Evening Shift

An air bubble of volume $2.9\ \mathrm{cm^3}$ rises from the bottom of a swimming pool of 5 m deep. At the bottom of the pool water temperature is $17^{\circ}\mathrm{C}$. The volume of the bubble when it reaches the surface, where the water temperature is $27^{\circ}\mathrm{C}$, is ____ $\mathrm{cm^3}$.

($\mathrm{g} = 10\ \mathrm{m/s^2}$, density of water $= 10^3\ \mathrm{kg/m^3}$, and 1 atm pressure is $10^5\ \mathrm{Pa}$)

Open solution

(A) 2.0 (B) 4.2 (C) 3.0 (D) 4.5

Q24. JEE Main 2026 (Online) 22nd January Evening Shift

Consider two boxes containing ideal gases A and B such that their temperatures, pressures and number densities are same. The molecular size of A is half of that of B and mass of molecule A is four times that of B. If the collision frequency in gas B is $32 \times 10^{18}/\mathrm{s}$ then collision frequency in gas A is ____ /s.

Open solution

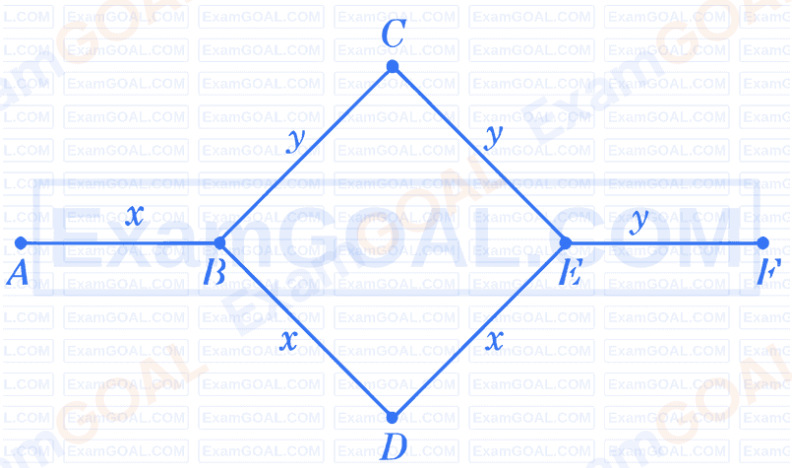
(A) 8×10^{18} (B) 2×10^{18} (C) 32×10^{18} (D) 4×10^{18}

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Q25. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Rods x and y of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points A and F are maintained at $100^\circ\mathrm{C}$ and $40^\circ\mathrm{C}$ respectively. Given the thermal conductivity of $\mathrm{rod}\, x$ is three times of that of $\mathrm{rod}\, y$, the temperature at junction points B and E are (close to):



(A) $60^\circ\mathrm{C}$ and $45^\circ\mathrm{C}$ respectively

(B) $80^\circ\mathrm{C}$ and $70^\circ\mathrm{C}$ respectively

(C) $89^\circ\mathrm{C}$ and $73^\circ\mathrm{C}$ respectively

(D) $80^\circ\mathrm{C}$ and $60^\circ\mathrm{C}$ respectively

Q26. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

The volume of an ideal gas increases 8 times and temperature becomes $(1/4)^{\text{th}}$ of initial temperature during a reversible change. If there is no exchange of heat in this process $(\Delta Q = 0)$ then identify the gas from the following options (Assuming the gases given in the options are ideal gases) :

(A) NH_3 (B) O_2 (C) CO_2 (D) He

Q27. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

The r.m.s. speed of oxygen molecules at 47 °C is equal to that of the hydrogen molecules kept at _____ °C. (Mass of oxygen molecule/mass of hydrogen molecule = 32/2)

(A) -100 (B) -253 (C) -20 (D) -235

Q28. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at $30^\circ\mathrm{C}$. The coefficient of linear expansion of aluminium and steel are $24 \times 10^{-6}/^\circ\mathrm{C}$ and $1.2 \times 10^{-5}/^\circ\mathrm{C}$, respectively. The length of this composite rod when its temperature is raised to $100^\circ\mathrm{C}$, is ____ cm.

(A) 120.20 (B) 120.06 (C) 120.15 (D) 120.03

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Q29. JEE Main 2026 (Online) 21st January Morning Shift

A gas based geyser heats water flowing at the rate of 5.0 litres per minute from $27^{\circ}\mathrm{C}$ to $87^{\circ}\mathrm{C}$. The rate of consumption of the gas is ____ $\mathrm{g/s}$.

(Take heat of combustion of gas $= 5.0 \times 10^4\ \mathrm{J/g}$, specific heat capacity of water $= 4200\ \mathrm{J/kg \cdot {}^{\circ}C}$)

Open solution

(A) 4.2 (B) 2.1 (C) 0.21 (D) 0.42

Q30. JEE Main 2025 (Online) 8th April Evening Shift

A monoatomic gas having $\gamma = \frac{5}{3}$ is stored in a thermally insulated container and the gas is suddenly compressed to $\left(\frac{1}{8}\right)^{\text{th}}$ of its initial volume. The ratio of final pressure and initial pressure is:

(γ is the ratio of specific heats of the gas at constant pressure and at constant volume)

Open solution

(A) 16 (B) 32 (C) 28 (D) 40

Q31. JEE Main 2025 (Online) 8th April Evening Shift

Water falls from a height of 200 m into a pool. Calculate the rise in temperature of the water assuming no heat dissipation from the water in the pool.

(Take g = 10 m/s^2, specific heat of water = 4200 J/(kg K))

Open solution

(A) 0.36 K (B) 0.23 K (C) 0.48 K (D) 0.14 K

Q32. JEE Main 2025 (Online) 7th April Evening Shift

The helium and argon are put in the flask at the same room temperature (300 K). The ratio of average kinetic energies (per molecule) of helium and argon is:

(Give: Molar mass of helium = 4 g/mol, Molar mass of argon = 40 g/mol)

Open solution

(A) $1 : \sqrt{10}$ (B) 10 : 1 (C) 1 : 10 (D) 1 : 1

Q33. JEE Main 2025 (Online) 7th April Evening Shift

Match **List - I** with **List - II**.

List - I	List - II
(A) Isothermal	(I) ΔW (work done) = 0
(B) Adiabatic	(II) ΔQ (supplied heat) = 0
(C) Isobaric	(III) ΔU (change in internal energy) $\neq$ 0
(D) Isochoric	(IV) ΔU = 0

Open solution

Choose the **correct** answer from the options given below :

(A) (A)-(III), (B)-(II), (C)-(I), (D)-(IV) (B) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

(C) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) (D) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

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Q34. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Match the List I with List II

	List - I		List - II
(A)	Triatomic rigid gas	(I)	$\frac{C_p}{C_v} = \frac{5}{3}$
(B)	Diatomic non-rigid gas	(II)	$\frac{C_p}{C_v} = \frac{7}{5}$
(C)	Monoatomic gas	(III)	$\frac{C_p}{C_v} = \frac{4}{3}$
(D)	Diatomic rigid gas	(IV)	$\frac{C_p}{C_v} = \frac{9}{7}$

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-II, B-IV, C-I, D-III

(C) A-IV, B-II, C-III, D-I

(D) A-III, B-II, C-IV, D-I

Q35. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Consider a rectangular sheet of solid material of length $l = 9\,\text{cm}$ and width $\text{d} = 4\,\text{cm}$. The coefficient of linear expansion is $\alpha = 3.1 \times 10^{-5}\,\text{K}^{-1}$ at room temperature and one atmospheric pressure. The mass of sheet $m = 0.1\,\text{kg}$ and the specific heat capacity $C_\text{v} = 900\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$. If the amount of heat supplied to the material is $8.1 \times 10^2\,\text{J}$ then change in area of the rectangular sheet is :

(A) $2.0 \times 10^{-6}\,\text{m}^2$

(B) $6.0 \times 10^{-7}\,\text{m}^2$

(C) $3.0 \times 10^{-7}\,\text{m}^2$

(D) $4.0 \times 10^{-7}\,\text{m}^2$

Q36. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

There are two vessels filled with an ideal gas where volume of one is double the volume of other. The large vessel contains the gas at 8 kPa at 1000 K while the smaller vessel contains the gas at 7 kPa at 500 K . If the vessels are connected to each other by a thin tube allowing the gas to flow and the temperature of both vessels is maintained at 600 K , at steady state the pressure in the vessels will be (in kPa).

(A) 24

(B) 4.4

(C) 18

(D) 6

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Q37. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Match List - I with List - II.

	List - I		List - II
(A)	Isobaric	(I)	$\Delta Q = \Delta W$
(B)	Isochoric	(II)	$\Delta Q = \Delta U$
(C)	Adiabatic	(III)	$\Delta Q =$ zero
(D)	Isothermal	(IV)	$\Delta Q = \Delta U + P\Delta V$

$\Delta Q =$ Heat supplied

$\Delta W =$ Work done by the system

$\Delta \mathrm{U} =$ Change in internal energy

$\mathrm{P} =$ Pressure of the system

$\Delta \mathrm{V} =$ Change in volume of the system

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(B) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(D) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Q38. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Consider the sound wave travelling in ideal gases of $\mathrm{He}, \mathrm{CH_4}$, and $\mathrm{CO_2}$. All the gases have the same ratio $\frac{P}{\rho}$, where P is the pressure and ρ is the density. The ratio of the speed of sound through the gases $\mathrm{V_{He}} : \mathrm{V_{CH_4}} : \mathrm{V_{CO_2}}$ is given by

(A) $\sqrt{\frac{7}{5}} : \sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}}$

(B) $\sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}} : \sqrt{\frac{4}{3}}$

(C) $\sqrt{\frac{5}{3}} : \sqrt{\frac{4}{3}} : \sqrt{\frac{7}{5}}$

(D) $\sqrt{\frac{4}{3}} : \sqrt{\frac{5}{3}} : \sqrt{\frac{7}{5}}$

Q39. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

The mean free path and the average speed of oxygen molecules at 300 K and 1 atm are $3 \times 10^{-7}\ \mathrm{m}$ and $600\ \mathrm{m/s}$, respectively. Find the frequency of its collisions.

(A) $5 \times 10^{8}/\mathrm{s}$

(B) $9 \times 10^{5}/\mathrm{s}$

(C) $2 \times 10^{10}/\mathrm{s}$

(D) $2 \times 10^{9}/\mathrm{s}$

Q40. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

An ideal gas exists in a state with pressure P_0, volume V_0. It is isothermally expanded to 4 times of its initial volume $(\mathrm{V_0})$, then isobarically compressed to its original volume. Finally the system is heated isochorically to bring it to its initial state. The amount of heat exchanged in this process is

(A) $\mathrm{P_0\,V_0}(\ln 2 - 0.75)$

(B) $\mathrm{P_0\,V_0}(2\ln 2 - 0.75)$

(C) $\mathrm{P_0\,V_0}(2\ln 2 - 0.25)$

(D) $\mathrm{P_0\,V_0}(\ln 2 - 0.25)$

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Q41. JEE Main 2025 (Online) 3rd April Evening Shift

Pressure of an ideal gas, contained in a closed vessel, is increased by 0.4% when heated by $1^{\circ}\mathrm{C}$. Its initial temperature must be:

(A) 2500 K (B) $25^{\circ}\mathrm{C}$ (C) $250^{\circ}\mathrm{C}$ (D) 250 K

Open solution

Q42. JEE Main 2025 (Online) 3rd April Morning Shift

A gas is kept in a container having walls which are thermally non-conducting. Initially the gas has a volume of $800\ \mathrm{cm}^3$ and temperature $27^{\circ}\mathrm{C}$. The change in temperature when the gas is adiabatically compressed to $200\ \mathrm{cm}^3$ is:

(Take $\gamma = 1.5$; γ is the ratio of specific heats at constant pressure and at constant volume)

(A) 300 K (B) 600 K (C) 327 K (D) 522 K

Open solution

Q43. JEE Main 2025 (Online) 3rd April Morning Shift

During the melting of a slab of ice at 273 K at atmospheric pressure :

(A) Internal energy of ice-water system remains unchanged.

(B) Positive work is done by the ice-water system on the atmosphere.

(C) Positive work is done on the ice-water system by the atmosphere.

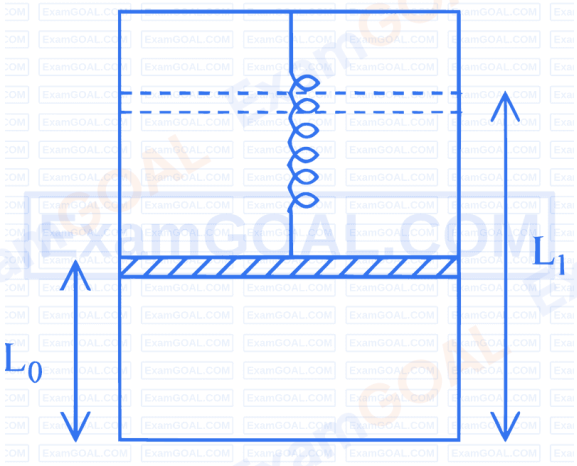
(D) Internal energy of the ice-water system decreases.

Open solution

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Q44. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution



A piston of mass M is hung from a massless spring whose restoring force law goes as $F = -kx^3$, where k is the spring constant of appropriate dimension. The piston separates the vertical chamber into two parts, where the bottom part is filled with ' n ' moles of an ideal gas. An external work is done on the gas isothermally (at a constant temperature T) with the help of a heating filament (with negligible volume) mounted in lower part of the chamber, so that the piston goes up from a height L_0 to L_1, the total energy delivered by the filament is:(Assume spring to be in its natural length before heating)

(A) $nRT\ln\left(\frac{L_1}{L_0}\right) + Mg(L_1 - L_0) + \frac{3k}{4}\left({L_1}^4 - {L_0}^4\right)$

(B) $nRT\ln\left(\frac{L_1}{L_0}\right) + Mg(L_1 - L_0) + \frac{k}{4}\left(L_1^4 - {L_0}^4\right)$

(C) $nRT\ln\left(\frac{L_1^2}{L_0^2}\right) + \frac{Mg}{2}(L_1 - L_0) + \frac{k}{4}\left(L_1^4 - {L_0}^4\right)$

(D) $3nRT\ln\left(\frac{L_1}{L_0}\right) + 2Mg(L_1 - L_0) + \frac{k}{3}\left({L_1}^3 - {L_0}^3\right)$

Q45. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Match List - I with List - II.

List - I		$List - II$
(A)	Heat capacity of body	(I) $\mathrm{J\,kg^{-1}}$
(B)	Specific heat capacity of body	(II) $\mathrm{J\,K^{-1}}$
(C)	Latent heat	(III) $\mathrm{J\,kg^{-1}\,K^{-1}}$
(D)	Thermal conductivity	(IV) $\mathrm{J\,m^{-1}\,K^{-1}\,s^{-1}}$

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(B) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

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Q46. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Identify the characteristics of an adiabatic process in a monoatomic gas.
(A) Internal energy is constant.
(B) Work done in the process is equal to the change in internal energy.
(C) The product of temperature and volume is a constant.
(D) The product of pressure and volume is a constant.
(E) The work done to change the temperature from T_1 to T_2 is proportional to $(T_2 - T_1)$.
Choose the correct answer from the options given below :

(A) (B), (D) only
(B) (B), (E) only
(C) (A), (C), (E) only
(D) (A), (C), (D) only

Q47. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

In an adiabatic process, which of the following statements is true?

(A) The internal energy of the gas decreases as the temperature increases
(B) The molar heat capacity is zero
(C) Work done by the gas equals the increase in internal energy
(D) The molar heat capacity is infinite

Q48. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

The difference of temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have

(A) low thermal conductivity and high electrical conductivity
(B) low thermal conductivity and low electrical conductivity
(C) high thermal conductivity and high electrical conductivity
(D) high thermal conductivity and low electrical conductivity

Q49. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process.

Reason (R) : In isothermal process, PV = constant, while in adiabatic process PV^{γ} = constant. Here γ is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) (A) is true but (R) is false
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(C) (A) is false but (R) is true
(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

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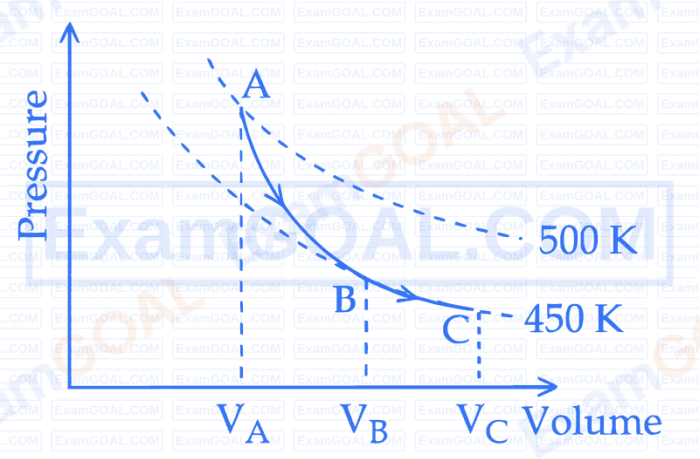
Q50. JEE Main 2025 (Online) 29th January Evening Shift

A cup of coffee cools from 90°C to 80°C in t minutes when the room temperature is 20°C. The time taken by the similar cup of coffee to cool from 80°C to 60°C at the same room temperature is:

(A) $\frac{13}{5}t$ (B) $\frac{10}{13}t$ (C) $\frac{5}{13}t$ (D) $\frac{13}{10}t$

Open solution

Q51. JEE Main 2025 (Online) 29th January Evening Shift



A poly-atomic molecule ($C_V = 3R, C_P = 4R$, where R is gas constant) goes from phase space point A $\left(P_A = 10^5\ Pa,\ V_A = 4 \times 10^{-6}\ m^3\right)$ to point B $\left(P_B = 5 \times 10^4\ Pa,\ V_B = 6 \times 10^{-6}\ m^3\right)$ to point C $\left(P_C = 10^4\ Pa, V_C = 8 \times 10^{-6}\ m^3\right)$. A to B is an adiabatic path and B to C is an isothermal path.

The net heat absorbed per unit mole by the system is :

(A) $500R(\ln 3 + \ln 4)$ (B) $450R(\ln 4 - \ln 3)$ (C) $500R \ln 2$ (D) $400R \ln 4$

Open solution

Q52. JEE Main 2025 (Online) 29th January Morning Shift

The work done in an adiabatic change in an ideal gas depends upon only :

(A) change in its pressure
(B) change in its temperature
(C) change in its specific heat
(D) change in its volume

Open solution

Q53. JEE Main 2025 (Online) 28th January Evening Shift

The ratio of vapour densities of two gases at the same temperature is $\frac{4}{25}$, then the ratio of r.m.s. velocities will be :

(A) $\frac{5}{2}$ (B) $\frac{25}{4}$ (C) $\frac{4}{25}$ (D) $\frac{2}{5}$

Open solution

Q54. JEE Main 2025 (Online) 28th January Evening Shift

The kinetic energy of translation of the molecules in 50 g of CO_2 gas at 17°C is :

(A) 4205.5 J (B) 3582.7 J (C) 3986.3 J (D) 4102.8 J

Open solution

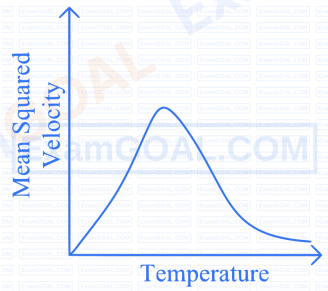
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Q55. JEE Main 2025 (Online) 28th January Morning Shift

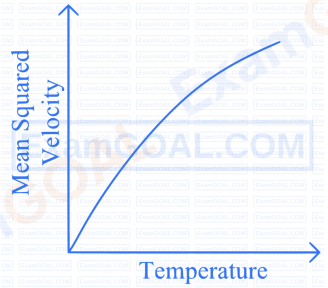
For a particular ideal gas which of the following graphs represents the variation of mean squared velocity of the gas molecules with temperature?

Open solution

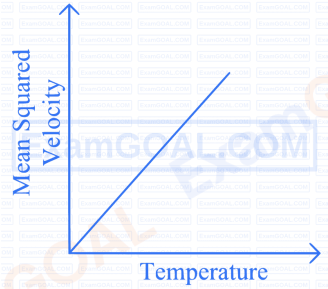
(A)



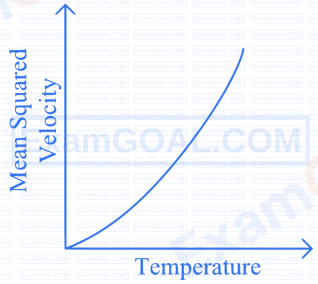
(B)



(C)



(D)



Q56. JEE Main 2025 (Online) 28th January Morning Shift

A Carnot engine (E) is working between two temperatures 473 K and 273 K . In a new system two engines - engine E_1 works between 473 K to 373 K and engine E_2 works between 373 K to 273 K . If η_{12}, η_1 and η_2 are the efficiencies of the engines E, E_1 and E_2, respectively, then

Open solution

(A) $\eta_{12} = \eta_1\eta_2$ (B) $\eta_{12} = \eta_1 + \eta_2$ (C) $\eta_{12} \geq \eta_1 + \eta_2$ (D) $\eta_{12} < \eta_1 + \eta_2$

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Q57. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

The temperature of a body in air falls from $40^{\circ}\mathrm{C}$ to $24^{\circ}\mathrm{C}$ in 4 minutes. The temperature of the air is $16^{\circ}\mathrm{C}$. The temperature of the body in the next 4 minutes will be :

(A) $\frac{28}{3}^{\circ}\mathrm{C}$

(B) $\frac{56}{3}^{\circ}\mathrm{C}$

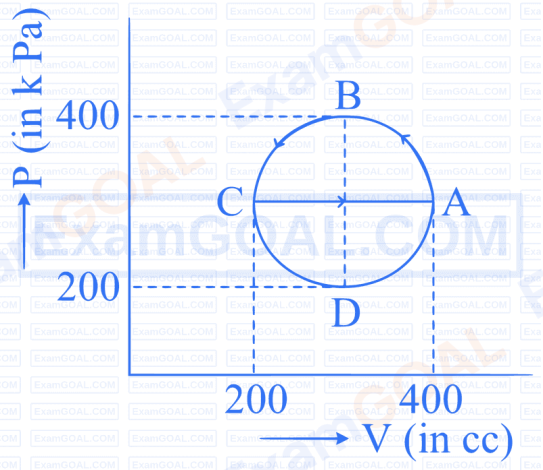
(C) $\frac{42}{3}^{\circ}\mathrm{C}$

(D) $\frac{14}{3}^{\circ}\mathrm{C}$

Q58. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

The magnitude of heat exchanged by a system for the given cyclic process ABCA (as shown in figure) is (in SI unit) :



(A) zero

(B) 40π

(C) 5π

(D) 10π

Q59. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In an insulated container, a gas is adiabatically shrunk to half of its initial volume. The temperature of the gas decreases.

Reason (R) : Free expansion of an ideal gas is an irreversible and an adiabatic process.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both $(\mathbf{A})$ and $(\mathbf{R})$ are true and $(\mathbf{R})$ is the correct explanation of $(\mathbf{A})$

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is false but (R) is true

(D) (A) is true but $(\mathbf{R})$ is false

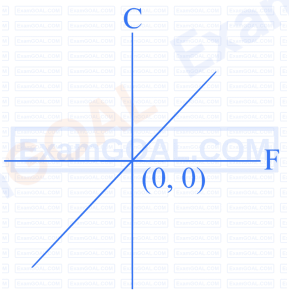
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Q60. JEE Main 2025 (Online) 24th January Evening Shift

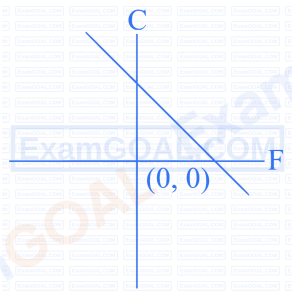
Open solution

Which of the following figure represents the relation between Celsius and Fahrenheit temperatures?

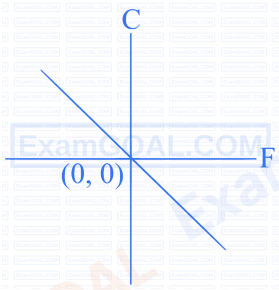
(A)



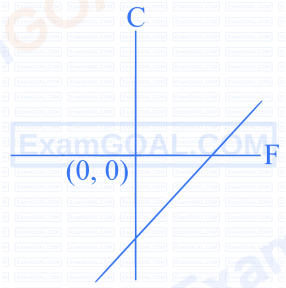
(B)



(C)



(D)



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Q61. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.

A. The work done by gas during the process is zero.

B. The heat added to gas is different from change in its internal energy.

C. The volume of the gas is increased.

D. The internal energy of the gas is increased.

E. The process is isochoric (constant volume process)

Choose the correct answer from the options given below:

(A) A, C Only (B) A, D, E Only (C) E Only (D) A, B, C, D Only

Q62. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

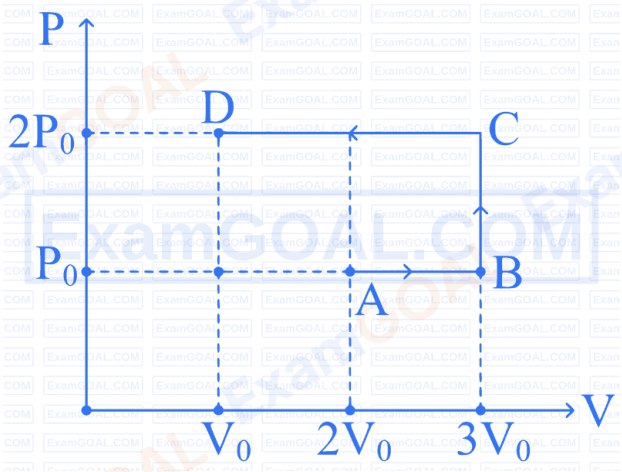
Water of mass m gram is slowly heated to increase the temperature from T_1 to T_2. The change in entropy of the water, given specific heat of water is $1 \mathrm{Jkg}^{-1} \mathrm{K}^{-1}$, is :

(A) $\mathrm{m}\left(\mathrm{T}_2-\mathrm{T}_1\right)$ (B) zero

(C) $\mathrm{m} \ln \left(\frac{\mathrm{T}_1}{\mathrm{T}_2}\right)$ (D) $\mathrm{m} \ln \left(\frac{\mathrm{T}_2}{\mathrm{T}_1}\right)$

Q63. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution



Using the given $P-V$ diagram, the work done by an ideal gas along the path $ABCD$ is :

(A) $3\mathrm{P}_0 \mathrm{V}_0$ (B) $-3\mathrm{P}_0 \mathrm{V}_0$ (C) $-4\mathrm{P}_0 \mathrm{V}_0$ (D) $4\mathrm{P}_0 \mathrm{V}_0$

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Q64. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Match the List - I with List - II

	List - I		List - II
(A)	Pressure varies inversely with volume of an ideal gas.	(I)	Adiabatic process
(B)	Heat absorbed goes partly to increase internal energy and partly to do work.	(II)	Isochoric process
(C)	Heat is neither absorbed nor released by a system.	(III)	Isothermal process
(D)	No work is done on or by a gas.	(IV)	Isobaric process

Choose the correct answer from the options given below:

(A) A-I, B-III, C-II, D-IV

(B) A-I, B-IV, C-II, D-III

(C) A-III, B-I, C-IV, D-II

(D) A-III, B-IV, C-I, D-II

Q65. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

A gun fires a lead bullet of temperature 300 K into a wooden block. The bullet having melting temperature of 600 K penetrates into the block and melts down. If the total heat required for the process is 625 J , then the mass of the bullet is _____ grams. (Latent heat of fusion of lead $= 2.5 \times 10^4 \mathrm{JKg}^{-1}$ and specific heat capacity of lead $= 125 \mathrm{JKg}^{-1}\,\mathrm{K}^{-1}$)

(A) 20 **(B)** 15 **(C)** 5 **(D)** 10

Q66. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given are statements for certain thermodynamic variables,

(A) Internal energy, volume (V) and mass (M) are extensive variables.

(B) Pressure (P), temperature (T) and density (ρ) are intensive variables.

(C) Volume (V), temperature (T) and density (ρ) are intensive variables.

(D) Mass (M), temperature (T) and internal energy are extensive variables.

Choose the correct answer from the options given below :

(A) (C) and (D) Only **(B)** (A) and (B) Only **(C)** (D) and (A) Only **(D)** (B) and (C) Only

Q67. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

For a diatomic gas, if $\gamma_1 = \left(\frac{Cp}{Cv}\right)$ for rigid molecules and $\gamma_2 = \left(\frac{Cp}{Cv}\right)$ for another diatomic molecules, but also having vibrational modes. Then, which one of the following options is correct? (Cp and Cv are specific heats of the gas at constant pressure and volume)

(A) $\gamma_2 < \gamma_1$ **(B)** $\gamma_2 > \gamma_1$ **(C)** $\gamma_2 = \gamma_1$ **(D)** $2\gamma_2 = \gamma_1$

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Q68. JEE Main 2025 (Online) 22nd January Morning Shift

Two spherical bodies of same materials having radii 0.2 m and 0.8 m are placed in same atmosphere. The temperature of the smaller body is 800 K and temperature of the bigger body is 400 K . If the energy radiated from the smaller body is E, the energy radiated from the bigger body is (assume, effect of the surrounding temperature to be negligible),

Open solution

(A) 64 E (B) 16 E (C) E (D) 256 E

Q69. JEE Main 2025 (Online) 22nd January Morning Shift

An amount of ice of mass $10^{-3}\ \mathrm{kg}$ and temperature $-10^{\circ}\mathrm{C}$ is transformed to vapour of temperature $110^{\circ}\mathrm{C}$ by applying heat. The total amount of work required for this conversion is, (Take, specific heat of ice $= 2100\mathrm{Jkg}^{-1}\ \mathrm{K}^{-1}$, specific heat of water $= 4180\mathrm{Jkg}^{-1}\ \mathrm{K}^{-1}$, specific heat of steam $= 1920\mathrm{Jkg}^{-1}\ \mathrm{K}^{-1}$, Latent heat of ice $= 3.35 \times 10^{5}\mathrm{Jkg}^{-1}$ and Latent heat of steam $= 2.25 \times 10^{6}\ \mathrm{Jkg}^{-1}$)

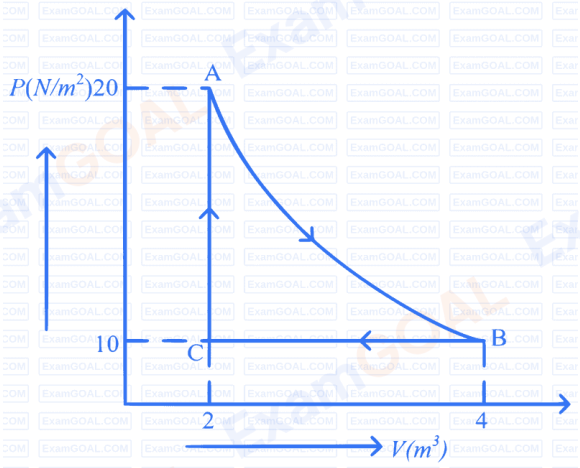
Open solution

(A) 3022 J (B) 3043 J (C) 3003 J (D) 3024 J

Q70. JEE Main 2024 (Online) 9th April Evening Shift

A real gas within a closed chamber at $27^{\circ}\mathrm{C}$ undergoes the cyclic process as shown in figure. The gas obeys $PV^3 = RT$ equation for the path A to B. The net work done in the complete cycle is (assuming $R = 8\ \mathrm{J/molK}$):

Open solution



(A) $-20\mathrm{J}$ (B) 205J (C) 225J (D) 20J

Q71. JEE Main 2024 (Online) 9th April Evening Shift

The temperature of a gas is $-78^{\circ}\mathrm{C}$ and the average translational kinetic energy of its molecules is K. The temperature at which the average translational kinetic energy of the molecules of the same gas becomes $2\ \mathrm{K}$ is :

Open solution

(A) $-78^{\circ}\mathrm{C}$ (B) $127^{\circ}\mathrm{C}$ (C) $-39^{\circ}\mathrm{C}$ (D) $117^{\circ}\mathrm{C}$

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Q72. JEE Main 2024 (Online) 9th April Morning Shift

The volume of an ideal gas $(\gamma = 1.5)$ is changed adiabatically from 5 litres to 4 litres. The ratio of initial pressure to final pressure is :

(A) $\frac{4}{5}$ (B) $\frac{8}{5\sqrt{5}}$ (C) $\frac{2}{\sqrt{5}}$ (D) $\frac{16}{25}$

Open solution

Q73. JEE Main 2024 (Online) 9th April Morning Shift

A sample of 1 mole gas at temperature T is adiabatically expanded to double its volume. If adiabatic constant for the gas is $\gamma = \frac{3}{2}$, then the work done by the gas in the process is :

(A) $\mathrm{RT}[2+\sqrt{2}]$ (B) $\mathrm{RT}[2-\sqrt{2}]$

(C) $\frac{\mathrm{R}}{\mathrm{T}}[2-\sqrt{2}]$ (D) $\frac{T}{R}[2+\sqrt{2}]$

Open solution

Q74. JEE Main 2024 (Online) 8th April Evening Shift

A diatomic gas $(\gamma = 1.4)$ does $100\ \mathrm{J}$ of work in an isobaric expansion. The heat given to the gas is :

(A) 150 J (B) 490 J (C) 350 J (D) 250 J

Open solution

Q75. JEE Main 2024 (Online) 8th April Evening Shift

Given below are two statements :

Statement (I) : The mean free path of gas molecules is inversely proportional to square of molecular diameter.

Statement (II) : Average kinetic energy of gas molecules is directly proportional to absolute temperature of gas.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Open solution

Q76. JEE Main 2024 (Online) 8th April Morning Shift

A mixture of one mole of monoatomic gas and one mole of a diatomic gas (rigid) are kept at room temperature $(27^{\circ}\mathrm{C})$. The ratio of specific heat of gases at constant volume respectively is:

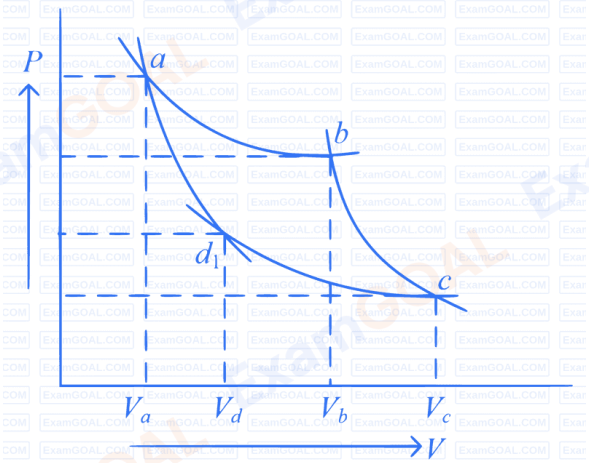
(A) $\frac{3}{2}$ (B) $\frac{3}{5}$ (C) $\frac{7}{5}$ (D) $\frac{5}{3}$

Open solution

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Q77. JEE Main 2024 (Online) 8th April Morning Shift

Two different adiabatic paths for the same gas intersect two isothermal curves as shown in P-V diagram. The relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is:



(A) $\frac{V_a}{V_d} \neq \frac{V_b}{V_c}$

(B) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^{-1}$

(C) $\frac{V_a}{V_d} = \frac{V_b}{V_c}$

(D) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^{2}$

Open solution

Q78. JEE Main 2024 (Online) 6th April Evening Shift

Energy of 10 non rigid diatomic molecules at temperature T is :

(A) 35 RT

(B) $\frac{7}{2}$ RT

(C) 70 K_BT

(D) 35 K_BT

Open solution

Q79. JEE Main 2024 (Online) 6th April Evening Shift

A total of $48\ \mathrm{J}$ heat is given to one mole of helium kept in a cylinder. The temperature of helium increases by $2^{\circ}\mathrm{C}$. The work done by the gas is: Given, $\mathrm{R} = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$.

(A) 23.1 J

(B) 48 J

(C) 24.9 J

(D) 72.9 J

Open solution

Q80. JEE Main 2024 (Online) 6th April Morning Shift

The specific heat at constant pressure of a real gas obeying $PV^2 = RT$ equation is:

(A) R

(B) $C_V + R$

(C) $C_V + \frac{R}{2V}$

(D) $\frac{R}{3} + C_V$

Open solution

Q81. JEE Main 2024 (Online) 6th April Morning Shift

A sample contains mixture of helium and oxygen gas. The ratio of root mean square speed of helium and oxygen in the sample, is :

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{4}$

(C) $\frac{2\sqrt{2}}{1}$

(D) $\frac{1}{32}$

Open solution

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Q82. JEE Main 2024 (Online) 5th April Evening Shift

During an adiabatic process, if the pressure of a gas is found to be proportional to the cube of its absolute temperature, then the ratio of $\frac{C_P}{C_V}$ for the gas is :

(A) $\frac{5}{3}$ (B) $\frac{3}{2}$ (C) $\frac{7}{5}$ (D) $\frac{9}{7}$

Open solution

Q83. JEE Main 2024 (Online) 5th April Evening Shift

If n is the number density and d is the diameter of the molecule, then the average distance covered by a molecule between two successive collisions (i.e. mean free path) is represented by :

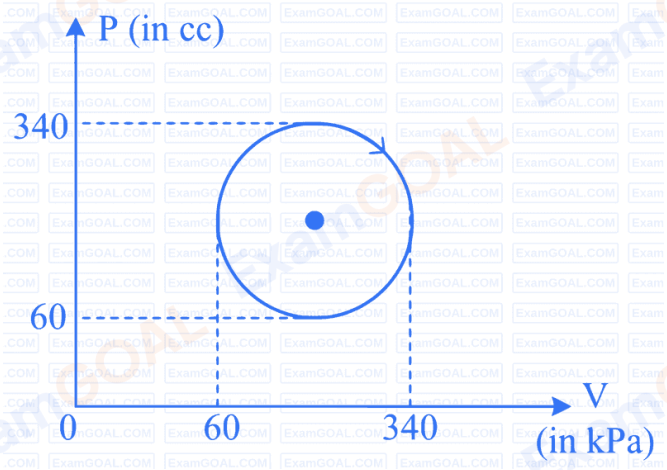
(A) $\frac{1}{\sqrt{2}\mathrm{n}\pi\mathrm{d}^2}$ (B) $\frac{1}{\sqrt{2}n^2\pi^2d^2}$

(C) $\frac{1}{\sqrt{2}n\pi d^2}$ (D) $\sqrt{2}\mathrm{n}\pi\mathrm{d}^2$

Open solution

Q84. JEE Main 2024 (Online) 5th April Morning Shift

The heat absorbed by a system in going through the given cyclic process is :



(A) 61.6 J (B) 431.2 J (C) 19.6 J (D) 616 J

Open solution

Q85. JEE Main 2024 (Online) 5th April Morning Shift

If the collision frequency of hydrogen molecules in a closed chamber at $27^{\circ}\mathrm{C}$ is Z, then the collision frequency of the same system at $127^{\circ}\mathrm{C}$ is :

(A) $\frac{\sqrt{3}}{2}\mathrm{Z}$ (B) $\frac{2}{\sqrt{3}}\mathrm{Z}$ (C) $\frac{3}{4}\mathrm{Z}$ (D) $\frac{4}{3}\mathrm{Z}$

Open solution

Q86. JEE Main 2024 (Online) 4th April Evening Shift

A sample of gas at temperature T is adiabatically expanded to double its volume. Adiabatic constant for the gas is $\gamma = 3/2$. The work done by the gas in the process is:

$(\mu = 1$ mole $)$

(A) $RT[2\sqrt{2}-1]$ (B) $RT[2-\sqrt{2}]$ (C) $RT[1-2\sqrt{2}]$ (D) $RT[\sqrt{2}-2]$

Open solution

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Q87. JEE Main 2024 (Online) 4th April Evening Shift

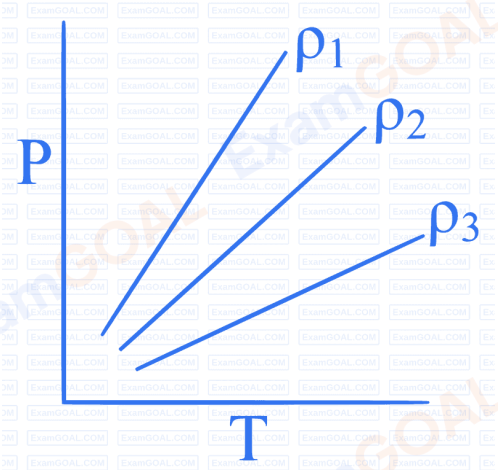
The translational degrees of freedom (f_t) and rotational degrees of freedom (f_r) of CH_4 molecule are:

(A) $f_t = 2$ and $f_r = 2$ (B) $f_t = 3$ and $f_r = 3$ (C) $f_t = 3$ and $f_r = 2$ (D) $f_t = 2$ and $f_r = 3$

Open solution

Q88. JEE Main 2024 (Online) 4th April Morning Shift

P-T diagram of an ideal gas having three different densities ρ_1, ρ_2, ρ_3 (in three different cases) is shown in the figure. Which of the following is correct :



(A) $\rho_1 > \rho_2$ (B) $\rho_2 < \rho_3$ (C) $\rho_1 = \rho_2 = \rho_3$ (D) $\rho_1 < \rho_2$

Open solution

Q89. JEE Main 2024 (Online) 4th April Morning Shift

The resistances of the platinum wire of a platinum resistance thermometer at the ice point and steam point are 8Ω and 10Ω respectively. After inserting in a hot bath of temperature $400^\circ C$, the resistance of platinum wire is :

(A) 10 Ω (B) 16 Ω (C) 8 Ω (D) 2 Ω

Open solution

Q90. JEE Main 2024 (Online) 4th April Morning Shift

On Celsius scale the temperature of body increases by $40^\circ C$. The increase in temperature on Fahrenheit scale is :

(A) $75^\circ F$ (B) $70^\circ F$ (C) $72^\circ F$ (D) $68^\circ F$

Open solution

Q91. JEE Main 2024 (Online) 1st February Evening Shift

A diatomic gas $(\gamma = 1.4)$ does 200 J of work when it is expanded isobarically. The heat given to the gas in the process is :

(A) 800 J (B) 600 J (C) 700 J (D) 850 J

Open solution

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Q92. JEE Main 2024 (Online) 1st February Evening Shift

If the root mean square velocity of hydrogen molecule at a given temperature and pressure is $2\ \mathrm{km/s}$, the root mean square velocity of oxygen at the same condition in $\mathrm{km/s}$ is :

(A) 1.0 (B) 1.5 (C) 2.0 (D) 0.5

Open solution

Q93. JEE Main 2024 (Online) 1st February Morning Shift

Two moles of a monoatomic gas is mixed with six moles of a diatomic gas. The molar specific heat of the mixture at constant volume is :

(A) $\frac{3}{2}\mathrm{R}$ (B) $\frac{7}{4}\mathrm{R}$ (C) $\frac{5}{2}\mathrm{R}$ (D) $\frac{9}{4}\mathrm{R}$

Open solution

Q94. JEE Main 2024 (Online) 1st February Morning Shift

The pressure and volume of an ideal gas are related as $\mathrm{PV}^{\frac{3}{2}} = \mathrm{K}$ (Constant). The work done when the gas is taken from state $A\,(P_1, V_1, T_1)$ to state $B\,(P_2, V_2, T_2)$ is :

(A) $2\left(\mathrm{P_2}\sqrt{\mathrm{V_2}} - \mathrm{P_1}\sqrt{\mathrm{V_1}}\right)$

(B) $2\left(\sqrt{\mathrm{P_1}}\,\mathrm{V_1} - \sqrt{\mathrm{P_2}}\,\mathrm{V_2}\right)$

(C) $2\left(\mathrm{P_2\,V_2} - \mathrm{P_1\,V_1}\right)$

(D) $2\left(\mathrm{P_1\,V_1} - \mathrm{P_2\,V_2}\right)$

Open solution

Q95. JEE Main 2024 (Online) 31st January Evening Shift

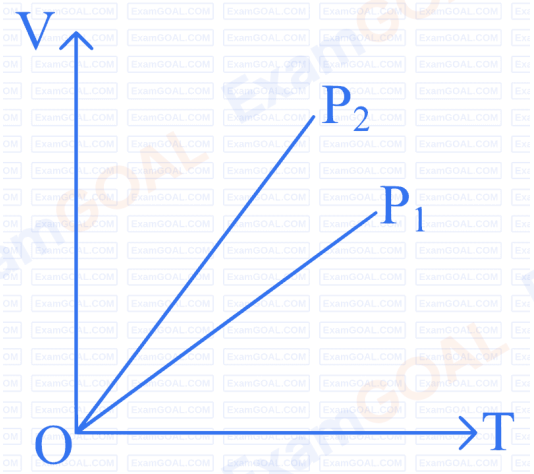
A gas mixture consists of 8 moles of argon and 6 moles of oxygen at temperature T. Neglecting all vibrational modes, the total internal energy of the system is:

(A) 29 RT (B) 27 RT (C) 20 RT (D) 21 RT

Open solution

Q96. JEE Main 2024 (Online) 31st January Morning Shift

The given figure represents two isobaric processes for the same mass of an ideal gas, then



(A) $P_2 > P_1$ (B) $P_1 > P_2$ (C) $P_1 = P_2$ (D) $P_2 \geq P_1$

Open solution

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Q97. JEE Main 2024 (Online) 31st January Morning Shift

The parameter that remains the same for molecules of all gases at a given temperature is :

(A) kinetic energy (B) mass (C) momentum (D) speed

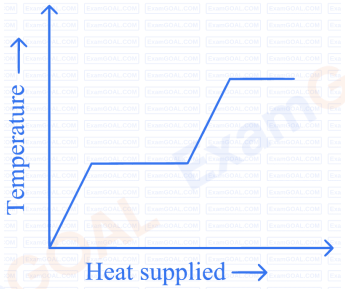
Open solution

Q98. JEE Main 2024 (Online) 30th January Evening Shift

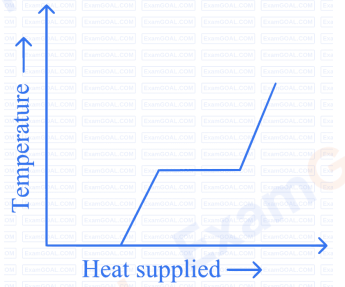
A block of ice at $-10^{\circ}\mathrm{C}$ is slowly heated and converted to steam at $100^{\circ}\mathrm{C}$. Which of the following curves represent the phenomenon qualitatively:

Open solution

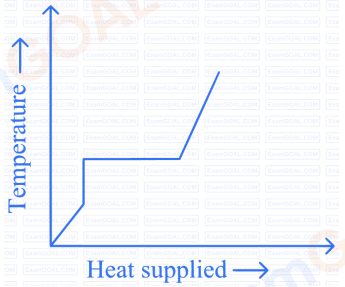
(A)



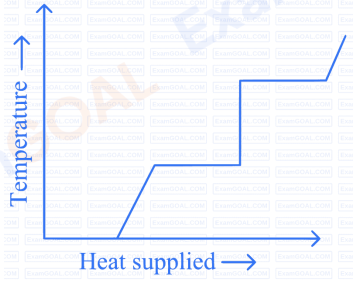
(B)



(C)



(D)



Q99. JEE Main 2024 (Online) 30th January Evening Shift

If three moles of monoatomic gas $\left(\gamma=\frac{5}{3}\right)$ is mixed with two moles of a diatomic gas $\left(\gamma=\frac{7}{5}\right)$, the value of adiabatic exponent γ for the mixture is

(A) 1.35 (B) 1.52 (C) 1.40 (D) 1.75

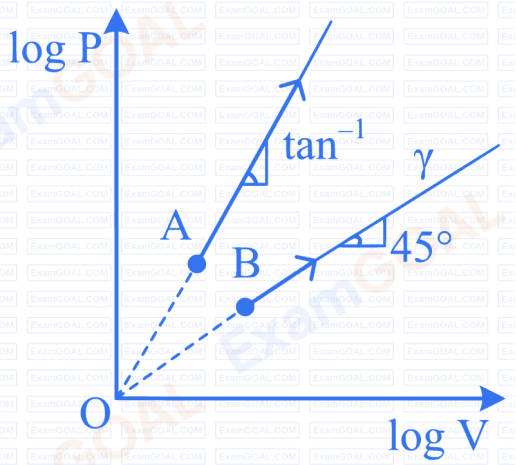
Open solution

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Q100. JEE Main 2024 (Online) 30th January Morning Shift

Two thermodynamical processes are shown in the figure. The molar heat capacity for process A and B are C_A and C_B. The molar heat capacity at constant pressure and constant volume are represented by C_P and C_V, respectively. Choose the correct statement.

Open solution



(A) $C_P > C_B > C_A > C_V$

(B) $C_P > C_V > C_A = C_B$

(C) $C_A = 0$ and $C_B = \infty$

(D) $C_A = \infty, C_B = 0$

Q101. JEE Main 2024 (Online) 30th January Morning Shift

At which temperature is the r.m.s. velocity of a hydrogen molecule equal to that of an oxygen molecule at $47^\circ C$?

Open solution

(A) 20 K (B) 80 K (C) 4 K (D) -73 K

Q102. JEE Main 2024 (Online) 29th January Evening Shift

The temperature of a gas having 2.0×10^{25} molecules per cubic meter at $1.38\ atm$ (Given, $k = 1.38 \times 10^{-23} JK^{-1}$) is :

Open solution

(A) 500 K (B) 300 K (C) 200 K (D) 100 K

Q103. JEE Main 2024 (Online) 29th January Evening Shift

N moles of a polyatomic gas $(f = 6)$ must be mixed with two moles of a monoatomic gas so that the mixture behaves as a diatomic gas. The value of N is :

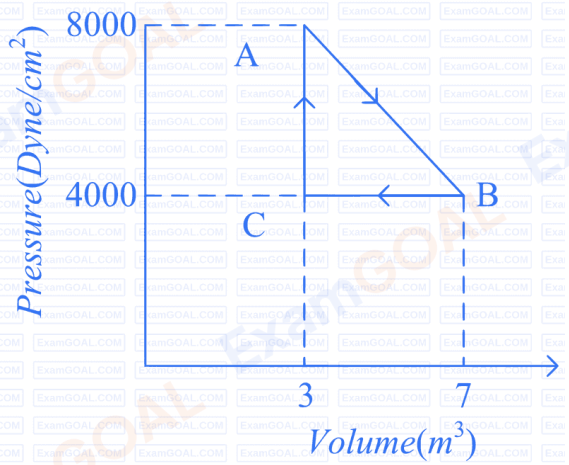
Open solution

(A) 6 (B) 2 (C) 4 (D) 3

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Q104. JEE Main 2024 (Online) 29th January Morning Shift

A thermodynamic system is taken from an original state A to an intermediate state B by a linear process as shown in the figure. Its volume is then reduced to the original value from B to C by an isobaric process. The total work done by the gas from A to B and B to C would be :



Open solution

(A) 800 J (B) 2200 J (C) 33800 J (D) 1200 J

Q105. JEE Main 2024 (Online) 29th January Morning Shift

Two vessels A and B are of the same size and are at same temperature. A contains 1 g of hydrogen and B contains 1 g of oxygen. $\mathrm{P_A}$ and $\mathrm{P_B}$ are the pressures of the gases in A and B respectively, then $\frac{P_A}{P_B}$ is:

Open solution

(A) 4 (B) 32 (C) 8 (D) 16

Q106. JEE Main 2024 (Online) 27th January Evening Shift

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio of $\frac{\mathrm{Cp}}{\mathrm{Cv}}$ for the gas is :

Open solution

(A) $\frac{7}{5}$ (B) $\frac{3}{2}$ (C) $\frac{9}{7}$ (D) $\frac{5}{3}$

Q107. JEE Main 2024 (Online) 27th January Evening Shift

The total kinetic energy of 1 mole of oxygen at $27^{\circ}\mathrm{C}$ is : [Use universal gas constant $(R) = 8.31\ \mathrm{J}/$ mole K]

Open solution

(A) 6232.5 J (B) 5670.5 J (C) 6845.5 J (D) 5942.0 J

Q108. JEE Main 2024 (Online) 27th January Morning Shift

0.08 kg air is heated at constant volume through $5^{\circ}\mathrm{C}$. The specific heat of air at constant volume is $0.17\ \mathrm{kcal/kg^{\circ}C}$ and $\mathrm{J} = 4.18$ joule/ cal. The change in its internal energy is approximately.

Open solution

(A) 318 J (B) 298 J (C) 284 J (D) 142 J

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Q109. JEE Main 2024 (Online) 27th January Morning Shift

The average kinetic energy of a monatomic molecule is $0.414\ \text{eV}$ at temperature :

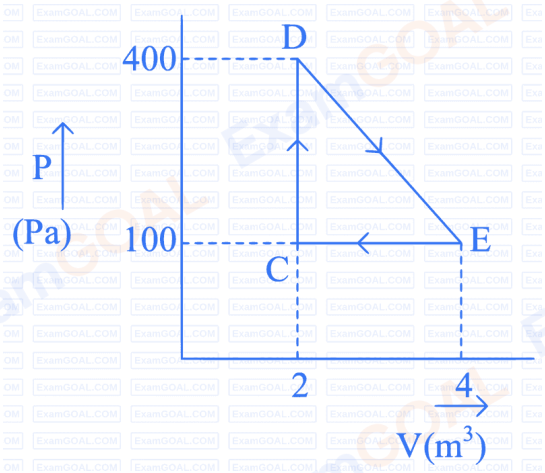
(Use $K_B = 1.38 \times 10^{-23}\ \text{J/mol} - \text{K}$)

(A) 3000 K (B) 3200 K (C) 1600 K (D) 1500 K

Open solution

Q110. JEE Main 2023 (Online) 15th April Morning Shift

A thermodynamic system is taken through cyclic process. The total work done in the process is :



(A) $100\ \text{J}$ (B) Zero (C) $300\ \text{J}$ (D) $200\ \text{J}$

Open solution

Q111. JEE Main 2023 (Online) 15th April Morning Shift

A flask contains Hydrogen and Argon in the ratio $2:1$ by mass. The temperature of the mixture is 30°C. The ratio of average kinetic energy per molecule of the two gases (K argon/K hydrogen) is :

(Given: Atomic Weight of $\text{Ar} = 39.9$)

(A) $\frac{39.9}{2}$ (B) 2 (C) 39.9 (D) 1

Open solution

Q112. JEE Main 2023 (Online) 13th April Evening Shift

The initial pressure and volume of an ideal gas are P_0 and V_0. The final pressure of the gas when the gas is suddenly compressed to volume $\frac{V_0}{4}$ will be :

(Given γ = ratio of specific heats at constant pressure and at constant volume)

(A) $P_0(4)^{\frac{1}{\gamma}}$ (B) P_0 (C) $4P_0$ (D) $P_0(4)^{\gamma}$

Open solution

Q113. JEE Main 2023 (Online) 13th April Evening Shift

The mean free path of molecules of a certain gas at STP is $1500\ \text{d}$, where d is the diameter of the gas molecules. While maintaining the standard pressure, the mean free path of the molecules at $373\ \text{K}$ is approximately:

(A) $750\ \text{d}$ (B) $1500\ \text{d}$ (C) $2049\ \text{d}$ (D) $1098\ \text{d}$

Open solution

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Q114. JEE Main 2023 (Online) 13th April Morning Shift

The rms speed of oxygen molecule in a vessel at particular temperature is $\left(1+\frac{5}{x}\right)^{\frac{1}{2}} v$, where v is the average speed of the molecule. The value of x will be:

(Take $\pi=\frac{22}{7}$)

(A) 4 (B) 8 (C) 28 (D) 27

Open solution

Q115. JEE Main 2023 (Online) 12th April Morning Shift

If the r. m.s speed of chlorine molecule is $490\ \mathrm{m/s}$ at $27^{\circ}\mathrm{C}$, the r. m. s speed of argon molecules at the same temperature will be (Atomic mass of argon $= 39.9\mathrm{u}$, molecular mass of chlorine $= 70.9\mathrm{u}$)

(A) $451.7\ \mathrm{m/s}$ (B) $751.7\ \mathrm{m/s}$ (C) $551.7\ \mathrm{m/s}$ (D) $651.7\ \mathrm{m/s}$

Open solution

Q116. JEE Main 2023 (Online) 11th April Evening Shift

The Thermodynamic process, in which internal energy of the system remains constant is

(A) Isobaric (B) Isochoric (C) Adiabatic (D) Isothermal

Open solution

Q117. JEE Main 2023 (Online) 11th April Evening Shift

The root mean square speed of molecules of nitrogen gas at $27^{\circ}\mathrm{C}$ is approximately : (Given mass of a nitrogen molecule $= 4.6\times10^{-26}\ \mathrm{kg}$ and take Boltzmann constant $\mathrm{k_B} = 1.4\times10^{-23}\mathrm{JK}^{-1}$)

(A) 91 m/s (B) 1260 m/s (C) 27.4 m/s (D) 523 m/s

Open solution

Q118. JEE Main 2023 (Online) 11th April Morning Shift

$1\ \mathrm{kg}$ of water at $100^{\circ}\mathrm{C}$ is converted into steam at $100^{\circ}\mathrm{C}$ by boiling at atmospheric pressure. The volume of water changes from $1.00\times10^{-3}\ \mathrm{m}^3$ as a liquid to $1.671\ \mathrm{m}^3$ as steam. The change in internal energy of the system during the process will be

(Given latent heat of vaporisation $= 2257\ \mathrm{kJ/kg}$, Atmospheric pressure = $1\times10^{5}\ \mathrm{Pa}$)

(A) + 2090 kJ (B) $-$ 2426 kJ (C) + 2476 kJ (D) $-$ 2090 kJ

Open solution

Q119. JEE Main 2023 (Online) 11th April Morning Shift

On a temperature scale 'X', the boiling point of water is $65^{\circ}\mathrm{X}$ and the freezing point is $-15^{\circ}\mathrm{X}$. Assume that the X scale is linear. The equivalent temperature corresponding to $-95^{\circ}\mathrm{X}$ on the Fahrenheit scale would be:

(A) $-148^{\circ}\mathrm{F}$ (B) $-48^{\circ}\mathrm{F}$ (C) $-63^{\circ}\mathrm{F}$ (D) $-112^{\circ}\mathrm{F}$

Open solution

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Q120. JEE Main 2023 (Online) 11th April Morning Shift

Three vessels of equal volume contain gases at the same temperature and pressure. The first vessel contains neon (monoatomic), the second contains chlorine (diatomic) and third contains uranium hexafluoride (polyatomic). Arrange these on the basis of their root mean square speed (v_{rms}) and choose the correct answer from the options given below:

(A) $\mathrm{v_{rms}}(\text{ mono }) = \mathrm{v_{rms}}(\text{ dia }) = \mathrm{v_{rms}}(\text{ poly })$

(B) $\mathrm{v_{rms}}$ (mono) $> \mathrm{v_{rms}}(\text{ dia }) > \mathrm{v_{rms}}$ (poly)

(C) $\mathrm{v_{rms}}$ (dia) $< \mathrm{v_{rms}}$ (poly) $< \mathrm{v_{rms}}$ (mono)

(D) $\mathrm{v_{rms}}$ (mono) $< \mathrm{v_{rms}}$ (dia) $< \mathrm{v_{rms}}$ (poly)

Open solution

Q121. JEE Main 2023 (Online) 10th April Evening Shift

A gas mixture consists of 2 moles of oxygen and 4 moles of neon at temperature T. Neglecting all vibrational modes, the total internal energy of the system will be,

(A) 4RT (B) 16RT (C) 8RT (D) 11RT

Open solution

Q122. JEE Main 2023 (Online) 10th April Evening Shift

A gas is compressed adiabatically, which one of the following statements is NOT true?

(A) There is no heat supplied to the system

(B) The temperature of the gas increases.

(C) There is no change in the internal energy

(D) The change in the internal energy is equal to the work done on the gas.

Open solution

Q123. JEE Main 2023 (Online) 10th April Morning Shift

Consider two containers A and B containing monoatomic gases at the same Pressure (P), Volume (V) and Temperature (T). The gas in A is compressed isothermally to $\frac{1}{8}$ of its original volume while the gas in B is compressed adiabatically to $\frac{1}{8}$ of its original volume. The ratio of final pressure of gas in B to that of gas in A is

(A) $\frac{1}{8}$ (B) $8^{\frac{3}{2}}$ (C) 4 (D) 8

Open solution

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Q124. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

Match List I with List II :

	List I		List II
(A)	3 Translational degrees of freedom	(I)	Monoatomic gases
(B)	3 Translational, 2 rotational degrees of freedom	(II)	Polyatomic gases
(C)	3 Translational, 2 rotational and 1 vibrational degrees of freedom	(III)	Rigid diatomic gases
(D)	3 Translational, 3 rotational and more than one vibrational degrees of freedom	(IV)	Nonrigid diatomic gases

Choose the correct answer from the options given below:

(A) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(B) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(C) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(D) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Q125. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

The temperature at which the kinetic energy of oxygen molecules becomes double than its value at $27^{\circ}\mathrm{C}$ is

(A) $627^{\circ}\mathrm{C}$ **(B)** $927^{\circ}\mathrm{C}$ **(C)** $327^{\circ}\mathrm{C}$ **(D)** $1227^{\circ}\mathrm{C}$

Q126. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

Given below are two statements:

Statement I: If heat is added to a system, its temperature must increase.

Statement II: If positive work is done by a system in a thermodynamic process, its volume must increase.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q127. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

The temperature of an ideal gas is increased from $200\ \mathrm{K}$ to $800\ \mathrm{K}$. If r.m.s. speed of gas at $200\ \mathrm{K}$ is v_0. Then, r.m.s. speed of the gas at $800\ \mathrm{K}$ will be:

(A) v_0 **(B)** $2v_0$ **(C)** $4v_0$ **(D)** $\frac{v_0}{4}$

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Q128. JEE Main 2023 (Online) 6th April Morning Shift

A source supplies heat to a system at the rate of $1000\ \mathrm{W}$. If the system performs work at a rate of $200\ \mathrm{W}$. The rate at which internal energy of the system increases is

(A) 600 W (B) 1200 W (C) 500 W (D) 800 W

Open solution

Q129. JEE Main 2023 (Online) 6th April Morning Shift

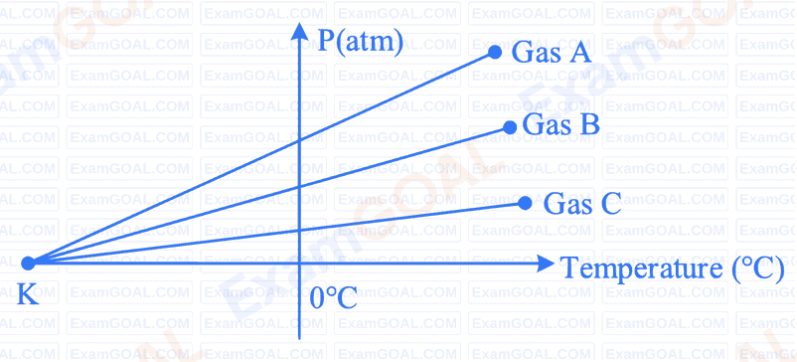
The number of air molecules per cm^3 increased from 3×10^{19} to 12×10^{19}. The ratio of collision frequency of air molecules before and after the increase in number respectively is:

(A) 1.25 (B) 0.25 (C) 0.50 (D) 0.75

Open solution

Q130. JEE Main 2023 (Online) 1st February Evening Shift

For three low density gases A, B, C pressure versus temperature graphs are plotted while keeping them at constant volume, as shown in the figure.



The temperature corresponding to the point 'K' is :

(A) $-273^{\circ}\mathrm{C}$ (B) $-373^{\circ}\mathrm{C}$ (C) $-100^{\circ}\mathrm{C}$ (D) $-40^{\circ}\mathrm{C}$

Open solution

Q131. JEE Main 2023 (Online) 1st February Morning Shift

A sample of gas at temperature T is adiabatically expanded to double its volume. The work done by the gas in the process is $\left(\text{given}, \gamma = \frac{3}{2}\right)$:

(A) $W = TR[\sqrt{2} - 2]$ (B) $W = \frac{T}{R}[\sqrt{2} - 2]$ (C) $W = \frac{R}{T}[2 - \sqrt{2}]$ (D) $W = RT[2 - \sqrt{2}]$

Open solution

Q132. JEE Main 2023 (Online) 1st February Morning Shift

The average kinetic energy of a molecule of the gas is

(A) proportional to volume (B) dependent on the nature of the gas

(C) proportional to absolute temperature (D) proportional to pressure

Open solution

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Q133. JEE Main 2023 (Online) 31st January Evening Shift

Heat energy of $735\ \mathrm{J}$ is given to a diatomic gas allowing the gas to expand at constant pressure. Each gas molecule rotates around an internal axis but do not oscillate. The increase in the internal energy of the gas will be :

Open solution

(A) $572\ \mathrm{J}$ (B) $441\ \mathrm{J}$ (C) $525\ \mathrm{J}$ (D) $735\ \mathrm{J}$

Q134. JEE Main 2023 (Online) 31st January Evening Shift

A hypothetical gas expands adiabatically such that its volume changes from 8 litres to 27 litres. If the ratio of final pressure of the gas to initial pressure of the gas is $\frac{16}{81}$. Then the ratio of $\frac{\mathrm{Cp}}{\mathrm{Cv}}$ will be.

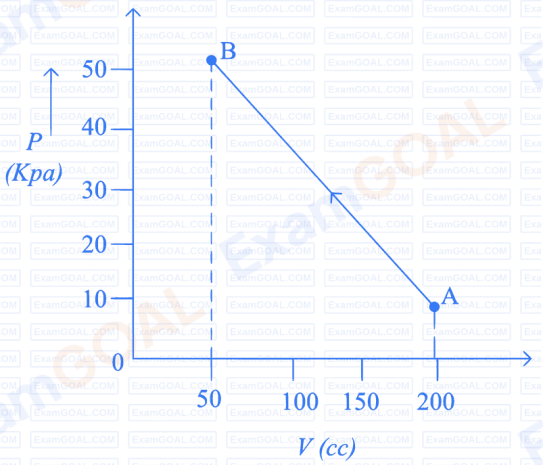
Open solution

(A) $\frac{3}{1}$ (B) $\frac{4}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Q135. JEE Main 2023 (Online) 31st January Morning Shift

The pressure of a gas changes linearly with volume from A to B as shown in figure. If no heat is supplied to or extracted from the gas then change in the internal energy of the gas will be

Open solution



(A) 6 J (B) 4.5 J (C) zero (D) -4.5 J

Q136. JEE Main 2023 (Online) 31st January Morning Shift

The correct relation between $\gamma = \frac{c_p}{c_v}$ and temperature T is :

Open solution

(A) $\gamma \propto T$ (B) $\gamma \propto \frac{1}{\sqrt{T}}$

(C) $\gamma \propto \frac{1}{T}$ (D) $\gamma \propto T^0$

Q137. JEE Main 2023 (Online) 30th January Evening Shift

A flask contains hydrogen and oxygen in the ratio of $2:1$ by mass at temperature $27^{\circ}\mathrm{C}$. The ratio of average kinetic energy per molecule of hydrogen and oxygen respectively is:

Open solution

(A) 1 : 1 (B) 4 : 1 (C) 1 : 4 (D) 2 : 1

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Q138. JEE Main 2023 (Online) 30th January Morning Shift

The pressure (P) and temperature (T) relationship of an ideal gas obeys the equation $\mathrm{PT}^2 =$ constant. The volume expansion coefficient of the gas will be :

(A) $3T^2$ (B) $\frac{3}{T^2}$ (C) $\frac{3}{T^3}$ (D) $\frac{3}{T}$

Open solution

Q139. JEE Main 2023 (Online) 30th January Morning Shift

Heat is given to an ideal gas in an isothermal process.

A. Internal energy of the gas will decrease.

B. Internal energy of the gas will increase.

C. Internal energy of the gas will not change.

D. The gas will do positive work.

E. The gas will do negative work.

Choose the correct answer from the options given below :

(A) B and D only (B) C and E only (C) A and E only (D) C and D only

Open solution

Q140. JEE Main 2023 (Online) 29th January Evening Shift

Heat energy of 184 kJ is given to ice of mass 600 g at $-12^{\circ}\mathrm{C}$. Specific heat of ice is $2222.3\ \mathrm{J\ kg^{-1^{\circ}}\ C^{-1}}$ and latent heat of ice is $336\ \mathrm{kJ\ kg^{-1}}$

A. Final temperature of system will be 0°C.

B. Final temperature of the system will be greater than 0°C.

C. The final system will have a mixture of ice and water in the ratio of 5 : 1.

D. The final system will have a mixture of ice and water in the ratio of 1 : 5.

E. The final system will have water only.

Choose the correct answer from the options given below :

(A) A and E only (B) B and D only (C) A and C only (D) A and D only

Open solution

Q141. JEE Main 2023 (Online) 29th January Evening Shift

At 300 K, the rms speed of oxygen molecules is $\sqrt{\frac{\alpha+5}{\alpha}}$ times to that of its average speed in the gas. Then, the value of α will be

(used $\pi = \frac{22}{7}$)

(A) 27 (B) 28 (C) 24 (D) 32

Open solution

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Q142. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If dQ and dW represent the heat supplied to the system and the work done on the system respectively. Then according to the first law of thermodynamics $dQ = dU - dW$.

Reason R: First law of thermodynamics is based on law of conservation of energy.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are correct but R is not the correct explanation of A

(B) Both A and R are correct and R is the correct explanation of A

(C) A is correct but R is not correct

(D) A is not correct but R is correct

Q143. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

A bicycle tyre is filled with air having pressure of $270\ \text{kPa}$ at 27°C. The approximate pressure of the air in the tyre when the temperature increases to 36°C is

(A) 262 kPa (B) 360 kPa (C) 270 kPa (D) 278 kPa

Q144. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Match List I with List II

	List I		List II
A.	Isothermal Process	I.	Work done by the gas decreases internal energy
B.	Adiabatic Process	II.	No change in internal energy
C.	Isochoric Process	III.	The heat absorbed goes partly to increase internal energy and partly to do work
D.	Isobaric Process	IV.	No work is done on or by the gas

Choose the correct answer from the options given below :

(A) A-I, B-II, C-IV, D-III (B) A-II, B-I, C-III, D-IV

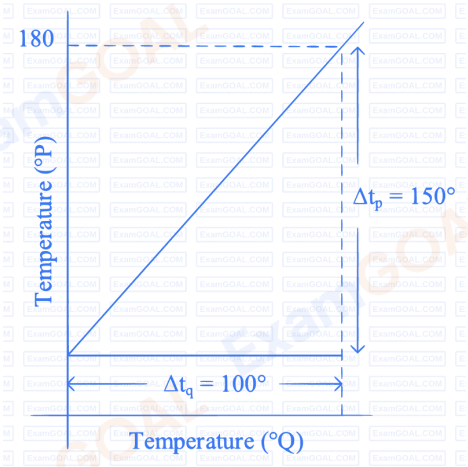
(C) A-II, B-I, C-IV, D-III (D) A-I, B-II, C-III, D-IV

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Q145. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

The graph between two temperature scales P and Q is shown in the figure. Between upper fixed point and lower fixed point there are 150 equal divisions of scale P and 100 divisions on scale Q. The relationship for conversion between the two scales is given by :-



(A) $\frac{t_P}{100} = \frac{t_Q - 180}{150}$

(B) $\frac{t_P}{180} = \frac{t_Q - 40}{100}$

(C) $\frac{t_Q}{150} = \frac{t_P - 180}{100}$

(D) $\frac{t_Q}{100} = \frac{t_P - 30}{150}$

Q146. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

According to law of equipartition of energy the molar specific heat of a diatomic gas at constant volume where the molecule has one additional vibrational mode is :-

(A) $\frac{9}{2}R$ (B) $\frac{5}{2}R$ (C) $\frac{3}{2}R$ (D) $\frac{7}{2}R$

Q147. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

The root mean square velocity of molecules of gas is

(A) Proportional to temperature (T)

(B) Inversely proportional to square root of temperature $\left(\sqrt{\frac{1}{T}}\right)$

(C) Proportional to square of temperature (T^2)

(D) Proportional to square root of temperature ($\sqrt{T}$)

Q148. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

Let γ_1 be the ratio of molar specific heat at constant pressure and molar specific heat at constant volume of a monoatomic gas and γ_2 be the similar ratio of diatomic gas. Considering the diatomic gas molecule as a rigid rotator, the ratio, $\frac{\gamma_1}{\gamma_2}$ is :

(A) $\frac{35}{27}$ (B) $\frac{25}{21}$ (C) $\frac{21}{25}$ (D) $\frac{27}{35}$

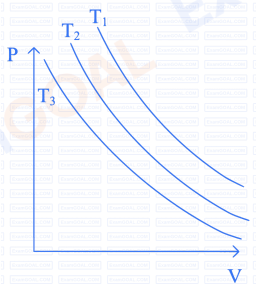
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Q149. JEE Main 2023 (Online) 24th January Evening Shift

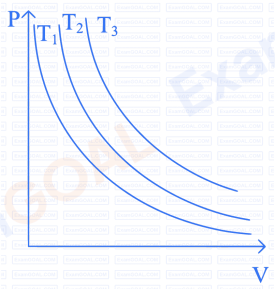
Open solution

In an Isothermal change, the change in pressure and volume of a gas can be represented for three different temperatures; $T_3 > T_2 > T_1$ as :

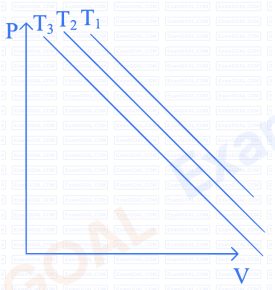
(A)



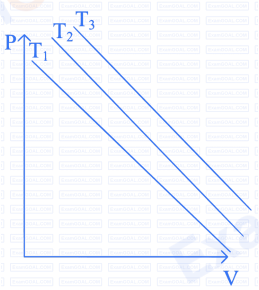
(B)



(C)



(D)



Q150. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

1 g of a liquid is converted to vapour at 3×10^5 Pa pressure. If 10% of the heat supplied is used for increasing the volume by 1600 cm^3 during this phase change, then the increase in internal energy in the process will be :

(A) 4800 J (B) 4320 J (C) 432000 J (D) 4.32×10^8 J

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Q151. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Given below are two statements :

Statement I : The temperature of a gas is -73°C. When the gas is heated to 527°C, the root mean square speed of the molecules is doubled.

Statement II : The product of pressure and volume of an ideal gas will be equal to translational kinetic energy of the molecules.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q152. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

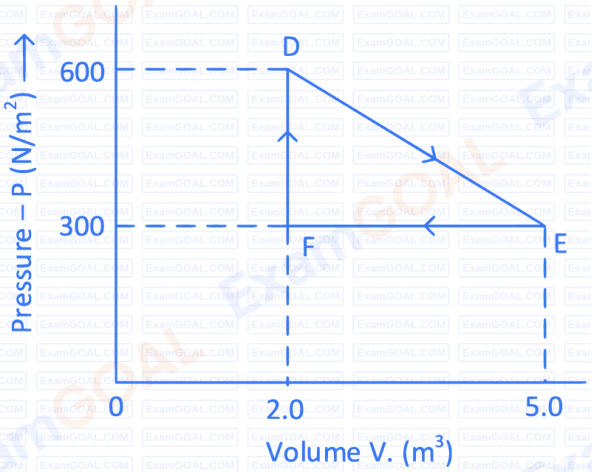
If the r.m.s speed of oxygen at room temperature is $\sqrt{56}\,\text{m/s}$. The r.m.s speed of nitrogen at the same temperature is _____.

(A) $8\sqrt{2}\,\text{m/s}$ (B) $8\,\text{m/s}$ (C) $16\,\text{m/s}$ (D) $7\,\text{m/s}$

Q153. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in the figure. Its volume is then reduced to the original volume from E to F by an isobaric process. The total work done by the gas from D to E to F will be



(A) -450 J (B) 450 J (C) 900 J (D) 1350 J

Q154. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

The root mean square speed of smoke particles of mass 5×10^{-17} kg in their Brownian motion in air at NTP is approximately. [Given $k = 1.38 \times 10^{-23} JK^{-1}$]

(A) $60\,\text{mm s}^{-1}$ (B) $12\,\text{mm s}^{-1}$ (C) $15\,\text{mm s}^{-1}$ (D) $36\,\text{mm s}^{-1}$

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Q155. JEE Main 2022 (Online) 28th July Evening Shift

A vessel contains $14\ \mathrm{g}$ of nitrogen gas at a temperature of $27^{\circ}\mathrm{C}$. The amount of heat to be transferred to the gas to double the r.m.s speed of its molecules will be :

Take $\mathrm{R} = 8.32\ \mathrm{J\ mol^{-1}\ k^{-1}}$.

Open solution

(A) 2229 J **(B)** 5616 J **(C)** 9360 J **(D)** 13,104 J

Q156. JEE Main 2022 (Online) 28th July Morning Shift

Given below are two statements :

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become $2v$.

In the light of the above statements, choose the correct answer from the options given below :

Open solution

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q157. JEE Main 2022 (Online) 27th July Evening Shift

Which statements are correct about degrees of freedom ?

(A) A molecule with n degrees of freedom has n^2 different ways of storing energy.

(B) Each degree of freedom is associated with $\frac{1}{2}$ RT average energy per mole.

(C) A monatomic gas molecule has 1 rotational degree of freedom whereas diatomic molecule has 2 rotational degrees of freedom.

(D) $\mathrm{CH_4}$ has a total of 6 degrees of freedom.

Choose the correct answer from the options given below :

Open solution

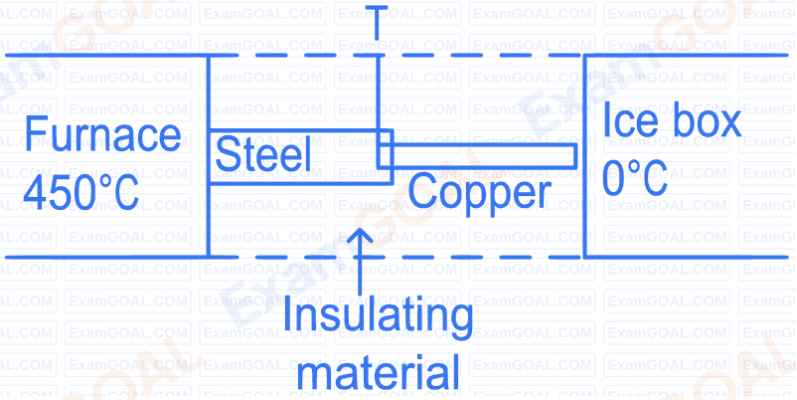
(A) (B) and (C) only **(B)** (B) and (D) only **(C)** (A) and (B) only **(D)** (C) and (D) only

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Q158. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

If K_1 and K_2 are the thermal conductivities, L_1 and L_2 are the lengths and A_1 and A_2 are the cross sectional areas of steel and copper rods respectively such that $\frac{K_2}{K_1} = 9$, $\frac{A_1}{A_2} = 2$, $\frac{L_1}{L_2} = 2$. Then, for the arrangement as shown in the figure, the value of temperature T of the steel - copper junction in the steady state will be:



(A) $18^\circ C$ (B) $14^\circ C$ (C) $45^\circ C$ (D) $150^\circ C$

Q159. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Read the following statements :

A. When small temperature difference between a liquid and its surrounding is doubled, the rate of loss of heat of the liquid becomes twice.

B. Two bodies P and Q having equal surface areas are maintained at temperature $10^\circ C$ and $20^\circ C$. The thermal radiation emitted in a given time by P and Q are in the ratio $1 : 1.15$.

C. A Carnot Engine working between $100\ K$ and $400\ K$ has an efficiency of 75%.

D. When small temperature difference between a liquid and its surrounding is quadrupled, the rate of loss of heat of the liquid becomes twice.

Choose the correct answer from the options given below :

(A) A, B, C only (B) A, B only (C) A, C only (D) B, C, D only

Q160. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Same gas is filled in two vessels of the same volume at the same temperature. If the ratio of the number of molecules is $1 : 4$, then

A. The r.m.s. velocity of gas molecules in two vessels will be the same.

B. The ratio of pressure in these vessels will be $1 : 4$.

C. The ratio of pressure will be $1 : 1$.

D. The r.m.s. velocity of gas molecules in two vessels will be in the ratio of $1 : 4$.

Choose the correct answer from the options given below :

(A) A and C only (B) B and D only (C) A and B only (D) C and D only

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Q161. JEE Main 2022 (Online) 26th July Evening Shift

An ice cube of dimensions $60\ \text{cm} \times 50\ \text{cm} \times 20\ \text{cm}$ is placed in an insulation box of wall thickness $1\ \text{cm}$. The box keeping the ice cube at 0°C of temperature is brought to a room of temperature 40°C. The rate of melting of ice is approximately :

(Latent heat of fusion of ice is $3.4 \times 10^5\ \text{J kg}^{-1}$ and thermal conducting of insulation wall is $0.05\ \text{Wm}^{-1\circ}\text{C}^{-1}$)

(A) $61 \times 10^{-3}\ \text{kg s}^{-1}$ (B) $61 \times 10^{-5}\ \text{kg s}^{-1}$

(C) $208\ \text{kg s}^{-1}$ (D) $30 \times 10^{-5}\ \text{kg s}^{-1}$

Open solution

Q162. JEE Main 2022 (Online) 26th July Evening Shift

A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be :

(A) $\frac{n}{n+2}$ (B) $\frac{n+2}{n}$ (C) $\frac{n}{2n+2}$ (D) $\frac{n}{n-2}$

Open solution

Q163. JEE Main 2022 (Online) 26th July Morning Shift

7 mol of a certain monoatomic ideal gas undergoes a temperature increase of $40\ \text{K}$ at constant pressure. The increase in the internal energy of the gas in this process is :

(Given $\text{R} = 8.3\ \text{JK}^{-1}\ \text{mol}^{-1}$)

(A) 5810 J (B) 3486 J (C) 11620 J (D) 6972 J

Open solution

Q164. JEE Main 2022 (Online) 26th July Morning Shift

A monoatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be :

(A) P (B) 8P (C) 32P (D) 64P

Open solution

Q165. JEE Main 2022 (Online) 25th July Evening Shift

Sound travels in a mixture of two moles of helium and n moles of hydrogen. If rms speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound, then the value of n will be :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q166. JEE Main 2022 (Online) 25th July Morning Shift

A certain amount of gas of volume V at 27°C temperature and pressure $2 \times 10^7\text{Nm}^{-2}$ expands isothermally until its volume gets doubled. Later it expands adiabatically until its volume gets redoubled. The final pressure of the gas will be (Use $\gamma = 1.5$) :

(A) $3.536 \times 10^5\ \text{Pa}$ (B) $3.536 \times 10^6\ \text{Pa}$ (C) $1.25 \times 10^6\ \text{Pa}$ (D) $1.25 \times 10^5\ \text{Pa}$

Open solution

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Q167. JEE Main 2022 (Online) 25th July Morning Shift

Open solution

Following statements are given :

(A) The average kinetic energy of a gas molecule decreases when the temperature is reduced.

(B) The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.

(C) The average kinetic energy of a gas molecule decreases with increase in volume.

(D) Pressure of a gas increases with increase in temperature at constant pressure.

(E) The volume of gas decreases with increase in temperature.

Choose the correct answer from the options given below :

(A) (A) and (D) only

(B) (A), (B) and (D) only

(C) (B) and (D) only

(D) (A), (B) and (E) only

Q168. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

The pressure of the gas in a constant volume gas thermometer is 100 cm of mercury when placed in melting ice at 1 atm. When the bulb is placed in a liquid, the pressure becomes 180 cm of mercury. Temperature of the liquid is :

(Given 0°C = 273 K)

(A) 300 K

(B) 400 K

(C) 600 K

(D) 491 K

Q169. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

A sample of monoatomic gas is taken at initial pressure of 75 kPa. The volume of the gas is then compressed from 1200 cm^3 to 150 cm^3 adiabatically. In this process, the value of workdone on the gas will be :

(A) 79 J

(B) 405 J

(C) 4050 J

(D) 9590 J

Q170. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

At what temperature a gold ring of diameter 6.230 cm be heated so that it can be fitted on a wooden bangle of diameter 6.241 cm ? Both the diameters have been measured at room temperature (27°C).

(Given : coefficient of linear thermal expansion of gold $\alpha_L = 1.4 \times 10^{-5}\ K^{-1}$)

(A) 125.7°C

(B) 91.7°C

(C) 425.7°C

(D) 152.7°C

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Q171. JEE Main 2022 (Online) 29th June Evening Shift

Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal, W_2, if the process is purely adiabatic and W_3 if the process is purely isobaric. Then, choose the correct option

Open solution

(A) $W_1 < W_2 < W_3$ (B) $W_2 < W_3 < W_1$ (C) $W_3 < W_1 < W_2$ (D) $W_2 < W_1 < W_3$

Q172. JEE Main 2022 (Online) 29th June Evening Shift

A vessel contains 16g of hydrogen and 128g of oxygen at standard temperature and pressure. The volume of the vessel in cm^3 is :

Open solution

(A) 72×10^5 (B) 32×10^5 (C) 27×10^4 (D) 54×10^4

Q173. JEE Main 2022 (Online) 29th June Morning Shift

A cylinder of fixed capacity of 44.8 litres contains helium gas at standard temperature and pressure. The amount of heat needed to raise the temperature of gas in the cylinder by 20.0°C will be :

(Given gas constant R = 8.3 JK^{-1}-mol^{-1})

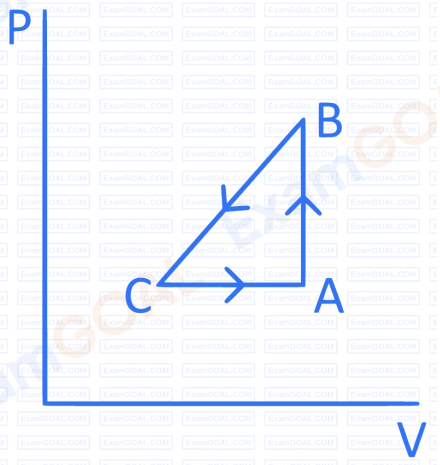
Open solution

(A) 249 J (B) 415 J (C) 498 J (D) 830 J

Q174. JEE Main 2022 (Online) 28th June Evening Shift

A sample of an ideal gas is taken through the cyclic process ABCA as shown in figure. It absorbs, 40 J of heat during the part AB, no heat during BC and rejects 60 J of heat during CA. A work of 50 J is done on the gas during the part BC. The internal energy of the gas at A is 1560 J. The workdone by the gas during the part CA is :

Open solution



(A) 20 J (B) 30 J (C) −30 J (D) −60 J

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Q175. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen?

(A) The velocity of atomic oxygen remains same

(B) The velocity of atomic oxygen doubles

(C) The velocity of atomic oxygen becomes half

(D) The velocity of atomic oxygen becomes four times

Q176. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Given below are two statements :

Statement I : When μ amount of an ideal gas undergoes adiabatic change from state (P_1, V_1, T_1) to state (P_2, V_2, T_2), then work done is $W = \frac{\mu R(T_2 - T_1)}{1-\gamma}$, where $\gamma = \frac{C_p}{C_v}$ and R = universal gas constant.

Statement II : In the above case, when work is done on the gas, the temperature of the gas would rise.

Choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

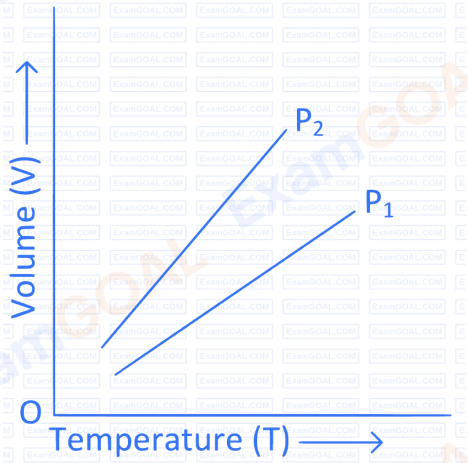
(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q177. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

For a perfect gas, two pressures P_1 and P_2 are shown in figure. The graph shows :



(A) $P_1 > P_2$

(B) $P_1 < P_2$

(C) $P_1 = P_2$

(D) Insufficient data to draw any conclusion

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Q178. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

According to kinetic theory of gases,

A. The motion of the gas molecules freezes at 0°C.

B. The mean free path of gas molecules decreases if the density of molecules is increased.

C. The mean free path of gas molecules increases if temperature is increased keeping pressure constant.

D. Average kinetic energy per molecule per degree of freedom is $\frac{3}{2}k_BT$ (for monoatomic gases).

Choose the most appropriate answer from the options given below :

(A) A and C only (B) B and C only (C) A and B only (D) C and D only

Q179. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

A lead bullet penetrates into a solid object and melts. Assuming that 40% of its kinetic energy is used to heat it, the initial speed of bullet is :

(Given : initial temperature of the bullet = 127°C, Melting point of the bullet = 327°C, Latent heat of fusion of lead = 2.5×10^4 J kg^{-1}, Specific heat capacity of lead = 125 J/kg K)

(A) 125 ms^{-1} (B) 500 ms^{-1} (C) 250 ms^{-1} (D) 600 ms^{-1}

Q180. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

A mixture of hydrogen and oxygen has volume 2000 cm^3, temperature 300 K, pressure 100 kPa and mass 0.76 g. The ratio of number of moles of hydrogen to number of moles of oxygen in the mixture will be:

[Take gas constant R = 8.3 $JK^{-1}mol^{-1}$]

(A) $\frac{1}{3}$ (B) $\frac{3}{1}$ (C) $\frac{1}{16}$ (D) $\frac{16}{1}$

Q181. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

A flask contains argon and oxygen in the ratio of 3 : 2 in mass and the mixture is kept at 27°C. The ratio of their average kinetic energy per molecule respectively will be :

(A) 3 : 2 (B) 9 : 4 (C) 2 : 3 (D) 1 : 1

Q182. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats 1.4. Vessel is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surrounding and vessel, temperature of the gas increases by :

(R = universal gas constant)

(A) $\frac{Mv^2}{7R}$ (B) $\frac{Mv^2}{5R}$ (C) $2\frac{Mv^2}{7R}$ (D) $7\frac{Mv^2}{5R}$

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Q183. JEE Main 2022 (Online) 25th June Evening Shift

A solid metallic cube having total surface area 24 m^2 is uniformly heated. If its temperature is increased by 10°C, calculate the increase in volume of the cube. (Given $\alpha = 5.0 \times 10^{-4}\ ^\circ C^{-1}$).

(A) $2.4 \times 10^6\ cm^3$ (B) $1.2 \times 10^5\ cm^3$ (C) $6.0 \times 10^4\ cm^3$ (D) $4.8 \times 10^5\ cm^3$

Open solution

Q184. JEE Main 2022 (Online) 25th June Evening Shift

A copper block of mass 5.0 kg is heated to a temperature of 500°C and is placed on a large ice block. What is the maximum amount of ice that can melt? [Specific heat of copper : 0.39 J $g^{-1}\ ^\circ C^{-1}$ and latent heat of fusion of water : 335 J g^{-1}]

(A) 1.5 kg (B) 5.8 kg (C) 2.9 kg (D) 3.8 kg

Open solution

Q185. JEE Main 2022 (Online) 25th June Evening Shift

The ratio of specific heats $\left(\frac{C_P}{C_V}\right)$ in terms of degree of freedom (f) is given by :

(A) $\left(1+\frac{f}{3}\right)$ (B) $\left(1+\frac{2}{f}\right)$ (C) $\left(1+\frac{f}{2}\right)$ (D) $\left(1+\frac{1}{f}\right)$

Open solution

Q186. JEE Main 2022 (Online) 25th June Morning Shift

The relation between root mean square speed (v_{rms}) and most probable speed (v_p) for the molar mass M of oxygen gas molecule at the temperature of 300 K will be :

(A) $v_{rms} = \sqrt{\frac{2}{3}} v_p$ (B) $v_{rms} = \sqrt{\frac{3}{2}} v_p$

(C) $v_{rms} = v_p$ (D) $v_{rms} = \sqrt{\frac{1}{3}} v_p$

Open solution

Q187. JEE Main 2022 (Online) 24th June Evening Shift

A 100 g of iron nail is hit by a 1.5 kg hammer striking at a velocity of 60 ms^{-1}. What will be the rise in the temperature of the nail if one fourth of energy of the hammer goes into heating the nail?

[Specific heat capacity of iron = 0.42 $Jg^{-1}\ ^\circ C^{-1}$]

(A) 675°C (B) 1600°C (C) 16.07°C (D) 6.75°C

Open solution

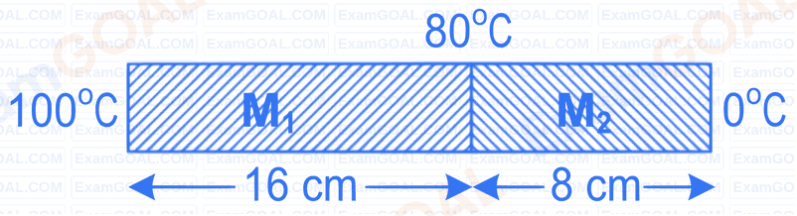
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Q188. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

Two metallic blocks M_1 and M_2 of same area of cross-section are connected to each other (as shown in figure). If the thermal conductivity of M_2 is K then the thermal conductivity of M_1 will be :

[Assume steady state heat conduction]



(A) 10 K (B) 8 K (C) 12.5 K (D) 2 K

Q189. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature $\theta_2(\theta_1 < \theta_2)$. The rate at which heat flows radially through the material is :-

(A) $\frac{4\pi K r_1 r_2(\theta_2-\theta_1)}{r_2-r_1}$ (B) $\frac{\pi r_1 r_2(\theta_2-\theta_1)}{r_2-r_1}$

(C) $\frac{K(\theta_2-\theta_1)}{r_2-r_1}$ (D) $\frac{K(\theta_2-\theta_1)(r_2-r_1)}{4\pi r_1 r_2}$

Q190. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

A mixture of hydrogen and oxygen has volume 500 cm^3, temperature 300 K, pressure 400 kPa and mass 0.76 g. The ratio of masses of oxygen to hydrogen will be :-

(A) 3 : 8 (B) 3 : 16 (C) 16 : 3 (D) 8 : 3

Q191. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

For an ideal gas the instantaneous change in pressure 'p' with volume 'v' is given by the equation $\frac{dp}{dv} = -ap$. If $p = p_0$ at $v = 0$ is the given boundary condition, then the maximum temperature one mole of gas can attain is : (Here R is the gas constant)

(A) $\frac{p_0}{aeR}$ (B) $\frac{ap_0}{eR}$ (C) infinity (D) 0°C

Q192. JEE Main 2021 (Online) 27th August Evening Shift

Open solution

if the rms speed of oxygen molecules at 0°C is 160 m/s, find the rms speed of hydrogen molecules at 0°C.

(A) 640 m/s (B) 40 m/s (C) 80 m/s (D) 332 m/s

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Q193. JEE Main 2021 (Online) 27th August Evening Shift

The height of Victoria falls is 63 m. What is the difference in temperature of water at the top and at the bottom of fall?

[Given 1 cal = 4.2 J and specific heat of water = 1 cal $g^{-1}\,{}^{\circ}C^{-1}$]

(A) 0.147° C (B) 14.76° C (C) 1.476° C (D) 0.014° C

Open solution

Q194. JEE Main 2021 (Online) 27th August Morning Shift

A balloon carries a total load of 185 kg at normal pressure and temperature of 27°C. What load will the balloon carry on rising to a height at which the barometric pressure is 45 cm of Hg and the temperature is −7°C. Assuming the volume constant?

(A) 181.46 kg (B) 214.15 kg (C) 219.07 kg (D) 123.54 kg

Open solution

Q195. JEE Main 2021 (Online) 27th August Morning Shift

An ideal gas is expanding such that PT^3 = constant. The coefficient of volume expansion of the gas is :

(A) $\frac{1}{T}$ (B) $\frac{2}{T}$ (C) $\frac{4}{T}$ (D) $\frac{3}{T}$

Open solution

Q196. JEE Main 2021 (Online) 26th August Evening Shift

The temperature of equal masses of three different liquids x, y and z are 10°C, 20°C and 30°C respectively. The temperature of mixture when x is mixed with y is 16°C and that when y is mixed with z is 26°C. The temperature of mixture when x and z are mixed will be :

(A) 28.32°C (B) 25.62°C (C) 23.84°C (D) 20.28°C

Open solution

Q197. JEE Main 2021 (Online) 26th August Evening Shift

A cylindrical container of volume 4.0×10^{-3} m^3 contains one mole of hydrogen and two moles of carbon dioxide. Assume the temperature of the mixture is 400 K. The pressure of the mixture of gases is :

[Take gas constant as 8.3 J mol^{-1} K^{-1}]

(A) 249×10^1 Pa (B) 24.9×10^3 Pa (C) 24.9×10^5 Pa (D) 24.9 Pa

Open solution

Q198. JEE Main 2021 (Online) 26th August Morning Shift

An electric appliance supplies 6000 J/min heat to the system. If the system delivers a power of 90W. How long it would take to increase the internal energy by 2.5×10^3 J ?

(A) 2.5×10^2 s (B) 4.1×10^1 s (C) 2.4×10^3 s (D) 2.5×10^1 s

Open solution

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Q199. JEE Main 2021 (Online) 26th August Morning Shift

The rms speeds of the molecules of Hydrogen, Oxygen and Carbon dioxide at the same temperature are V_H, V_O and V_C respectively then :

(A) $V_H > V_O > V_C$ (B) $V_C > V_O > V_H$ (C) $V_H = V_O > V_C$ (D) $V_H = V_O = V_C$

Open solution

Q200. JEE Main 2021 (Online) 27th July Evening Shift

One mole of an ideal gas is taken through an adiabatic process where the temperature rises from 27° C to 37° C. If the ideal gas is composed of polyatomic molecule that has 4 vibrational modes, which of the following is true? [R = 8.314 J mol^{-1} K^{-1}]

(A) work done by the gas is close to 332 J (B) work done on the gas is close to 582 J

(C) work done by the gas is close to 582 J (D) work done on the gas is close to 332 J

Open solution

Q201. JEE Main 2021 (Online) 27th July Morning Shift

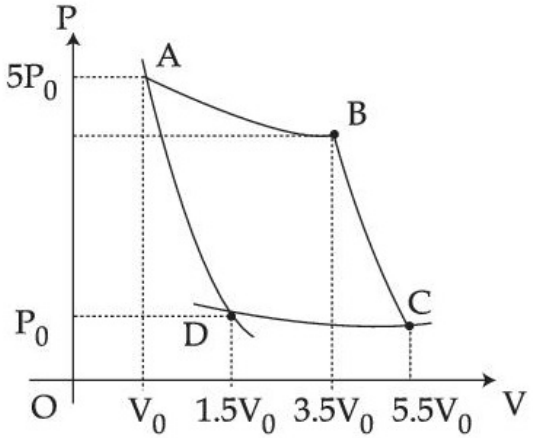
The number of molecules in one litre of an ideal gas at 300 K and 2 atmospheric pressure with mean kinetic energy 2×10^{-9} J per molecules is :

(A) 0.75×10^{11} (B) 3×10^{11} (C) 1.5×10^{11} (D) 6×10^{11}

Open solution

Q202. JEE Main 2021 (Online) 27th July Morning Shift

In the reported figure, there is a cyclic process ABCDA on a sample of 1 mol of a diatomic gas. The temperature of the gas during the process A $\rightarrow$ B and C $\rightarrow$ D are T_1 and T_2 ($T_1 > T_2$) respectively.



Choose the correct option out of the following for work done if processes BC and DA are adiabatic.

(A) $W_{AB} = W_{DC}$ (B) $W_{AD} = W_{BC}$ (C) $W_{BC} + W_{DA} > 0$ (D) $W_{AB} < W_{CD}$

Open solution

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Q203. JEE Main 2021 (Online) 25th July Morning Shift

For a gas $C_P - C_V = R$ in a state P and $C_P - C_V = 1.10\ R$ in a state Q, T_P and T_Q are the temperatures in two different states P and Q respectively. Then

(A) $T_P = T_Q$ (B) $T_P < T_Q$ (C) $T_P = 0.9\ T_Q$ (D) $T_P > T_Q$

Open solution

Q204. JEE Main 2021 (Online) 25th July Morning Shift

A monoatomic ideal gas, initially at temperature T_1 is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If l_1 and l_2 are the lengths of the gas column, before and after the expansion respectively, then the value of $\frac{T_1}{T_2}$ will be :

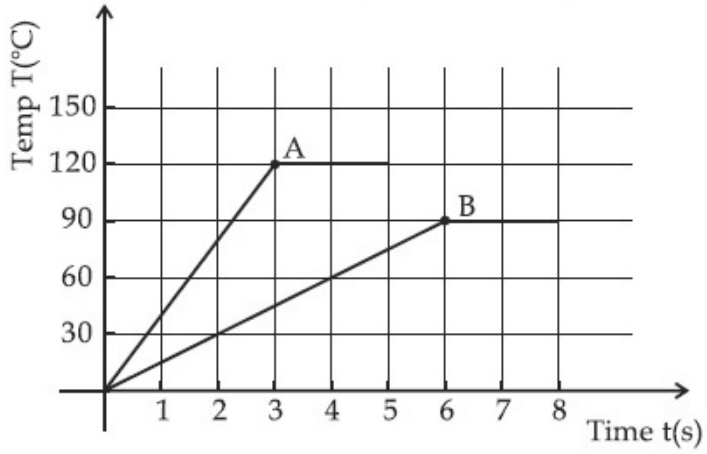
(A) $\left(\frac{l_1}{l_2}\right)^{\frac{2}{3}}$ (B) $\left(\frac{l_2}{l_1}\right)^{\frac{2}{3}}$

(C) $\frac{l_2}{l_1}$ (D) $\frac{l_1}{l_2}$

Open solution

Q205. JEE Main 2021 (Online) 25th July Morning Shift

Two different metal bodies A and B of equal mass are heated at a uniform rate under similar conditions. The variation of temperature of the bodies is graphically represented as shown in the figure. The ratio of specific heat capacities is :



(A) $\frac{8}{3}$ (B) $\frac{3}{8}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$

Open solution

Q206. JEE Main 2021 (Online) 22th July Evening Shift

What will be the average value of energy for a monoatomic gas in thermal equilibrium at temperature T?

(A) $\frac{3}{2}k_BT$ (B) k_BT (C) $\frac{2}{3}k_BT$ (D) $\frac{1}{2}k_BT$

Open solution

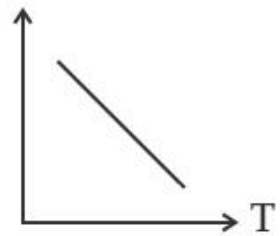
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Q207. JEE Main 2021 (Online) 20th July Evening Shift

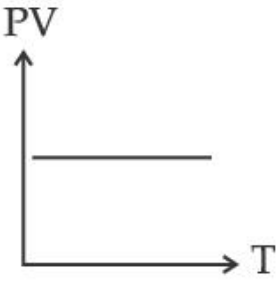
Open solution

Which of the following graphs represent the behavior of an ideal gas? Symbols have their usual meaning.

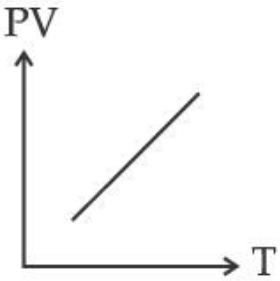
(A)



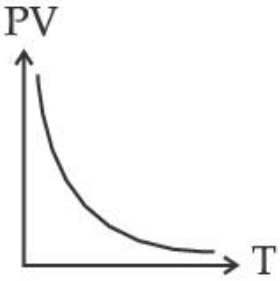
(B)



(C)



(D)



Q208. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

The correct relation between the degrees of freedom f and the ratio of specific heat γ is :

(A) $f = \frac{2}{\gamma - 1}$ (B) $f = \frac{2}{\gamma + 1}$ (C) $f = \frac{\gamma + 1}{2}$ (D) $f = \frac{1}{\gamma + 1}$

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Q209. JEE Main 2021 (Online) 20th July Morning Shift

Consider a mixture of gas molecule of types A, B and C having masses $m_A < m_B < m_C$. The ratio of their root mean square speeds at normal temperature and pressure is :

Open solution

(A) $v_A = v_B \neq v_C$

(B) $\frac{1}{v_A} > \frac{1}{v_B} > \frac{1}{v_C}$

(C) $\frac{1}{v_A} < \frac{1}{v_B} < \frac{1}{v_C}$

(D) $v_A = v_B = v_C = 0$

Q210. JEE Main 2021 (Online) 20th July Morning Shift

The amount of heat needed to raise the temperature of 4 moles of rigid diatomic gas from 0° C to 50° C when no work is done is _____. (R is the universal gas constant).

Open solution

(A) 500 R (B) 250 R (C) 750 R (D) 175 R

Q211. JEE Main 2021 (Online) 18th March Evening Shift

Consider a sample of oxygen behaving like an ideal gas. At 300 K, the ratio of root mean square (rms) velocity to the average velocity of gas molecule would be :

Open solution

(Molecular weight of oxygen is 32g/mol; R = 8.3 J K^{-1} mol^{-1})

(A) $\sqrt{\frac{3\pi}{8}}$ (B) $\sqrt{\frac{3}{3}}$ (C) $\sqrt{\frac{8}{3}}$ (D) $\sqrt{\frac{8\pi}{3}}$

Q212. JEE Main 2021 (Online) 18th March Evening Shift

For an adiabatic expansion of an ideal gas, the fractional change in its pressure is equal to (where γ is the ratio of specific heats) :

Open solution

(A) $-\frac{1}{\gamma}\frac{dV}{V}$

(B) $-\gamma\frac{V}{dV}$

(C) $-\gamma\frac{dV}{V}$

(D) $\frac{dV}{V}$

Q213. JEE Main 2021 (Online) 18th March Evening Shift

An ideal gas in a cylinder is separated by a piston in such a way that the entropy of one part is S_1 and that of the other part is S_2. Given that $S_1 > S_2$. If the piston is removed then the total entropy of the system will be :

Open solution

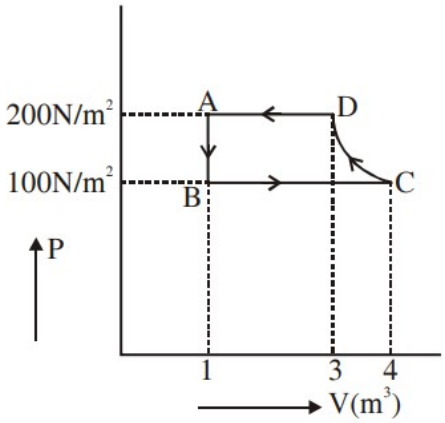
(A) $S_1 - S_2$ (B) $\frac{S_1}{S_2}$ (C) $S_1 \times S_2$ (D) $S_1 + S_2$

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Q214. JEE Main 2021 (Online) 18th March Morning Shift

The P-V diagram of a diatomic ideal gas system going under cyclic process as shown in figure. The work done during an adiabatic process CD is (use γ = 1.4) :

Open solution



(A) -500 J (B) -400 J (C) 400 J (D) 200 J

Q215. JEE Main 2021 (Online) 18th March Morning Shift

What will be the average value of energy along one degree of freedom for an ideal gas in thermal equilibrium at a temperature T? (k_B is Boltzmann constant)

Open solution

(A) $\frac{1}{2}k_BT$ (B) $\frac{2}{3}k_BT$ (C) $\frac{3}{2}k_BT$ (D) k_BT

Q216. JEE Main 2021 (Online) 17th March Evening Shift

If one mole of the polyatomic gas is having two vibrational modes and β is the ratio of molar specific heats for polyatomic gas $\left(\beta = \frac{C_P}{C_V}\right)$ then the value of β is :

Open solution

(A) 1.02 (B) 1.35 (C) 1.2 (D) 1.25

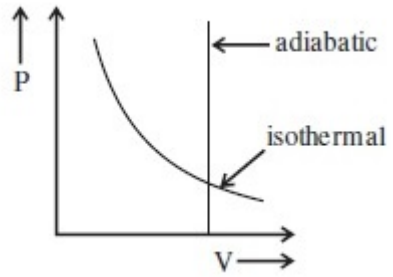
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Q217. JEE Main 2021 (Online) 17th March Evening Shift

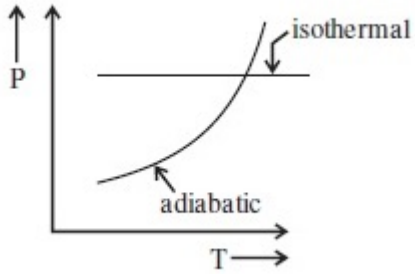
Open solution

Which one is the correct option for the two different thermodynamic processes?

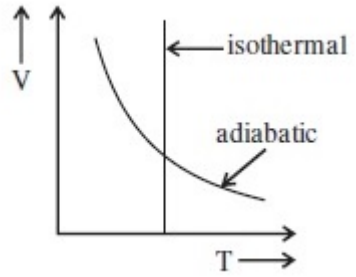
(a)



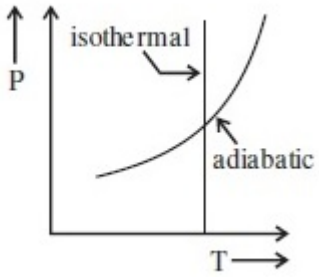
(b)



(c)



(d)



(A) (a) only **(B)** (c) and (d) **(C)** (b) and (c) **(D)** (c) and (a)

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Q218. JEE Main 2021 (Online) 17th March Morning Shift

A polyatomic ideal gas has 24 vibrational modes. What is the value of γ?

(A) 1.37 (B) 1.30 (C) 1.03 (D) 10.3

Open solution

Q219. JEE Main 2021 (Online) 17th March Morning Shift

Two ideal polyatomic gases at temperatures T_1 and T_2 are mixed so that there is no loss of energy. If F_1 and F_2, m_1 and m_2, n_1 and n_2 be the degrees of freedom, masses, number of molecules of the first and second gas respectively, the temperature of mixture of these two gases is :

(A) $\frac{n_1F_1T_1+n_2F_2T_2}{F_1+F_2}$ (B) $\frac{n_1F_1T_1+n_2F_2T_2}{n_1F_1+n_2F_2}$

(C) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$ (D) $\frac{n_1F_1T_1+n_2F_2T_2}{n_1+n_2}$

Open solution

Q220. JEE Main 2021 (Online) 17th March Morning Shift

Two identical metal wires of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is :

(A) $\frac{2K_1K_2}{K_1+K_2}$ (B) $\frac{K_1+K_2}{K_1K_2}$

(C) $\frac{K_1+K_2}{2K_1K_2}$ (D) $\frac{K_1K_2}{K_1+K_2}$

Open solution

Q221. JEE Main 2021 (Online) 16th March Evening Shift

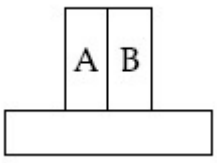
Calculate the value of mean free path (λ) for oxygen molecules at temperature 27°C and pressure 1.01×10^5 Pa. Assume the molecular diameter 0.3 nm and the gas is ideal. (k = 1.38×10^{-23} JK $^{-1}$)

(A) 32 nm (B) 58 nm (C) 86 nm (D) 102 nm

Open solution

Q222. JEE Main 2021 (Online) 16th March Evening Shift

A bimetallic strip consists of metals A and B. It is mounted rigidly as shown. The metal A has higher coefficient of expansion compared to that of metal B. When the bimetallic strip is placed in a cold bath, it will :



(A) Neither bend nor shrink (B) Bend towards the left

(C) Not bend but shrink (D) Bend towards the right

Open solution

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Q223. JEE Main 2021 (Online) 16th March Morning Shift

In thermodynamics, heat and work are :

(A) Path functions

(B) Point functions

(C) Extensive thermodynamics state variables

(D) Intensive thermodynamic state variables

Open solution

Q224. JEE Main 2021 (Online) 16th March Morning Shift

The volume V of an enclosure contains a mixture of three gases, 16 g of oxygen, 28 g of nitrogen and 44 g of carbon dioxide at absolute temperature T. Consider R as universal gas constant. The pressure of the mixture of gases is :

(A) $\frac{3RT}{V}$

(B) $\frac{4RT}{V}$

(C) $\frac{88RT}{V}$

(D) $\frac{5}{2}\frac{RT}{V}$

Open solution

Q225. JEE Main 2021 (Online) 26th February Evening Shift

The internal energy (U), pressure (P) and volume (V) of an ideal gas are related as U = 3PV + 4. The gas is :

(A) either monoatomic or diatomic.

(B) monoatomic only.

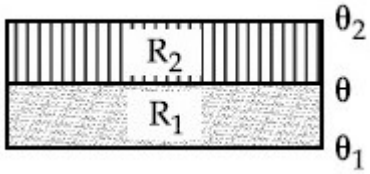
(C) polyatomic only.

(D) diatomic only.

Open solution

Q226. JEE Main 2021 (Online) 26th February Morning Shift

The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by :



(A) $\frac{\theta_1 R_2+\theta_2 R_1}{R_1+R_2}$

(B) $\frac{\theta_1 R_1+\theta_2 R_2}{R_1+R_2}$

(C) $\frac{\theta_1 R_2-\theta_2 R_1}{R_2-R_1}$

(D) $\frac{\theta_2 R_2-\theta_1 R_1}{R_2-R_1}$

Open solution

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Q227. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

Given below are two statements :

Statement I : In a diatomic molecule, the rotational energy at a given temperature obeys Maxwell's distribution.

Statement II : In a diatomic molecule, the rotational energy at a given temperature equals the translational kinetic energy for each molecule.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

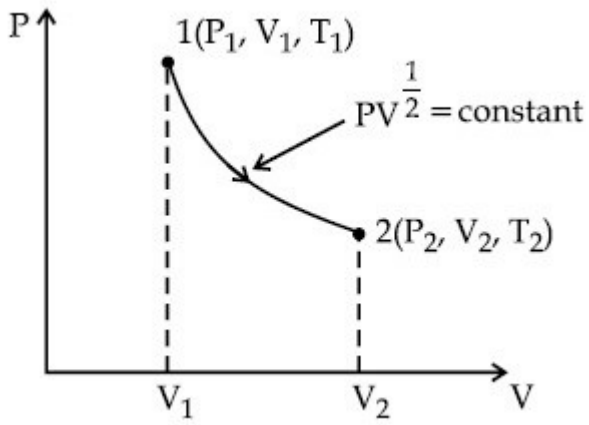
(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Q228. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

Thermodynamic process is shown below on a P-V diagram for one mole of an ideal gas. If $V_2 = 2V_1$ then the ratio of temperature T_2/T_1 is :



(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{2}$ (D) 2

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Q229. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : When a rod lying freely is heated, no thermal stress is developed in it.

Reason R : On heating, the length of the rod increases.

In the light of the above statements, choose the correct answer from the options given below :

(A) A is true but R is false

(B) A is false but R is true

(C) Both A and B are true but R is NOT the correct explanation of A

(D) Both A and R are true and R is the correct explanation of A

Q230. JEE Main 2021 (Online) 25th February Morning Shift

Open solution

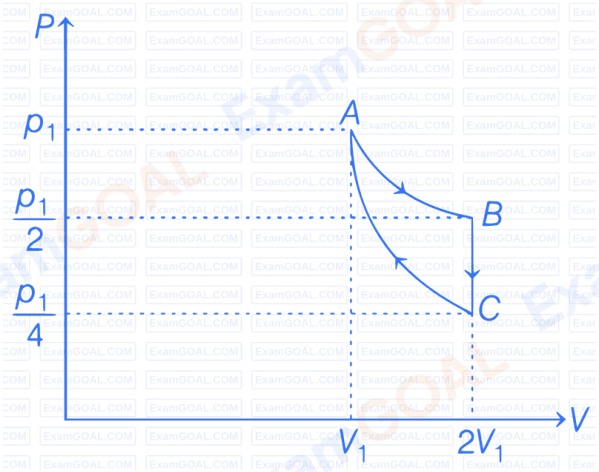
A diatomic gas, having $C_p = \frac{7}{2}R$ and $C_v = \frac{5}{2}R$, is heated at constant pressure. The ratio dU : dQ : dW :

(A) 5 : 7 : 3 (B) 3 : 7 : 2 (C) 5 : 7 : 2 (D) 3 : 5 : 2

Q231. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

If one mole of an ideal gas at (P_1, V_1) is allowed to expand reversibly and isothermally (A to B) its pressure is reduced to one-half of the original pressure (see figure). This is followed by a constant volume cooling till its pressure is reduced to one-fourth of the initial value (B $\rightarrow$ C). Then it is restored to its initial state by a reversible adiabatic compression (C to A). The net work done by the gas is equal to :



(A) $-\frac{RT}{2(\gamma-1)}$ (B) $RT\left(\ln 2 - \frac{1}{2(\gamma-1)}\right)$

(C) $RT \ln 2$ (D) 0

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Q232. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

On the basis of kinetic theory of gases, the gas exerts pressure because its molecules :

(A) continuously lose their energy till it reaches wall.

(B) are attracted by the walls of container.

(C) suffer change in momentum when impinge on the walls of container.

(D) continuously stick to the walls of container.

Q233. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

Match List I with List II.

	List I		List II
(a)	Isothermal	(i)	Pressure constant
(b)	Isochoric	(ii)	Temperature constant
(c)	Adiabatic	(iii)	Volume constant
(d)	Isobaric	(iv)	Heat content is constant

Choose the correct answer from the options given below :

(1) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(2) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(3) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(4) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

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Q234. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

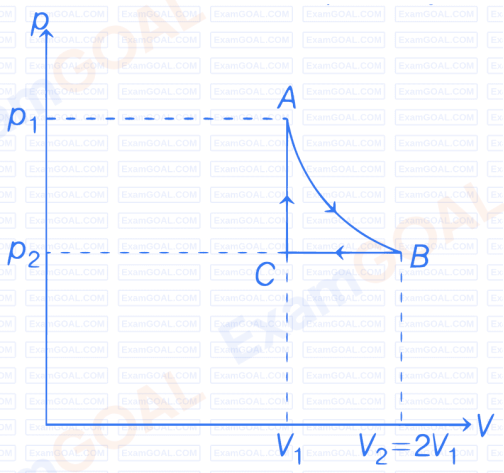
n mole of a perfect gas undergoes a cyclic process ABCA (see figure) consisting of the following processes.

A $\rightarrow$ B : Isothermal expansion at temperature T so that the volume is doubled from V_1 to $V_2 = 2V_1$ and pressure changes from P_1 to P_2

B $\rightarrow$ C : Isobaric compression at pressure P_2 to initial volume V_1.

C $\rightarrow$ A : Isochoric change leading to change of pressure from P_2 to P_1.

Total work done in the complete cycle ABCA is :



(A) nRTln 2

(B) 0

(C) $nRT\left(\ln 2 - \frac{1}{2}\right)$

(D) $nRT\left(\ln 2 + \frac{1}{2}\right)$

Q235. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

Each side of a box made of metal sheet in cubic shape is 'a' at room temperature 'T', the coefficient of linear expansion of the metal sheet is 'α'. The metal sheet is heated uniformly, by a small temperature ΔT, so that its new temperature is T + ΔT. Calculate the increase in the volume of the metal box.

(A) $3a^3\alpha\Delta T$

(B) $4\pi a^3\alpha\Delta T$

(C) $\frac{4}{3}\pi a^3\alpha\Delta T$

(D) $4a^3\alpha\Delta T$

Q236. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

In a dilute gas at pressure P and temperature T, the mean time between successive collisions of a molecule varies with T as :

(A) $\sqrt{T}$

(B) T

(C) $\frac{1}{T}$

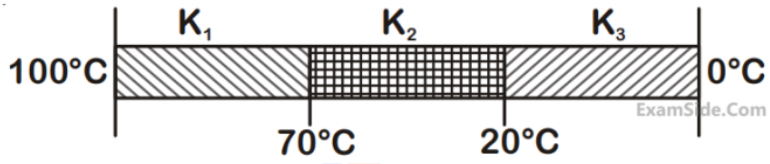
(D) $\frac{1}{\sqrt{T}}$

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Q237. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1 , K_2 and K_3 , respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at 100°C and the other at 0°C (see figure). If the joints of the rod are at 70°C and 20°C in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1 , K_2 and K_3 is :



(A) $K_1 : K_3 = 2 : 3$, $K_2 : K_3 = 2 : 5$

(B) $K_1 < K_2 < K_3$

(C) $K_1 : K_2 = 5 : 2$, $K_1 : K_3 = 3 : 5$

(D) $K_1 > K_2 > K_3$

Q238. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

Molecules of an ideal gas are known to have three translational degrees of freedom and two rotational degrees of freedom.The gas is maintained at a temperature of T. The total internal energy, U of a mole of this gas, and the value of $\gamma \left(= \frac{C_p}{C_v}\right)$ are given, respectively by:

(A) U = $\frac{5}{2}RT$ and $\gamma = \frac{7}{5}$

(B) U = 5RT and $\gamma = \frac{6}{5}$

(C) U = 5RT and $\gamma = \frac{7}{5}$

(D) U = $\frac{5}{2}RT$ and $\gamma = \frac{6}{5}$

Q239. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

Two different wires having lengths L_1 and L_2, and respective temperature coefficient of linear expansion α_1 and α_2, are joined end-to-end. Then the effective temperature coefficient of linear expansion is :

(A) $2\sqrt{\alpha_1\alpha_2}$

(B) $4\frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}\frac{L_2L_1}{(L_2+L_1)^2}$

(C) $\frac{\alpha_1+\alpha_2}{2}$

(D) $\frac{\alpha_1L_1+\alpha_2L_2}{L_1+L_2}$

Q240. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

In an adiabatic process, the density of a diatomic gas becomes 32 times its initial value. The final pressure of the gas is found to be n times the initial pressure. The value of n is :

(A) 128 (B) 32 (C) 326 (D) $\frac{1}{32}$

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Q241. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

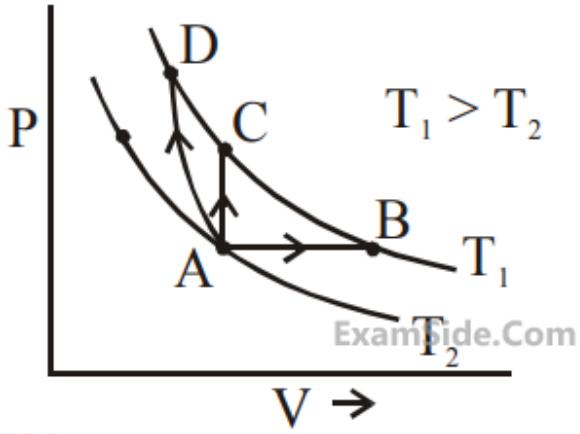
A bullet of mass 5 g, travelling with a speed of 210 m/s, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is 0.030 cal/(g – ^{o}C) (1 cal = 4.2×10^{7} ergs) close to :

(A) 87.5 ^{o}C (B) 83.3 ^{o}C (C) 38.4 ^{o}C (D) 119.2 ^{o}C

Q242. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

Three different processes that can occur in an ideal monoatomic gas are shown in the P vs V diagram. The paths are labelled as A $\rightarrow$ B, A $\rightarrow$ C and A $\rightarrow$ D. The change in internal energies during these process are taken as E_{AB}, E_{AC} and E_{AD} and the work done as W_{AB}, W_{AC} and W_{AD}.
The correct relation between these parameters are :



(A) $E_{AB} < E_{AC} < E_{AD}$, $W_{AB} > 0$, $W_{AC} > W_{AD}$

(B) $E_{AB} = E_{AC} = E_{AD}$, $W_{AB} > 0$, $W_{AC} = 0$, $W_{AD} < 0$

(C) $E_{AB} > E_{AC} > E_{AD}$, $W_{AB} < W_{AC} < W_{AD}$

(D) $E_{AB} = E_{AC} < E_{AD}$, $W_{AB} > 0$, $W_{AC} = 0$, $W_{AD} < 0$

Q243. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

Number of molecules in a volume of 4 cm^3 of a perfect monoatomic gas at some temperature T and at a pressure of 2 cm of mercury is close to?
(Given, mean kinetic energy of a molecule (at T) is 4×10^{-14} erg, g = 980 cm/s^2, density of mercury = 13.6 g/cm^3)

(A) 5.8×10^{18} (B) 4.0×10^{16} (C) 5.8×10^{16} (D) 4.0×10^{18}

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Q244. JEE Main 2020 (Online) 4th September Evening Slot

Match the thermodynamic processes taking place in a system with the correct conditions. In the table : ΔQ is the heat supplied, ΔW is the work done and ΔU is change in internal energy of the system.

Open solution

Process	Condition
(I) Adiabatic	(1) $\Delta W = 0$
(II) Isothermal	(2) $\Delta Q = 0$
(III) Isochoric	(3) $\Delta U \neq 0, \Delta W \neq 0, \Delta Q \neq 0$
(IV) Isobaric	(4) $\Delta U = 0$

(A) (I) - (1), (II) - (1), (III) - (2), (IV) - (3)

(B) (I) - (2), (II) - (4), (III) - (1), (IV) - (3)

(C) (I) - (1), (II) - (2), (III) - (4), (IV) - (4)

(D) (I) - (2), (II) - (1), (III) - (4), (IV) - (3)

Q245. JEE Main 2020 (Online) 4th September Morning Slot

The specific heat of water = 4200 J $kg^{-1}K^{-1}$ and the latent heat of ice = 3.4×10^5 J kg^{-1}. 100 grams of ice at 0^oC is placed in 200 g of water at 25^oC. The amount of ice that will melt as the temperature of water reaches 0^oC is close to (in grams) :

Open solution

(A) 63.8 **(B)** 61.7 **(C)** 69.3 **(D)** 64.6

Q246. JEE Main 2020 (Online) 4th September Morning Slot

Match the $\frac{C_P}{C_V}$ ratio for ideal gases with different type of molecules :

Open solution

Molecule Type	C_P/C_V
(A) Monatomic	(I) 7/5
(B) Diatomic rigid molecules	(II) 9/7
(C) Diatomic non-rigid molecules	(III) 4/3
(D) Triatomic rigid molecules	(IV) 5/3

(A) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(B) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

(C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

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Q247. JEE Main 2020 (Online) 3rd September Evening Slot

To raise the temperature of a certain mass of gas by 50°C at a constant pressure, 160 calories of heat is required. When the same mass of gas is cooled by 100°C at constant volume, 240 calories of heat is released. How many degrees of freedom does each molecule of this gas have (assume gas to be ideal)?

Open solution

(A) 6 (B) 7 (C) 5 (D) 3

Q248. JEE Main 2020 (Online) 3rd September Evening Slot

A calorimeter of water equivalent 20 g contains 180 g of water at 25°C. 'm' grams of steam at 100°C is mixed in it till the temperature of the mixure is 31°C. The value of 'm' is close to : (Latent heat of water = 540 cal g^{-1}, specific heat of water = 1 cal g^{-1} $^oC^{-1}$)

Open solution

(A) 2.6 (B) 2 (C) 4 (D) 3.2

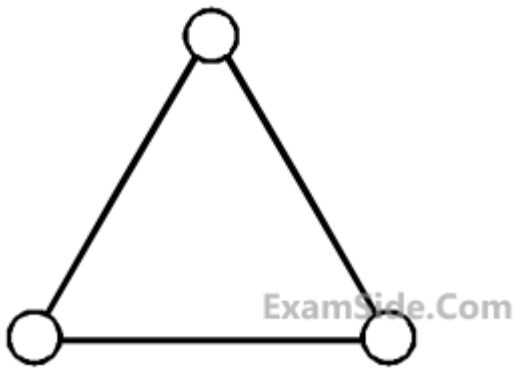
Q249. JEE Main 2020 (Online) 3rd September Morning Slot

A balloon filled with helium (32°C and 1.7 atm.) bursts. Immediately afterwards the expansion of helium can be considered as

Open solution

(A) Irreversible adiabatic (B) Reversible adiabatic
(C) Irreversible isothermal (D) Reversible isothermal

Q250. JEE Main 2020 (Online) 3rd September Morning Slot



Open solution

Consider a gas of triatomic molecules. The molecules are assumed to be triangular and made of massless rigid rods whose vertices are occupied by atoms. The internal energy of a mole of the gas at temperature T is :

(A) $\frac{3}{2}RT$ (B) $\frac{9}{2}RT$ (C) $\frac{5}{2}RT$ (D) 3RT

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Q251. JEE Main 2020 (Online) 2nd September Evening Slot

When the temperature of a metal wire is increased from 0°C to 10°C, its length increases by 0.02%. The percentage change in its mass density will be closest to :

(A) 0.008 **(B)** 0.06 **(C)** 0.8 **(D)** 2.3

Open solution

Q252. JEE Main 2020 (Online) 2nd September Evening Slot

An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are true?
(A) the mean free path of the molecules decreases.
(B) the mean collision time between the molecules decreases.
(C) the mean free path remains unchanged.
(D) the mean collision time remains unchanged.

(A) (C) and (D) **(B)** (A) and (D) **(C)** (B) and (C) **(D)** (A) and (B)

Open solution

Q253. JEE Main 2020 (Online) 2nd September Morning Slot

A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Assuming the gases to be ideal and the oxygen bond to be rigid, the total internal energy (in units of RT) of the mixture is :

(A) 11 **(B)** 20 **(C)** 15 **(D)** 13

Open solution

Q254. JEE Main 2020 (Online) 9th January Evening Slot

Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to :

(A) 2.3 **(B)** 1.83 **(C)** 4.67 **(D)** 3.67

Open solution

Q255. JEE Main 2020 (Online) 9th January Morning Slot

Consider two ideal diatomic gases A and B at some temperature T. Molecules of the gas A are rigid, and have a mass m. Molecules of the gas B have an additional vibrational mode, and have a mass $\frac{m}{4}$. The ratio of the specific heats (C_V^A and C_V^B) of gas A and B, respectively is :

(A) 7 : 9 **(B)** 5 : 7 **(C)** 3 : 5 **(D)** 5 : 9

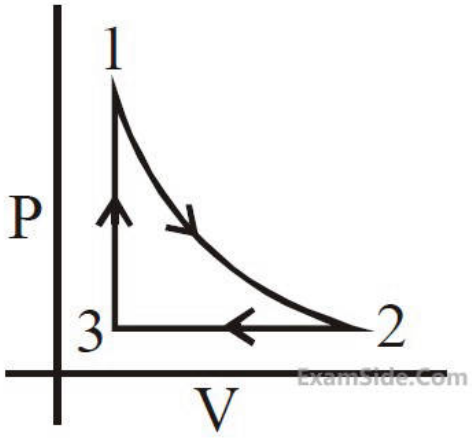
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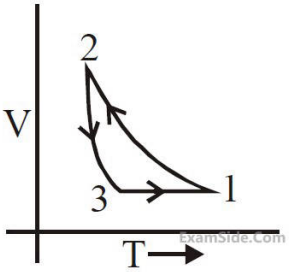
Q256. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

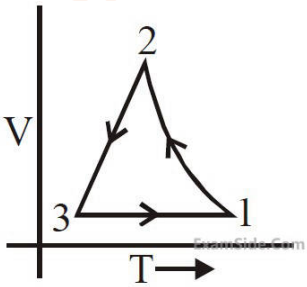
Which of the following is an equivalent cyclic process corresponding to the thermodynamic cycle given in the figure ? where, 1 $\rightarrow$ 2 is adiabatic.
(Graphs are schematic and are not to scale)



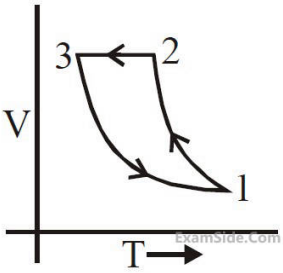
(A)



(B)

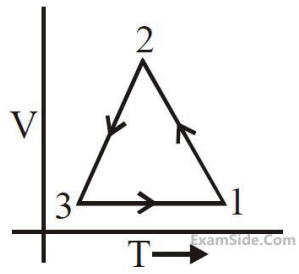


(C)



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(D)



Q257. JEE Main 2020 (Online) 8th January Evening Slot

Consider a mixture of n moles of helium gas and 2n moles of oxygen gas (molecules taken to be rigid) as an ideal gas. Its C_P/C_V value will be :

(A) 23/15 (B) 67/45 (C) 40/27 (D) 19/13

Open solution

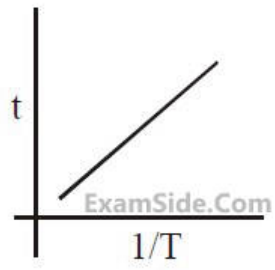
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Q258. JEE Main 2020 (Online) 8th January Morning Slot

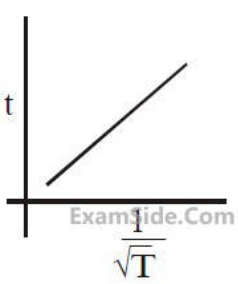
The plot that depicts the behavior of the mean free time t (time between two successive collisions) for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is:
(Graphs are schematic and not drawn to scale)

Open solution

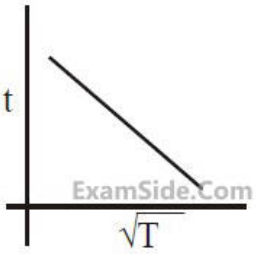
(A)



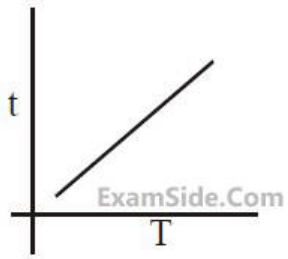
(B)



(C)



(D)

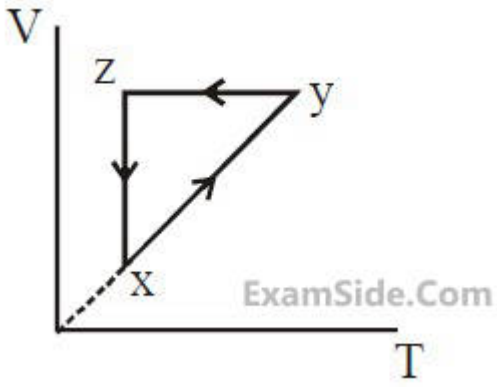


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Q259. JEE Main 2020 (Online) 8th January Morning Slot

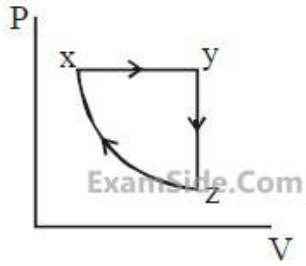
Open solution

A thermodynamic cycle xyzx is shown on a V-T diagram.

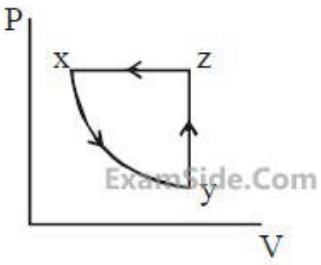


The P-V diagram that best describes this cycle is :
(Diagrams are schematic and not to scale)

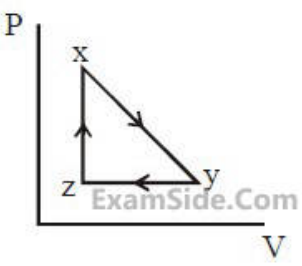
(A)



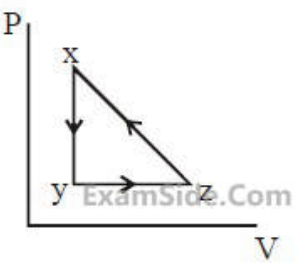
(B)



(C)



(D)



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Q260. JEE Main 2020 (Online) 7th January Evening Slot

Under an adiabatic process, the volume of an ideal gas gets doubled. Consequently the mean collision time between the gas molecule changes from τ_1 to τ_2 . If $\frac{C_p}{C_v} = \gamma$ for this gas then a good estimate for $\frac{\tau_2}{\tau_1}$ is given by :

Open solution

(A) $(2)^{\frac{1+\gamma}{2}}$

(B) 2

(C) $\left(\frac{1}{2}\right)^{\frac{1+\gamma}{2}}$

(D) $\left(\frac{1}{2}\right)^{\gamma}$

Q261. JEE Main 2020 (Online) 7th January Morning Slot

Two moles of an ideal gas with $\frac{C_P}{C_V} = \frac{5}{3}$ are mixed with 3 moles of another ideal gas with $\frac{C_P}{C_V} = \frac{4}{3}$. The value of $\frac{C_P}{C_V}$ for the mixture is :

Open solution

(A) 1.50

(B) 1.45

(C) 1.47

(D) 1.42

Q262. JEE Main 2020 (Online) 7th January Morning Slot

A litre of dry air at STP expands adiabatically to a volume of 3 litres. If γ = 1.40, the work done by air is : ($3^{1.4}$ = 4.6555) [Take air to be an ideal gas]

Open solution

(A) 60.7 J

(B) 100.8 J

(C) 90.5 J

(D) 48 J

Q263. JEE Main 2019 (Online) 12th April Evening Slot

One kg of water, at 20°C, heated in an electric kettle whose heating element has a mean (temperature averaged) resistance of 20 Ω. The rms voltage in the mains is 200 V. Ignoring heat loss from the kettle, time taken for water to evaporate fully, is close to : [Specific heat of water = 4200 J/(kg ^{o}C), Latent heat of water = 2260 kJ/kg]

Open solution

(A) 10 minutes

(B) 22 minutes

(C) 3 minutes

(D) 16 minutes

Q264. JEE Main 2019 (Online) 12th April Evening Slot

A diatomic gas with rigid molecules does 10 J of work when expanded at constant pressure. What would be the heat energy absorbed by the gas, in this process ?

Open solution

(A) 35 J

(B) 30 J

(C) 25 J

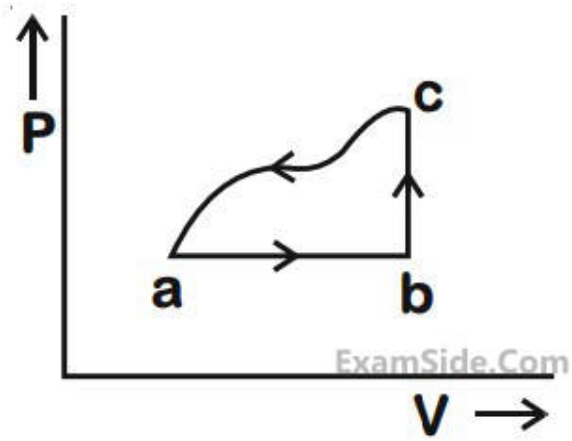
(D) 40 J

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Q265. JEE Main 2019 (Online) 12th April Morning Slot

A sample of an ideal gas is taken through the cyclic process abca as shown in the figure. The change in the internal energy of the gas along the path ca is −180 J. The gas absorbs 250 J of heat along the path ab and 60 J along the path bc. The work done by the gas along the path abc is:

Open solution



(A) 120 J (B) 130 J (C) 100 J (D) 140 J

Q266. JEE Main 2019 (Online) 12th April Morning Slot

At 40° C, a brass wire of 1 mm radius is hung from the ceiling. A small mass, M is hung from the free end of the wire. When the wire is cooled down from 40°C to 20°C it regains its original length of 0.2 m. The value of M is close to : (Coefficient of linear expansion and Young's modulus of brass are 10^{-5} /^{o}C and 10^{11} N/m ^{2} , respectively; g= 10 ms^{-2})

Open solution

(A) 1.5 kg (B) 0.5 kg (C) 9 kg (D) 0.9 kg

Q267. JEE Main 2019 (Online) 12th April Morning Slot

Two moles of helium gas is mixed with three moles of hydrogen molecules (taken to be rigid). What is the molar specific heat of mixture at constant volume ? (R = 8.3 J/mol K)

Open solution

(A) 21.6 J/mol K (B) 17.4 J/mol K (C) 15.7 J/mol K (D) 19.7 J/mol K

Q268. JEE Main 2019 (Online) 12th April Morning Slot

When M_1 gram of ice at −10°C (specific heat = 0.5 cal g^{-1} ^{o}C^{-1}) is added to M_2 gram of water at 50°C, finally no ice is left and the water is at 0°C. The value of latent heat of ice, in cal g^{-1} is :

Open solution

(A) $\frac{50M_2}{M_1} - 5$ (B) $\frac{50M_2}{M_1}$ (C) $\frac{5M_2}{M_1} - 5$ (D) $\frac{5M_1}{M_2} - 50$

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Q269. JEE Main 2019 (Online) 10th April Evening Slot

One mole of ideal gas passes through a process where pressure and volume obey the relation $P = P_0\left[1 - \frac{1}{2}\left(\frac{V_0}{V}\right)^2\right]$. Here P_0 and V_0 are constants. Calculate the change in the temperature of the gas if its volume changes form V_0 to $2V_0$

Open solution

(A) $\frac{3}{4}\frac{P_0V_0}{R}$ (B) $\frac{1}{2}\frac{P_0V_0}{R}$

(C) $\frac{5}{4}\frac{P_0V_0}{R}$ (D) $\frac{1}{4}\frac{P_0V_0}{R}$

Q270. JEE Main 2019 (Online) 10th April Evening Slot

When heat Q is supplied to a diatomic gas of rigid molecules, at constant volume its temperature increases by ΔT. the heat required to produce the same change in temperature, at a constant pressure is :

Open solution

(A) $\frac{7}{5}Q$ (B) $\frac{3}{2}Q$ (C) $\frac{2}{3}Q$ (D) $\frac{5}{3}Q$

Q271. JEE Main 2019 (Online) 10th April Morning Slot

A cylinder with fixed capacity of 67.2 lit contains helium gas at STP. The amount of heat needed to raise the temperature of the gas by 20°C is : [Given that R = 8.31 J mol^{-1} K^{-1}]

Open solution

(A) 374 J (B) 700 J (C) 748 J (D) 350 J

Q272. JEE Main 2019 (Online) 10th April Morning Slot

A 25×10^{-3} m^3 volume cylinder is filled with 1 mol of O_2 gas at room temperature (300K). The molecular diameter of O_2, and its root mean square speed, are found to be 0.3 nm, and 200 m/s, respectively. What is the average collision rate (per second) for an O_2 molecule ?

Open solution

(A) $\sim 10^{13}$ (B) $\sim 10^{12}$ (C) $\sim 10^{11}$ (D) $\sim 10^{10}$

Q273. JEE Main 2019 (Online) 10th April Morning Slot

n moles of an ideal gas with constant volume heat capcity C_V undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is :

Open solution

(A) $\frac{nR}{C_V - nR}$ (B) $\frac{4nR}{C_V - nR}$ (C) $\frac{4nR}{C_V + nR}$ (D) $\frac{nR}{C_V + nR}$

Q274. JEE Main 2019 (Online) 9th April Evening Slot

A massless spring (k = 800 N/m), attached with a mass (500 g) is completely immersed in 1 kg of water. The spring is stretched by 2 cm and released so that it starts vibrating. What would be the order of magnitude of the change in the temperature of water when the vibrations stop completely ? (Assume that the water container and spring receive negligible heat and specific heat of mass = 400 J/kg K, specific heat of water = 4184 J/kg K)

Open solution

(A) 10^{-3} K (B) 10^{-1} K (C) 10^{-5}K (D) 10^{-4} K

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Q275. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

The specific heats, C_P and C_V of a gas of diatomic molecules, A, are given (in units of J mol^{-1} K^{-1}) by 29 and 22, respectively. Another gas of diatomic molecules, B, has the corresponding values 30 and 21. If they are treated as ideal gases, then :-

(A) A is rigid but B has a vibrational mode

(B) A has a vibrational mode but B has none

(C) A has one vibrational mode and B has two

(D) Both A and B have a vibrational mode each

Q276. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

Two materials having coefficients of thermal conductivity '3K' and 'K' and thickness 'd' and '3d', respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are 'θ_2' and 'θ_1' respectively, ($\theta_2 > \theta_1$). The temperature at the interface is :-



(A) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$

(B) $\frac{\theta_2+\theta_1}{2}$

(C) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$

(D) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

Q277. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

An HCl molecule has rotational, translational and vibrational motions. If the rms velocity of HCl molecules in its gaseous phase is $\overline{v}$, m is its mass and k_B is Boltzmann constant, then its temperature will be :

(A) $\frac{m\overline{v}^2}{5k_B}$

(B) $\frac{m\overline{v}^2}{6k_B}$

(C) $\frac{m\overline{v}^2}{7k_B}$

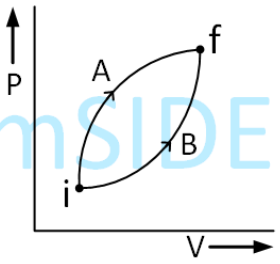
(D) $\frac{m\overline{v}^2}{3k_B}$

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Q278. JEE Main 2019 (Online) 9th April Morning Slot

Following figure shows two processes A and B for a gas. If ΔQ_A and ΔQ_B are the amount of heat absorbed by the system in two cases, and ΔU_A and ΔU_B are changes in internal energies, respectively, then :

Open solution



ExamSIDE.com

(A) $\Delta Q_A > \Delta Q_B$; $\Delta U_A > \Delta U_B$

(B) $\Delta Q_A < \Delta Q_B$; $\Delta U_A < \Delta U_B$

(C) $\Delta Q_A > \Delta Q_B$; $\Delta U_A = \Delta U_B$

(D) $\Delta Q_A = \Delta Q_B$; $\Delta U_A = \Delta U_B$

Q279. JEE Main 2019 (Online) 9th April Morning Slot

For a given gas at 1 atm pressure, rms speed of the molecule is 200 m/s at 127°C. At 2 atm pressure and at 227°C, the rms speed of the molecules will be :

Open solution

(A) 100 m/s

(B) $100\sqrt{5}$ m/s

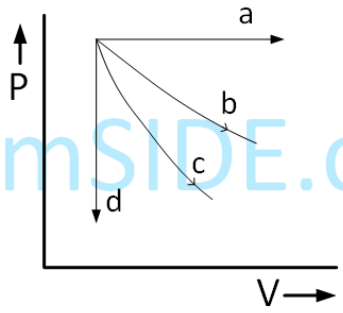
(C) $80\sqrt{5}$ m/s

(D) 80 m/s

Q280. JEE Main 2019 (Online) 8th April Evening Slot

The given diagram shows four processes i.e., isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in the same order is given by :-

Open solution



ExamSIDE.com

(A) d a b c

(B) a d b c

(C) d a c b

(D) a d c b

Q281. JEE Main 2019 (Online) 8th April Evening Slot

The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to :

[Boltzmann Constant $k_B = 1.38 \times 10^{-23}$ J/K Avogadro Number $N_A = 6.02 \times 10^{26}$ /kg Radius of Earth : 6.4×10^6 m Gravitational acceleration on Earth = $10 ms^{-2}$]

Open solution

(A) 3×10^5 K

(B) 10^4 K

(C) 650 K

(D) 800 K

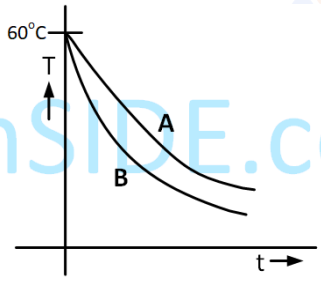
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Q282. JEE Main 2019 (Online) 8th April Morning Slot

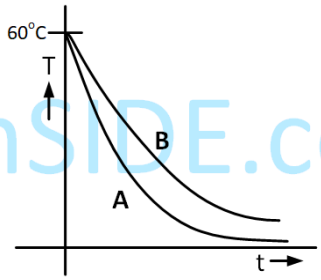
Open solution

Two identical beakers A and B contain equal volumes of two different liquids at 60°C each and left to cool down. Liquid in A has density of 8×10^2 kg/m^3 and specific heat of 2000 J kg^{-1} K^{-1} while liquid in B has density of 10^3 kg m^{-3} and specific heat of 4000 J kg^{-1} K^{-1}. Which of the following best describes their temperature versus time graph schematically? (assume the emissivity of both the beakers to be the same)

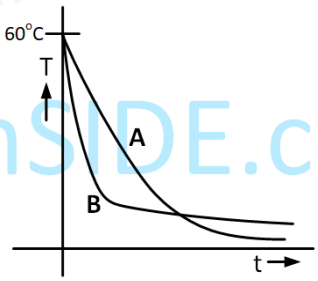
(A)



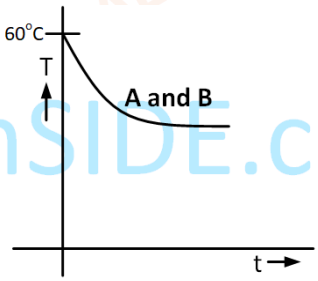
(B)



(C)



(D)



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Q283. JEE Main 2019 (Online) 8th April Morning Slot

Open solution

A thermally insulated vessel contains 150g of water at 0°C. Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closest to : (Latent heat of vaporization of water = 2.10×10^6 J kg^{-1} and Latent heat of Fusion of water = 3.36×10^5 J kg^{-1})

(A) 35 g (B) 130 g (C) 20 g (D) 150 g

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Q284. JEE Main 2019 (Online) 8th April Morning Slot

If 10^{22} gas molecules each of mass 10^{-26} kg collide with a surface (perpendicular to it) elastically per second over an area 1 m^2 with a speed 10^4 m/s, the pressure exerted by the gas molecules will be of the order of :

(A) 10^8 N/m^2 (B) 10^{16} N/m^2 (C) 10^4 N/m^2 (D) 2 N/m^2

Open solution

Q285. JEE Main 2019 (Online) 12th January Evening Slot

An ideal gas is enclosed in a cylinder at pressure of 2 atm and temperature 300 K. The mean time between two successive collisions is 6×10^{-8} s. If the pressure is doubled and temperature is increased to 500 K, the mean time between two successive collisions will be close to

(A) 0.5×10^{-8} s (B) 4×10^{-8} s (C) 3×10^{-6} s (D) 2×10^{-7} s

Open solution

Q286. JEE Main 2019 (Online) 12th January Evening Slot

A vertical closed cylinder is separated into two parts by a frictionless piston of mass m and of negligible thickness. The piston is free to move along the length of the cylinder. The length of the cylinder above the piston is ℓ_1, and that below the piston is ℓ_2, such that $\ell_1 > \ell_2$. Each part of the cylinder contains n moles of an ideal gas at equal temperature T. If the piston is stationary, its mass, m, will be given by :
(R is universal gas constant and g is the acceleration due to gravity)

(A) $\frac{nRT}{g}\left[\frac{\ell_1-\ell_2}{\ell_1\ell_2}\right]$

(B) $\frac{RT}{g}\left[\frac{2\ell_1+\ell_2}{\ell_1\ell_2}\right]$

(C) $\frac{nRT}{g}\left[\frac{1}{\ell_2}+\frac{1}{\ell_1}\right]$

(D) $\frac{RT}{ng}\left[\frac{\ell_1-3\ell_2}{\ell_1\ell_2}\right]$

Open solution

Q287. JEE Main 2019 (Online) 12th January Morning Slot

An ideal gas occupies a volume of $2m^3$ at a pressure of 3×10^6 Pa. The energy of the gas is :

(A) 6×10^4 J (B) 9×10^6 J (C) 3×10^2 J (D) 10^8 J

Open solution

Q288. JEE Main 2019 (Online) 12th January Morning Slot

A cylinder of radius R is surrounded by a cylindrical shell of inner radius R and outer radius 2R. The thermal conductivity of the material of the inner cylinder is K_1 and the of the outer cylinder is K_2. Assuming no loss of heat, the effective thermal conductivity of the system for heat flowing along the length of the cylinder is :

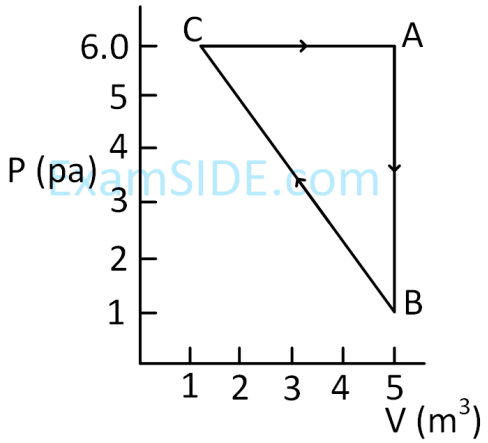
(A) $K_1 + K_2$ (B) $\frac{K_1+3K_2}{4}$ (C) $\frac{K_1+K_2}{2}$ (D) $\frac{2K_1+3K_2}{5}$

Open solution

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Q289. JEE Main 2019 (Online) 12th January Morning Slot

For the given cyclic process CAB as shown for a gas, the work done is :



Open solution

(A) 1 J (B) 10 J (C) 5 J (D) 30 J

Q290. JEE Main 2019 (Online) 11th January Evening Slot

Two rods A and B of identical dimensions are at temperature 30°C. If A is heated upto 180°C and B upto T°C, then the new lengths are the same. If the ratio of the coefficients of linear expansion of A and B is 4 : 3, then the value of T is

Open solution

(A) 200°C (B) 270°C (C) 230°C (D) 250°C

Q291. JEE Main 2019 (Online) 11th January Evening Slot

When 100 g of a liquid A at 100°C is added to 50 g of a liquid B at temperature 75°C, the temperature of the mixture becomes 90°C. The temperature of the mixture, if 100 g of liquid A at 100°C is added to 50 g of liquid B at 50°C, will be :

Open solution

(A) 60°C (B) 70°C (C) 85°C (D) 80°C

Q292. JEE Main 2019 (Online) 11th January Evening Slot

A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in °C, if this thermometer in the contact with the object reads $x_0/2$?

Open solution

(A) 60 (B) 35 (C) 25 (D) 40

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Q293. JEE Main 2019 (Online) 11th January Evening Slot

A metal ball of mass 0.1 kg is heated upto 500°C and dropped into a vessel of heat capacity 800 JK^{-1} and containing 0.5 kg water. The initial temperature of water and vessel is 30°C. What is the approximate percentage increment in the temperature of the water? [Specific Heat Capacities of water and metal are, respectively, 4200 Jkg^{-1} K^{-1} and 400 Jkg^{-1} K^{-1}]

(A) 20% (B) 25% (C) 15% (D) 30%

Open solution

Q294. JEE Main 2019 (Online) 11th January Evening Slot

In a process, temperature and volume of one mole of an ideal monoatomic gas are varied according to the relation VT = K, where K is a constant. In this process the temperature of the gas is increased by ΔT. The amount of heat absorbed by gas is (R is gas constant) :

(A) $\frac{1}{2}$ KRΔT (B) $\frac{1}{2}$ RΔT (C) $\frac{3}{2}$ RΔT (D) $\frac{2K}{3}$ ΔT

Open solution

Q295. JEE Main 2019 (Online) 11th January Morning Slot

A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. considering only translational and rotational modes, the total internal energy of the system is :

(A) 12 RT (B) 20 RT (C) 4 RT (D) 15 RT

Open solution

Q296. JEE Main 2019 (Online) 11th January Morning Slot

Ice at –20°C is added to 50 g of water at 40°C. When the temperature of the mixture reaches 0°C, it is found that 20 g of ice is still unmelted. The amount of ice added to the water was close to (Specific heat of water = 4.2J/g/^{o}C Specific heat of Ice = 2.1J/g/^{o}C Heat of fusion of water at 0°C= 334J/g)

(A) 100 g (B) 60 g (C) 50 g (D) 40 g

Open solution

Q297. JEE Main 2019 (Online) 11th January Morning Slot

A rigid diatomic ideal gas undergoes an adiabatic process at room temperature. The relation between temperature and volume for this process is TV^{x} = constant, then x is :

(A) $\frac{5}{3}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{2}{3}$

Open solution

Q298. JEE Main 2019 (Online) 10th January Evening Slot

Half mole of an ideal monoatomic gas is heated at constant pressure of 1 atm from 20°C to 90°C. Work done by gas is close to – (Gas constant R = 8.31 J/mol.K)

(A) 581 J (B) 73 J (C) 146 J (D) 291 J

Open solution

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Q299. JEE Main 2019 (Online) 10th January Evening Slot

Two kg of a monoatomic gas is at a pressure of 4×10^4 N/m^2. The density of the gas is 8 kg/m^3. What is the order of energy of the gas due to its thermal motion ?

(A) 10^4 J (B) 10^3 J (C) 10^5 J (D) 10^6 J

Open solution

Q300. JEE Main 2019 (Online) 10th January Evening Slot

An unknown metal of mass 192 g heated to a temperature of 100°C was immersed into a brass calorimeter of mass 128 g containing 240 g of water at a temperature of 8.4^{o}C. Calculate the specific heat of the unknown metal if water temperature stabilizes at 21.5^{o}C. (Specific heat of brass is 394 J kg^{-1} K^{-1})

(A) 458 J kg^{-1} K^{-1} (B) 1232 J kg^{-1} K^{-1} (C) 654 J kg^{-1} K^{-1} (D) 916 J kg^{-1} K^{-1}

Open solution

Q301. JEE Main 2019 (Online) 10th January Morning Slot

A heat source at T = 10^3 K is connected to another heat reservoir at T = 10^2 K by a copper slab which is 1 m thick. Given that the thermal conductivity of copper is 0.1 WK^{-1}m^{-1}, the energy flux through it in the steady state is -

(A) 200 Wm^{-2} (B) 65 Wm^{-2} (C) 120 Wm^{-2} (D) 90 Wm^{-2}

Open solution

Q302. JEE Main 2019 (Online) 9th January Evening Slot

A 15 g mass of nitrogen gas is enclosed in a vessel at a temperature 27°C. Amount of heat transferred to the gas, so that rms velocity of molecules is doubled, is about : [Take R = 8.3 J/K mole]

(A) 0.9 kJ (B) 6 kJ (C) 10 kJ (D) 14 kJ

Open solution

Q303. JEE Main 2019 (Online) 9th January Morning Slot

A mixture of 2 moles of helium gas (atomic mass = 4 u), and 1 mole of argon gas (atomic mass = 40 u) is kept at 300 K in a container. The ratio of their rms speeds $\left[\frac{V_{rms}\,(helium)}{V_{rms}\,(\arg on)}\right]$, is close to :

(A) 3.16 (B) 0.32 (C) 0.45 (D) 2.24

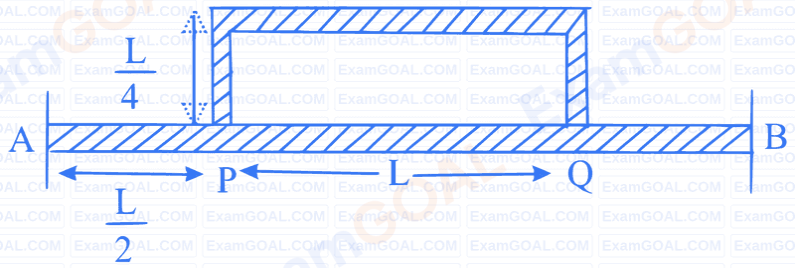
Open solution

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Q304. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

Temperature difference of 120°C is maintained between ends of a uniform rod AB of length 2L. Another bent rod PQ, of same cross-section as AB and length $\frac{3L}{2}$, is connected across AB (see figure). In steady state, temperature difference between P and Q will be close to :



(A) 45°C (B) 75°C (C) 60°C (D) 35°C

Q305. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

A rod, of length L at room temperature and uniform area of cross section A, is made of a metal having coefficient of linear expansion $\alpha/^{o}C$. It is observed that an external compressive force F, is applied on each of its ends, prevents any change in the length of the rod, when its temperature rises by ΔT K. Young's modulus, Y, for this metal is :

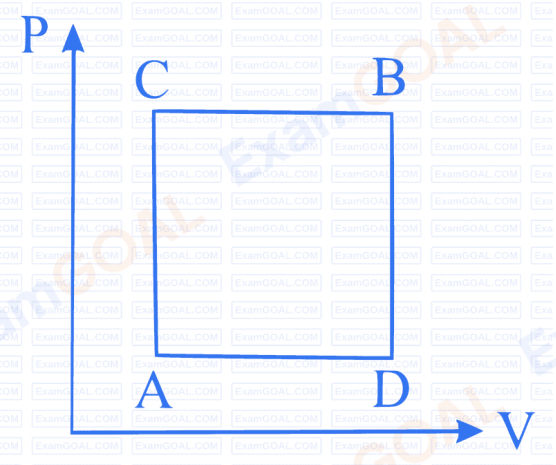
(A) $\frac{F}{A\alpha\Delta T}$ (B) $\frac{F}{A\alpha(\Delta T-273)}$

(C) $\frac{F}{2A\alpha\Delta T}$ (D) $\frac{2F}{A\alpha\Delta T}$

Q306. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

A gas can be taken from A to B via two different processes ACB and ADB.



When path ACB is used 60 J of heat flows into the system and 30 J of work is done by the system. If path ADB is used work done by the system is 10 J. The heat Flow into the system in path ADB is :

(A) 40 J (B) 80 J (C) 100 J (D) 20 J

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Q307. JEE Main 2018 (Online) 16th April Morning Slot

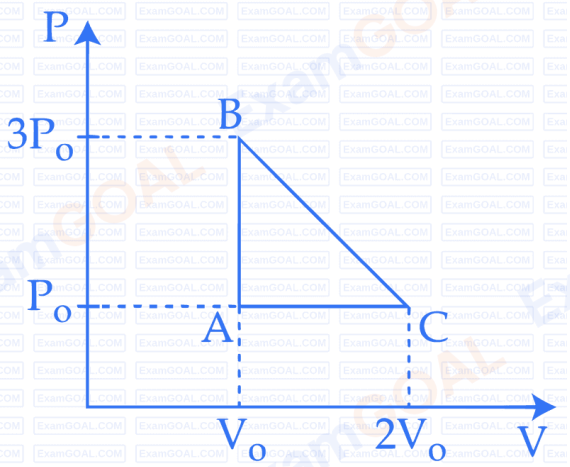
Two moles of helium are mixed with n moles of hydrogen. If $\frac{Cp}{Cv} = \frac{3}{2}$ for the mixture, then the value of n is :

(A) 1 (B) 3 (C) 2 (D) 3 / 2

Open solution

Q308. JEE Main 2018 (Online) 16th April Morning Slot

One mole of an ideal monoatomic gas is taken along the path ABCA as shown in the PV diagram. The maximum temperature attained by the gas along the path BC is given by :



(A) $\frac{25}{16}\frac{P_oV_o}{R}$ (B) $\frac{25}{8}\frac{P_oV_o}{R}$

(C) $\frac{25}{4}\frac{P_oV_o}{R}$ (D) $\frac{5}{8}\frac{P_oV_o}{R}$

Open solution

Q309. JEE Main 2018 (Offline)

Two moles of an ideal monatomic gas occupies a volume V at 27°C. The gas expands adiabatically to a volume 2 V. Calculate (a) the final temperature of the gas and (b) change in its internal energy.

(A) (a) 195 K (b) 2.7 kJ (B) (a) 189 K (b) 2.7 kJ

(C) (a) 195 K (b) −2.7 kJ (D) (a) 189 K (b) − 2.7 kJ

Open solution

Q310. JEE Main 2018 (Offline)

The mass of a hydrogen molecule is 3.32×10^{-27} kg. If 10^{23} hydrogen molecules strike, per second, a fixed wall of area 2 cm^2 at an angle of 45° to the normal, and rebound elastically with a speed of 10^3 m/s, then the pressure on the wall is nearly:

(A) 2.35×10^3 N m^{-2} (B) 4.70×10^3 N m^{-2}

(C) 2.35×10^2 N m^{-2} (D) 4.70×10^2 N m^{-2}

Open solution

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Q311. JEE Main 2018 (Online) 15th April Evening Slot

The value closest to the thermal velocity of a Helium atom at room temperature (300 K) in ms^{-1} is : [$k_B = 1.4 \times 10^{-23}$ J/K; $m_{He} = 7 \times 10^{-27}$ kg]

Open solution

(A) 1.3×10^4 (B) 1.3×10^3 (C) 1.3×10^5 (D) 1.3×10^2

Q312. JEE Main 2018 (Online) 15th April Morning Slot

One mole of an ideal monoatomic gas is compressed isothermally in a rigid vessel to double its pressure at room temperature, $27^{\circ}C$. The work done on the gas will be :

Open solution

(A) $300\ R$ (B) $300\ R\ ln\ 6$ (C) $300\ R\ ln\ 2$ (D) $300\ R\ ln\ 7$

Q313. JEE Main 2017 (Online) 9th April Morning Slot

A steel rail of length 5 m and area of cross section $40cm^2$ is prevented from expanding along its length while the temperature rises by 10^oC. If coefficient of linear expansion and Young's modulus of steel are $1.2 \times 10^{-5}\ K^{-1}$ and $2 \times 10^{11}\ Nm^{-2}$ respectively, the force developed in the rail is approximately :

Open solution

(A) 2×10^7 N (B) 1×10^5 N (C) 2×10^9 N (D) 3×10^{-5} N

Q314. JEE Main 2017 (Online) 9th April Morning Slot

N moles of a diatomic gas in a cylinder are at a temperature T. Heat is supplied to the cylinder such that the temperature remains constant but n moles of the diatomic gas get converted into monoatomic gas. What is the change in the total kinetic energy of the gas ?

Open solution

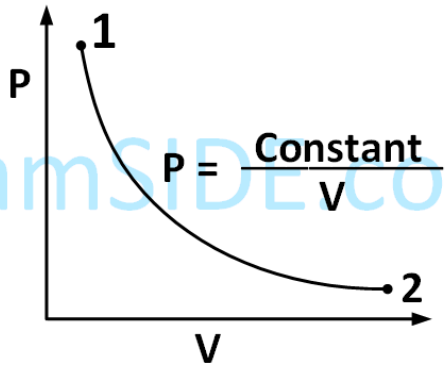
(A) $\frac{1}{2}$ nRT (B) 0 (C) $\frac{3}{2}$ nRT (D) $\frac{5}{2}$ nRT

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Q315. JEE Main 2017 (Online) 9th April Morning Slot

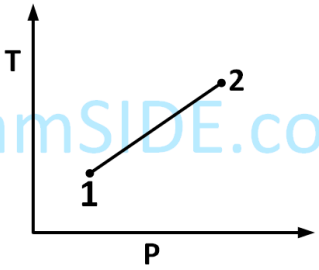
Open solution

For the P-V diagram given for an ideal gas,

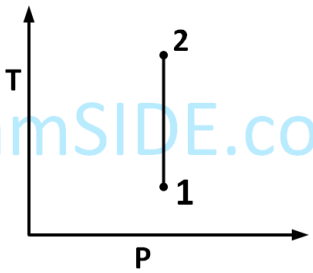


out of the following which one correctly represents the T-P diagram ?

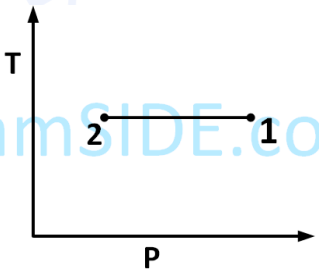
(A)



(B)

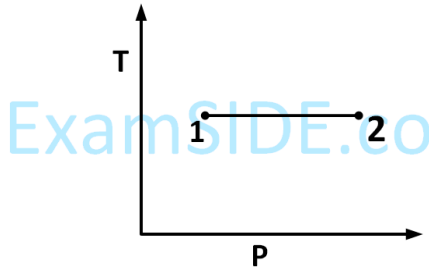


(C)



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(D)



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Q316. JEE Main 2017 (Online) 8th April Morning Slot

In an experiment, a sphere of aluminium of mass 0.20 kg is heated upto 150°C. Immediately, it is put into water of volume 150 cc at 27°C kept in a calorimeter of water equivalent to 0.025 kg. Final temperature of the system is 40°C. The specific heat of aluminium is : (take 4.2 Joule = 1 calorie)

Open solution

(A) 378 J/kg $-^{o}$C (B) 315 J/kg $-^{o}$C (C) 476 J/kg $-^{o}$C (D) 434 J/kg $-^{o}$C

Q317. JEE Main 2017 (Online) 8th April Morning Slot

A compressive force, F is applied at the two ends of a long thin steel rod. It is heated, simultaneously, such that its temperature increases by ΔT. The net change in its length is zero. Let ℓ be the length of the rod, A its area of cross-section,Y its Young's modulus, and α its coefficient of linear expansion. Then, F is equal to :

Open solution

(A) ℓ^2 Yα ΔT (B) ℓA Yα ΔT (C) A Yα ΔT (D) $\frac{AY}{\alpha \Delta T}$

Q318. JEE Main 2017 (Online) 8th April Morning Slot

An ideal gas has molecules with 5 degrees of freedom. The ratio of specific heats at constant pressure (C_p) and at constant volume (C_v) is :

Open solution

(A) 6 (B) $\frac{7}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{5}$

Q319. JEE Main 2017 (Offline)

A copper ball of mass 100 gm is at a temperature T. It is dropped in a copper calorimeter of mass 100 gm, filled with 170 gm of water at room temperature. Subsequently, the temperature of the system is found to be 75°C. T is given by: (Given : room temperature = 30°C, specific heat of copper = 0.1 cal/gm°C)

Open solution

(A) 825°C (B) 800°C (C) 885°C (D) 1250°C

Q320. JEE Main 2017 (Offline)

An external pressure P is applied on a cube at 0°C so that it is equally compressed from all sides. K is the bulk modulus of the material of the cube and α is its coefficient of linear expansion. Suppose we want to bring the cube to its original size by heating. The temperature should be raised by:

Open solution

(A) $\frac{P}{3\alpha K}$ (B) $\frac{P}{\alpha K}$ (C) $\frac{3\alpha}{PK}$ (D) 3PKα

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Q321. JEE Main 2017 (Offline)

Open solution

The temperature of an open room of volume 30 m^3 increases from 17°C to 27°C due to the sunshine. The atmospheric pressure in the room remains 1×10^5 Pa. If N_i and N_f are the number of molecules in the room before and after heating, then $N_f - N_i$ will be :

(A) -1.61×10^{23} (B) 1.38×10^{23} (C) 2.5×10^{25} (D) -2.5×10^{25}

Q322. JEE Main 2017 (Offline)

Open solution

C_P and C_v are specific heats at constant pressure and constant volume respectively. It is observed that

$C_P - C_v = a$ for hydrogen gas

$C_P - C_v = b$ for nitrogen gas

The correct relation between a and b is

(A) a = 28 b (B) a = 1/14 b (C) a = b (D) a = 14 b

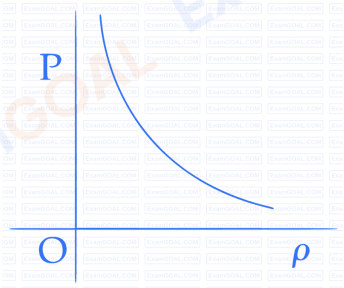
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Q323. JEE Main 2016 (Online) 10th April Morning Slot

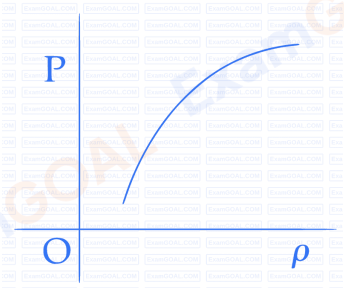
Which of the following shows the correct relationship between the pressure 'P' and density ρ of an ideal gas at constant temperature ?

Open solution

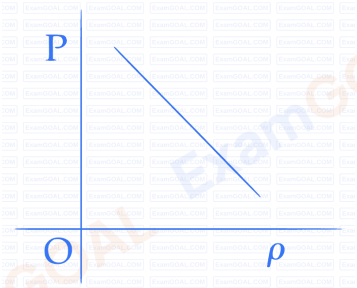
(A)



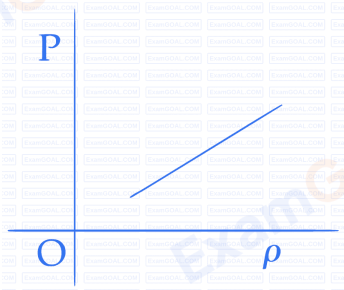
(B)



(C)



(D)



Q324. JEE Main 2016 (Online) 9th April Morning Slot

200 g water is heated from 40°C to 60°C. Ignoring the slight expansion of water, the change in its internal energy is close to (Given specific heat of water = 4184 J/kg/K) :

Open solution

(A) 8.4 kJ (B) 4.2 kJ (C) 16.7 kJ (D) 167.4 kJ

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Q325. JEE Main 2016 (Online) 9th April Morning Slot

The ratio of work done by an ideal monoatomic gas to the heat supplied to it in an isobaric process is :

(A) $\frac{3}{5}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{2}{5}$

Open solution

Q326. JEE Main 2016 (Online) 9th April Morning Slot

A simple pendulum made of a bob of mass *m* and a metallic wire of negligible mass has time period 2 s at T=0°C. If the temperature of the wire is increased and the corresponding change in its time period is plotted against its temperature, the resulting graph is a line of slope S. If the coefficient of linear expansion of metal is α then the value of S is :

(A) α (B) $\frac{\alpha}{2}$ (C) 2α (D) $\frac{1}{\alpha}$

Open solution

Q327. JEE Main 2016 (Offline)

A pendulum clock loses $12\ s$ a day if the temperature is $40^{\circ}C$ and gains $4\ s$ a day if the temperature is $20^{\circ}C$. The temperature at which the clock will show correct time, and the co-efficient of linear expansion (α) of the metal of the pendulum shaft are respectively :

(A) $30^{\circ}C;\ \alpha = 1.85 \times 10^{-3}/^{\circ}C$ (B) $55^{\circ}C;\ \alpha = 1.85 \times 10^{-2}/^{\circ}C$

(C) $25^{\circ}C;\ \alpha = 1.85 \times 10^{-5}/^{\circ}C$ (D) $60^{\circ}C;\ \alpha = 1.85 \times 10^{-4}/^{\circ}C$

Open solution

Q328. JEE Main 2016 (Offline)

An ideal gas undergoes a quasi static, reversible process in which its molar heat capacity C remains constant. If during this process the relation of pressure P and volume V is given by $PV^n =$ constant, then n is given by (Here C_p and C_v are molar specific heat at constant pressure and constant volume, respectively:

(A) $n = \frac{C_p - C}{C - C_v}$ (B) $n = \frac{C - C_v}{C - C_p}$

(C) $n = \frac{C_p}{C_v}$ (D) $n = \frac{C - C_p}{C - C_v}$

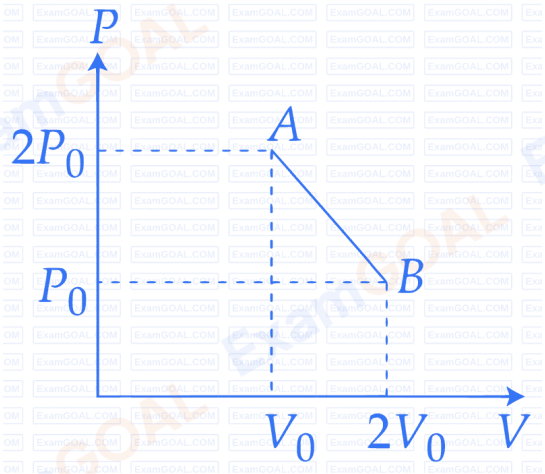
Open solution

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Q329. JEE Main 2016 (Offline)

Open solution

$'n'$ moles of an ideal gas undergoes a process $A \rightarrow B$ as shown in the figure. The maximum temperature of the gas during the process will be :



(A) $\frac{9P_0V_0}{2nR}$ (B) $\frac{9P_0V_0}{nR}$ (C) $\frac{9P_0V_0}{4nR}$ (D) $\frac{3P_0V_0}{2nR}$

Q330. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

An experiment takes 10 minutes to raise the temperature of water in a container from 0^oC to 100^oC and another 55 minutes to convert it totally into steam by a heater supplying heat at a uniform rate. Neglecting the specific heat of the container and taking specific heat of water to be 1 cal/g ^oC, the heat of vapourization according to this experiment will come out to be :

(A) 530 cal/g (B) 540 cal/g (C) 550 cal/g (D) 560 cal/g

Q331. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

Using equipartition of energy, the specific heat (in J kg $^{-1}$ K^{-1}) of aluminium at room temperature can be estimated to be (atomic weight of aluminium = 27)

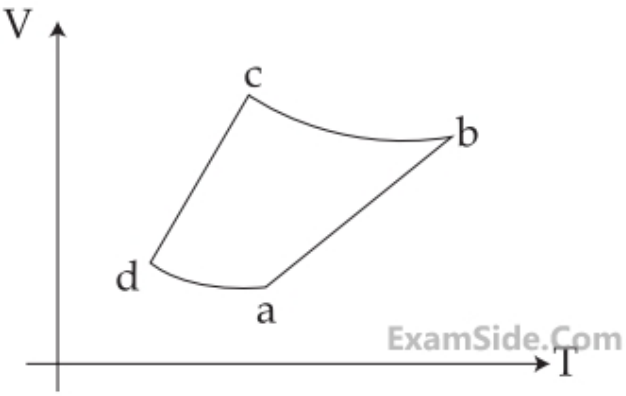
(A) 25 (B) 410 (C) 925 (D) 1850

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Q332. JEE Main 2015 (Online) 10th April Morning Slot

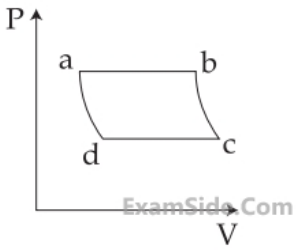
Open solution

An ideal gas goes through a reversible cycle a $\rightarrow$ b $\rightarrow$ c $\rightarrow$ d which has the V $-$ T diagram shown below. Process d $\rightarrow$ a and b $\rightarrow$ c are adiabatic.

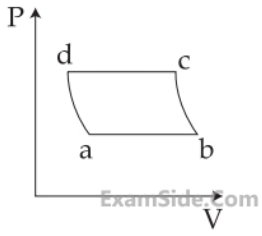


The corresponding P - V diagram for the process is (all figures are schematic and not drawn to scale) :

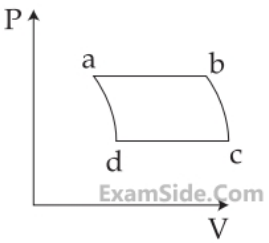
(A)



(B)

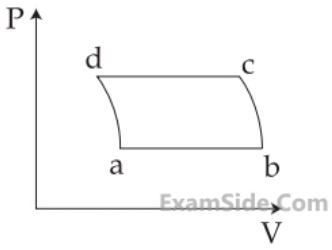


(C)



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(D)



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Q333. JEE Main 2015 (Online) 10th April Morning Slot

In an ideal gas at temperature T, the average force that a molecule applies on the walls of a closed container depends on T as T^q . A good estimate for q is :

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Open solution

Q334. JEE Main 2015 (Offline)

A solid body of constant heat capacity $1\ J/^\circ C$ is being heated by keeping it in contact with reservoirs in two ways:

(i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplies same amount of heat.

(ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat.

In both the cases body is brought from initial temperature $100^\circ C$ to final temperature $200^\circ C$. Entropy change of the body in the two cases respectively is :

(A) $ln2, 2ln2$ (B) $2ln2, 8ln2$ (C) $ln2, 4ln2$ (D) $ln2, ln2$

Open solution

Q335. JEE Main 2015 (Offline)

Consider a spherical shell of radius R at temperature T. The black body radiation inside it can be considered as an ideal gas of photons with internal energy per unit volume $u = \frac{U}{V} \propto T^4$ and pressure $p = \frac{1}{3}\left(\frac{U}{V}\right)$. If the shell now undergoes an adiabatic expansion the relation between T and R is:

(A) $T \propto \frac{1}{R}$ (B) $T \propto \frac{1}{R^3}$ (C) $T \propto e^{-R}$ (D) $T \propto e^{-3R}$

Open solution

Q336. JEE Main 2015 (Offline)

Consider an ideal gas confined in an isolated closed chamber. As the gas undergoes an adiabatic expansion, the average time of collision between molecules increases as V^q, where V is the volume of the gas. The value of q is: $\left(\gamma = \frac{C_p}{C_v}\right)$

(A) $\frac{\gamma+1}{2}$ (B) $\frac{\gamma-1}{2}$ (C) $\frac{3\gamma+5}{6}$ (D) $\frac{3\gamma-5}{6}$

Open solution

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Q337. JEE Main 2014 (Online) 19th April Morning Slot

A black coloured solid sphere of radius R and mass M is inside a cavity with vacuum inside. The walls of the cavity are maintained at a temperature T_0. The initial temperature of the sphere is $3T_0$.

If the specific heat of the material of the sphere varies as αT^3 per unit mass with the temperature T of the sphere, where α is a constant, then the time taken for the sphere to cool down to temperature $2T_0$ will be (σ is Stefan Boltzmann constant)

(A) $\frac{M\alpha}{4\pi R^2\sigma}\ln\left(\frac{3}{2}\right)$ (B) $\frac{M\alpha}{4\pi R^2\sigma}\ln\left(\frac{16}{3}\right)$

(C) $\frac{M\alpha}{16\pi R^2\sigma}\ln\left(\frac{16}{3}\right)$ (D) $\frac{M\alpha}{16\pi R^2\sigma}\ln\left(\frac{3}{2}\right)$

Open solution

Q338. JEE Main 2014 (Online) 19th April Morning Slot

A gas is compressed from a volume of 2 m^3 to a volume of 1 m^3 at a constant pressure of 100 N/m^2. Then it is heated at constant volume by supplying 150 J of energy. As a result, the internal energy of the gas :

(A) Increases by 250 J (B) Decreases by 250 J

(C) Increases by 50 J (D) Decreases by 50 J

Open solution

Q339. JEE Main 2014 (Online) 19th April Morning Slot

A gas molecule of mass M at the surface of the Earth has kinetic energy equivalent to 0^oC. If it were to go up straight without colliding with any other molecules, how high it would rise ? Assume that the height attained is much less than radius of the earth. (k_B is Boltzmann constant)

(A) 0 (B) $\frac{273k_B}{2Mg}$ (C) $\frac{546k_B}{3Mg}$ (D) $\frac{819k_B}{2Mg}$

Open solution

Q340. JEE Main 2014 (Online) 12th April Morning Slot

At room temperature a diatomic gas is found to have an r.m.s. speed of 1930 ms^{-1}. The gas is :

(A) H_2 (B) Cl_2 (C) O_2 (D) F_2

Open solution

Q341. JEE Main 2014 (Online) 11th April Morning Slot

During an adiabatic compression, 830 J of work is done on 2 moles of a diatomic ideal gas to reduce its volume by 50%. The change in its temperature is nearly : (R = 8.3 JK^{-1} mol^{-1})

(A) 40 K (B) 33 K (C) 20 K (D) 14 K

Open solution

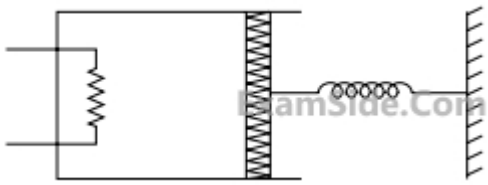
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Q342. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

An ideal monoatomic gas is confined in a cylinder by a spring loaded piston of cross section 8.0×10^{-3} m^2. Initially the gas is at 300K and occupies a volume of 2.4×10^{-3} m^3 and the spring is in its relaxed state as shown in figure. The gas is heated by a small heater until the piston moves out slowly by 0.1 m. The force constant of the spring is 8000 N/m and the atmospheric pressure is 1.0×10^5 N/m^2. The cylinder and the piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and the cylinder. The final temperature of the gas will be :

(Neglect the heat loss through the lead wires of the heater. The heat capacity of the heater coil is also negligible)



(A) 300 K (B) 800 K (C) 500 K (D) 1000 K

Q343. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Water of volume 2 L in a closed container is heated with a coil of 1 kW. While water is heated, the container loses energy at a rate of 160J/s. In how much time will the temperature of water rise from 27°C to 77°C ? (Specific heat of water is 4.2 kJ/kg K and that of the container is negligible).

(A) 8 min 20 s (B) 6 min 2 s (C) 7 min (D) 14 min

Q344. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

The equation of state for a gas is given by PV = nRT + αV, where n is the number of moles and α is a positive constant. The initial temperature and pressure of one mole of the gas contained in a cylinder are T_o and P_o respectively. The work done by the gas when its temperature doubles isobarically will be :

(A) $\frac{P_o T_o R}{P_o - \alpha}$ (B) $\frac{P_o T_o R}{P_o + \alpha}$

(C) $P_o T_o R \ln 2$ (D) $P_o T_o R$

Q345. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Modern vacuum pumps can evacuate a vessel down to a pressure of 4.0×10^{-15} atm. at room temperature (300 K). Taking R = 8.3 JK^{-1} $mole^{-1}$, 1 atm = 10^5 Pa and $N_{Avogadro} = 6 \times 10^{23}$ $mole^{-1}$, the mean distance between molecules of gas in an evacuated vessel will be of the order of :

(A) 0.2 μm (B) 0.2 mm (C) 0.2 cm (D) 0.2 nm

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Q346. JEE Main 2014 (Offline)

Open solution

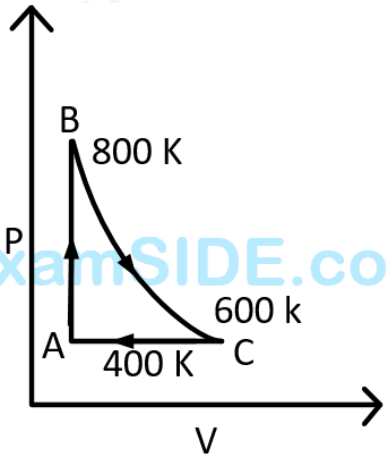
Three rods of Copper, Brass and Steel are welded together to form a Y shaped structure. Area of cross - section of each rod $= 4cm^2$. End of copper rod is maintained at $100^\circ C$ where as ends of brass and steel are kept at $0^\circ C$. Lengths of the copper, brass and steel rods are 46, 13 and 12 cms respectively. The rods are thermally insulated from surroundings excepts at ends. Thermal conductivities of copper, brass and steel are $0.92, 0.26$ and 0.12 CGS units respectively. Rate of heat flow through copper rod is:

(A) 1.2 cal/s (B) 2.4 cal/s (C) 4.8 cal/s (D) 6.0 cal/s

Q347. JEE Main 2014 (Offline)

Open solution

One mole of a diatomic ideal gas undergoes a cyclic process ABC as shown in figure. The process BC is adiabatic. The temperatures at A, B and C are 400 K, 800 K and 600 K respectively. Choose the correct statement :



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(A) The change in internal energy in whole cyclic process is 250 R.

(B) The change in internal energy in the process CA is 700 R.

(C) The change in internal energy in the process AB is - 350 R.

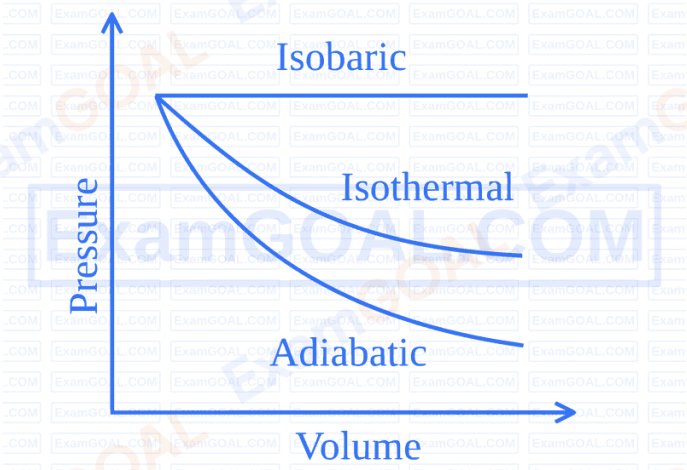
(D) The change in internal energy in the process BC is - 500 R.

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Q348. JEE Main 2013 (Online) 25th April Morning Slot

A sample of gas expands from V_1 to V_2. In which of the following, the work done will be greatest?

Open solution



(A) Same in all processes
(B) Isobaric process
(C) Isothermal process
(D) Adiabatic process

Q349. JEE Main 2013 (Online) 25th April Morning Slot

In the isothermal expansion of 10 g of gas from volume V to 2V the work done by the gas is 575 J. What is the root mean square speed of the molecules of the gas at that temperature?

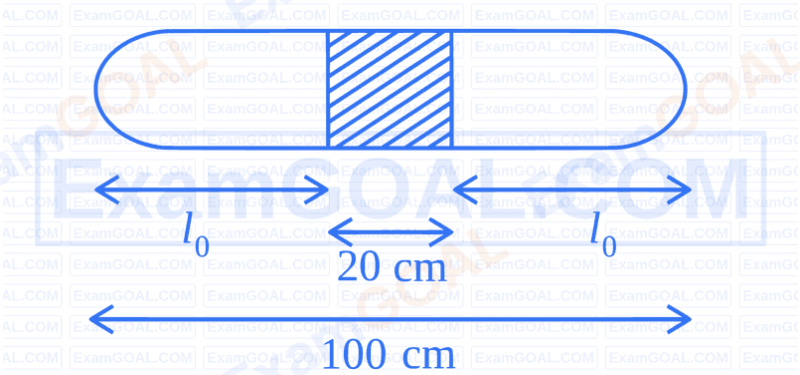
Open solution

(A) 398 m/s
(B) 520 m/s
(C) 499 m/s
(D) 532 m/s

Q350. JEE Main 2013 (Online) 23th April Morning Slot

A thin tube sealed at both ends is 100 cm long. It lies horizontally, the middle 20 cm containing mercury and two equal ends containing air at standard atmospheric pressure. If the tube is now turned to a vertical position, by what amount will the mercury be displaced?
(Given : cross-section of the tube can be assumed to be uniform)

Open solution



(A) 2.95 cm
(B) 5.18 cm
(C) 8.65 cm
(D) 0.0 cm

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Q351. JEE Main 2013 (Online) 23th April Morning Slot

The ratio of the coefficient of volume expansion of a glass container to that of a viscous liquid kept inside the container is 1 : 4. What fraction of the inner volume of the container should the liquid occupy so that the volume of the remaining vacant space will be same at all temperatures?

Open solution

(A) 2 : 5 (B) 1 : 4 (C) 1 : 64 (D) 1 : 8

Q352. JEE Main 2013 (Online) 23th April Morning Slot

500 g of water and 100 g of ice at 0°C are in a calorimeter whose water equivalent is 40 g. 10 g of steam at 100°C is added to it. Then water in the calorimeter is : (Latent heat of ice = 80 cal/g, Latent heat of steam = 540 cal/g)

Open solution

(A) 580 g (B) 590 g (C) 600 g (D) 610 g

Q353. JEE Main 2013 (Online) 23th April Morning Slot

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: The internal energy of a perfect gas is entirely kinetic and depends only on absolute temperature of the gas and not on its pressure or volume.
Statement 2: A perfect gas is heated keeping pressure constant and later at constant volume. For the same amount of heat the temperature of the gas at constant pressure is lower than that at constant volume.

Open solution

(A) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1.

(B) Statement-1 is true, Statement-2 is false.

(C) Statement-1 is true, Statement-2 is true but Statement-2 is not the correct explanation of Statement-1.

(D) Statement-1 is false, Statement-2 is true.

Q354. JEE Main 2013 (Online) 22th April Morning Slot

Given that 1 g of water in liquid phase has volume 1 cm^3 and in vapour phase 1671 cm^3 at atmospheric pressure and the latent heat of vaporization of water is 2256 J/g; the change in the internal energy in joules for 1 g of water at 373 K when it changes from liquid phase to vapour phase at the same temperature is :

Open solution

(A) 2256 (B) 167 (C) 2089 (D) 1

Q355. JEE Main 2013 (Online) 22th April Morning Slot

An ideal gas at atmospheric pressure is adiabatically compressed so that its density becomes 32 times of its initial value. If the final pressure of gas is 128 atmospheres, the value of 'γ' of the gas is :

Open solution

(A) 1.5 (B) 1.4 (C) 1.3 (D) 1.6

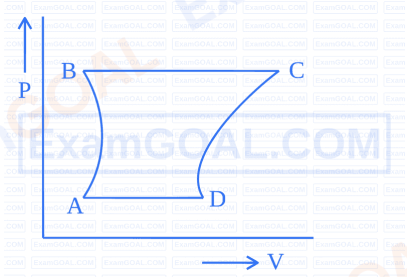
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Q356. JEE Main 2013 (Online) 22th April Morning Slot

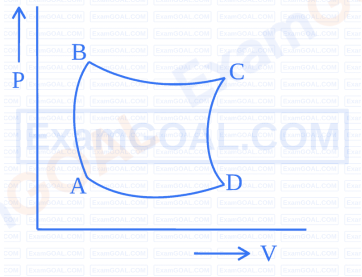
A certain amount of gas is taken through a cyclic process (A B C D A) that has two isobars, one isochore and one isothermal. The cycle can be represented on a P - V indicator diagram as :

Open solution

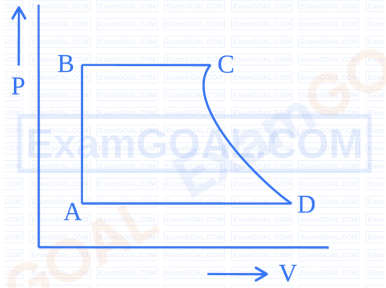
(A)



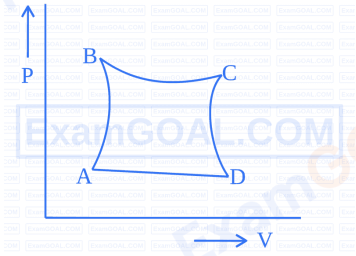
(B)



(C)



(D)



Q357. JEE Main 2013 (Online) 9th April Morning Slot

On a linear temperature scale Y, water freezes at -160°Y and boils at -50°Y. On this Y scale, a temperature of 340 K would be read as : (water freezes at 273 K and boils at 373 K)

(A) -73.7°Y (B) -233.7°Y (C) -86.3°Y (D) -106.3°Y

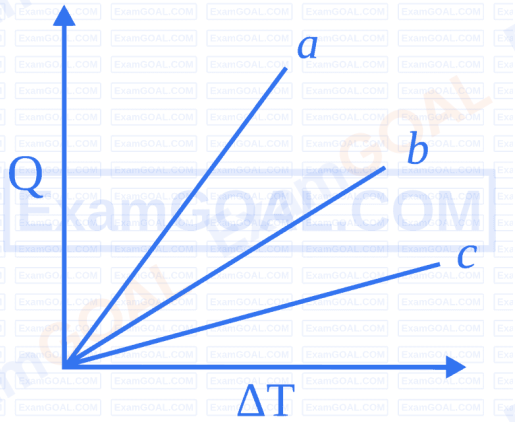
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Q358. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

Figure shows the variation in temperature (ΔT) with the amount of heat supplied (Q) in an isobaric process corresponding to a monoatomic (M), diatomic (D) and a polyatomic (P) gas. The initial state of all the gases are the same and the scales for the two axes coincide. Ignoring vibrational degrees of freedom, the lines a, b and c respectively correspond to :



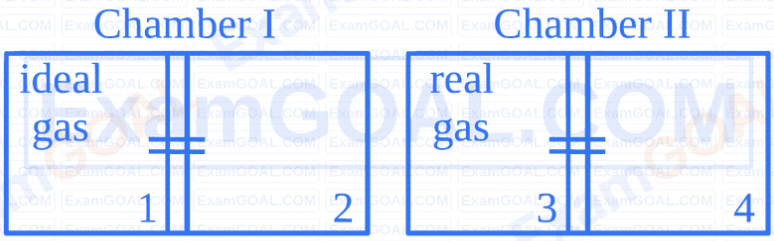
(A) P, M and D (B) M, D and P (C) P, D and M (D) D, M and P

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Q359. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

There are two identical chambers, completely thermally insulated from surroundings. Both chambers have a partition wall dividing the chambers in two compartments. Compartment 1 is filled with an ideal gas and Compartment 3 is filled with a real gas. Compartments 2 and 4 are vacuum. A small hole (orifice) is made in the partition walls and the gases are allowed to expand in vacuum.
Statement - 1 : No change in the temperature of the gas takes place when ideal gas expands in vacuum. However, the temperature of real gas goes down (cooling) when it expands in vacuum.
Statement - 2 : The internal energy of an ideal gas is only kinetic. The internal energy of a real gas is kinetic as well as potential.



(A) Statement-1 is false and Statement-2 is true.

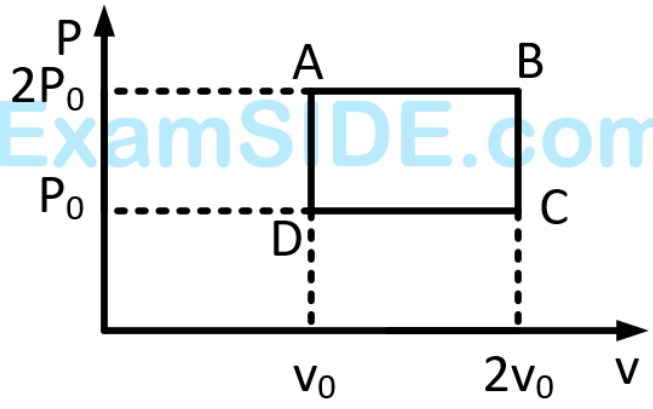
(B) Statement-1 and Statement-2 both are true. Statement-2 is the correct explanation of Statement-1.

(C) Statement-1 is true and Statement-2 is false.

(D) Statement-1 and Statement-2 both are true. Statement-2 is not correct explanation of Statement-1.

Q360. JEE Main 2013 (Offline)

Open solution



The above p-v diagram represents the thermodynamic cycle of an engine, operating with an ideal monatomic gas. The amount of heat, extracted from the source in a single cycle is

(A) p_0v_0

(B) $\left(\frac{13}{2}\right)p_0v_0$

(C) $\left(\frac{11}{2}\right)p_0v_0$

(D) $4p_0v_0$

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Q361. AIEEE 2012 (Online) 26th May

An ideal monatomic gas with pressure P, volume V and temperature T is expanded isothermally to a volume $2V$ and a final pressure P_i. If the same gas is expanded adiabatically to a volume $2V$, the final pressure is P_a. The ratio $\frac{P_a}{P_i}$ is

Open solution

(A) $2^{-1/3}$ (B) $2^{1/3}$ (C) $2^{2/3}$ (D) $2^{-2/3}$

Q362. AIEEE 2012 (Online) 19th May

A large cylindrical rod of length L is made by joining two identical rods of copper and steel of length $\left(\frac{L}{2}\right)$ each. The rods are completely insulated from the surroundings. If the free end of copper rod is maintained at $100^{\circ}\mathrm{C}$ and that of steel at $0^{\circ}\mathrm{C}$ then the temperature of junction is (Thermal conductivity of copper is 9 times that of steel)

Open solution

(A) $90^{\circ}\mathrm{C}$ (B) $50^{\circ}\mathrm{C}$ (C) $10^{\circ}\mathrm{C}$ (D) $67^{\circ}\mathrm{C}$

Q363. AIEEE 2012 (Online) 19th May

The pressure of an ideal gas varies with volume as $P = \alpha V$, where α is a constant. One mole of the gas is allowed to undergo expansion such that its volume becomes 'm' times its initial volume. The work done by the gas in the process is

Open solution

(A) $\frac{\alpha V}{2}\left(m^2-1\right)$ (B) $\frac{\alpha^2 V^2}{2}\left(m^2-1\right)$
(C) $\frac{\alpha}{2}\left(m^2-1\right)$ (D) $\frac{\alpha V^2}{2}\left(m^2-1\right)$

Q364. AIEEE 2012 (Online) 12th May

A given ideal gas with $\gamma = \frac{C_p}{C_v} = 1.5$ at a temperature T. If the gas is compressed adiabatically to one-fourth of its initial volume, the final temperature will be

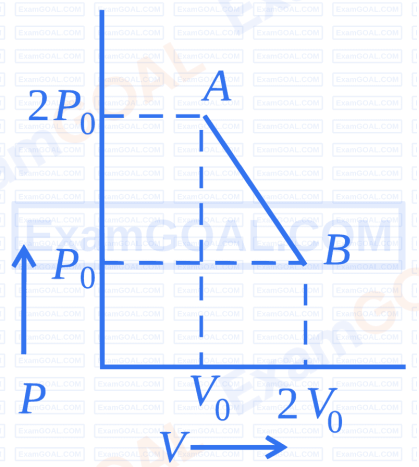
Open solution

(A) $2\sqrt{2}T$ (B) $4T$ (C) $2T$ (D) $8T$

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Q365. AIEEE 2012 (Online) 12th May

n moles of an ideal gas undergo a process $A \rightarrow B$ as shown in the figure. Maximum temperature of the gas during the process is



(A) $\frac{9P_0V_0}{nR}$ (B) $\frac{3P_0V_0}{2nR}$ (C) $\frac{9P_0V_0}{2nR}$ (D) $\frac{9P_0V_0}{4nR}$

Open solution

Q366. AIEEE 2012 (Online) 7th May

The heat radiated per unit area in 1 hour by a furnace whose temperature is $3000\ \mathrm{K}$ is $\left(\sigma = 5.7 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}\right)$

(A) $1.7 \times 10^{10}\ \mathrm{J}$ (B) $1.1 \times 10^{12}\ \mathrm{J}$ (C) $2.8 \times 10^{8}\ \mathrm{J}$ (D) $4.6 \times 10^{6}\ \mathrm{J}$

Open solution

Q367. AIEEE 2012 (Online) 7th May

A perfect gas at $27^\circ\mathrm{C}$ is heated at constant pressure so as to double its volume. The final temperature of the gas will be, close to

(A) $327^\circ\mathrm{C}$ (B) $200^\circ\mathrm{C}$ (C) $54^\circ\mathrm{C}$ (D) $300^\circ\mathrm{C}$

Open solution

Q368. AIEEE 2012 (Online) 7th May

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: In an adiabatic process, change in internal energy of a gas is equal to work done on/by the gas in the process.
Statement 2: The temperature of a gas remains constant in an adiabatic process.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation of Statement 1.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is false, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

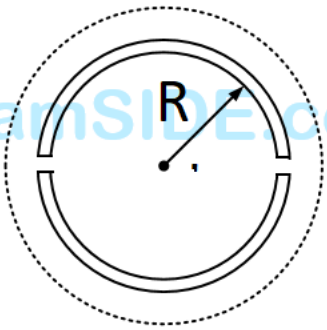
Open solution

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Q369. AIEEE 2012

Open solution

A wooden wheel of radius R is made of two semicircular part (see figure). The two parts are held together by a ring made of a metal strip of cross sectional area S and length L. L is slightly less than $2\pi R$. To fit the ring on the wheel, it is heated so that its temperature rises by ΔT and it just steps over the wheel. As it cools down to surrounding temperature, it process the semicircular parts together. If the coefficient of linear expansion of the metal is α, and its Young's modulus is Y, the force that one part of the wheel applies on the other part is :

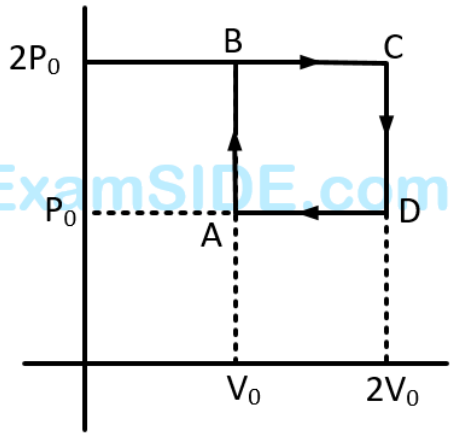


(A) $2\pi SY\alpha\Delta T$ (B) $SY\alpha\Delta T$ (C) $\pi SY\alpha\Delta T$ (D) $2SY\alpha\Delta T$

Q370. AIEEE 2012

Open solution

Helium gas goes through a cycle $ABCD$ (consisting of two isochoric and isobaric lines) as shown in figure efficiency of this cycle is nearly : (Assume the gas to be close to ideal gas)



(A) 15.4% (B) 9.1% (C) 10.5% (D) 12.5%

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Q371. AIEEE 2011

A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats γ. It is moving with speed v and it's suddenly brought to rest. Assuming no heat is lost to the surroundings, Its temperature increases by:

Open solution

(A) $\frac{(\gamma-1)}{2\gamma R}Mv^2K$

(B) $\frac{\gamma M^2v}{2R}K$

(C) $\frac{(\gamma-1)}{2R}Mv^2K$

(D) $\frac{(\gamma-1)}{2(\gamma+1)R}Mv^2K$

Q372. AIEEE 2011

Three perfect gases at absolute temperatures T_1, T_2 and T_3 are mixed. The masses of molecules are m_1, m_2 and m_3 and the number of molecules are n_1, n_2 and n_3 respectively. Assuming no loss of energy, the final temperature of the mixture is:

Open solution

(A) $\frac{n_1T_1+n_2T_2+n_3T_3}{n_1+n_2+n_3}$

(B) $\frac{n_1T_1^2+n_2T_2^2+n_3T_3^2}{n_1T_1+n_2T_2+n_3T_3}$

(C) $\frac{n_1^2T_1^2+n_2^2T_2^2+n_3^2T_3^2}{n_1T_1+n_2T_2+n_3T_3}$

(D) $\frac{(T_1+T_2+T_3)}{3}$

Q373. AIEEE 2011

$100g$ of water is heated from $30^\circ C$ to $50^\circ C$. Ignoring the slight expansion of the water, the change in its internal energy is (specific heat of water is $4184\ J/kg/K$):

Open solution

(A) $8.4\ kJ$ (B) $84\ kJ$ (C) $2.1\ kJ$ (D) $4.2\ kJ$

Q374. AIEEE 2009

One kg of a diatomic gas is at a pressure of $8\times10^4\ N/m^2$. The density of the gas is $4kg/m^3$. What is the energy of the gas due to its thermal motion ?

Open solution

(A) $5\times10^4\ J$ (B) $6\times10^4\ J$ (C) $7\times10^4\ J$ (D) $3\times10^4\ J$

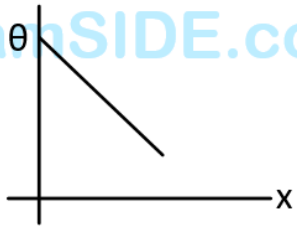
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Q375. AIEEE 2009

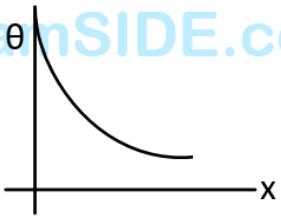
A long metallic bar is carrying heat from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figures?

Open solution

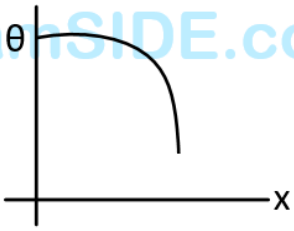
(A)



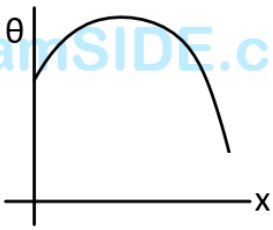
(B)



(C)



(D)



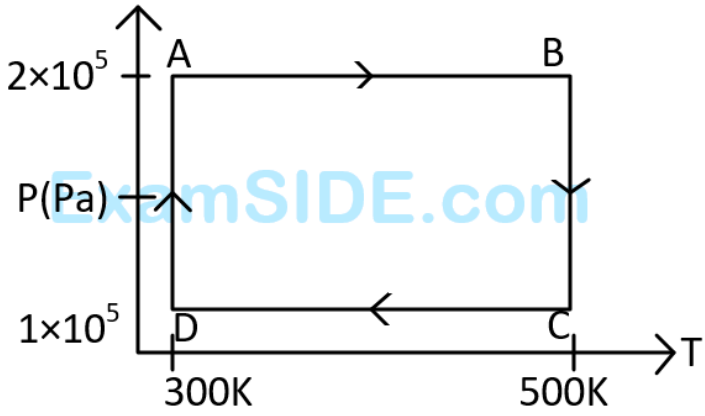
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Q376. AIEEE 2009

Two moles of helium gas are taken over the cycle $ABCD$, as shown in the P-T diagram.

Open solution



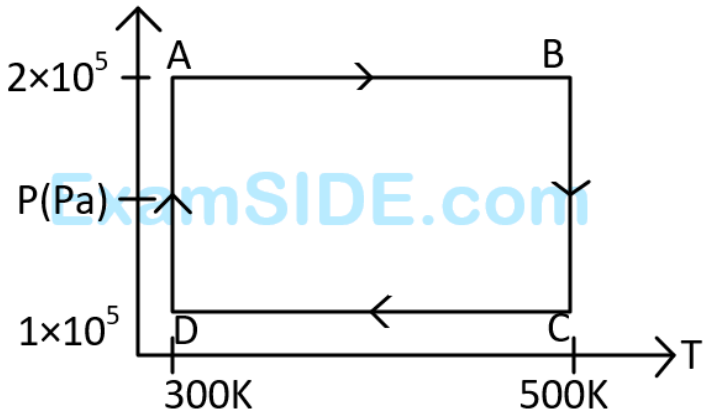
Assuming the gas to be ideal the work done on the gas in taking it from A to B is :

(A) $300\ R$ (B) $400\ R$ (C) $500\ R$ (D) $200\ R$

Q377. AIEEE 2009

Two moles of helium gas are taken over the cycle $ABCD$, as shown in the P-T diagram.

Open solution



The work done on the gas in taking it from D to A is :

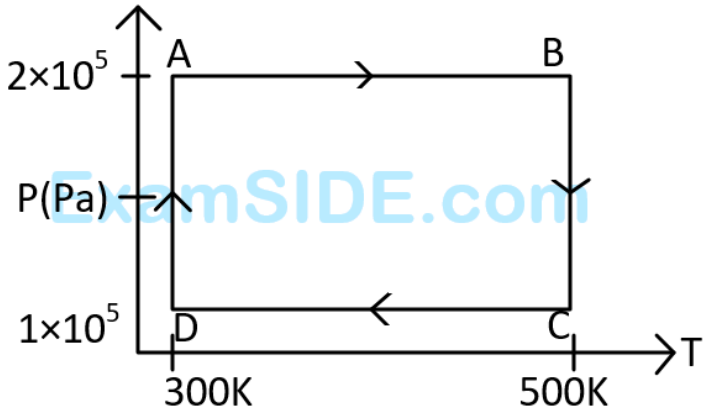
(A) $+414\ R$ (B) $-690\ R$ (C) $+690\ R$ (D) $-414\ R$

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Q378. AIEEE 2009

Two moles of helium gas are taken over the cycle $ABCD$, as shown in the P-T diagram.

Open solution



The net work done on the gas in the cycle $ABCDA$ is:

(A) $276\ R$ (B) $1076\ R$ (C) $1904\ R$ (D) zero

Q379. AIEEE 2008

The speed of sound in oxygen (O_2) at a certain temperature is $460\ ms^{-1}$. The speed of sound in helium (He) at the same temperature will be (assume both gases to be ideal)

(A) $1421\ ms^{-1}$ (B) $500\ ms^{-1}$ (C) $650\ ms^{-1}$ (D) $300\ ms^{-1}$

Open solution

Q380. AIEEE 2008

An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume V_1 and contains ideal gas at pressure P_1 and temperature T_1. The other chamber has volume V_2 and contains ideal gas at pressure P_2 and temperature T_2. If the partition is removed without doing any work on the gas, the final equilibrium temperature of the gas in the container will be

(A) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_2+P_2V_2T_1}$

(B) $\frac{P_1V_1T_1+P_2V_2T_2}{P_1V_1+P_2V_2}$

(C) $\frac{P_1V_1T_2+P_2V_2T_1}{P_1V_1+P_2V_2}$

(D) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_1+P_2V_2T_2}$

Open solution

Q381. AIEEE 2007

If C_p and C_v denote the specific heats of nitrogen per unit mass at constant pressure and constant volume respectively, then

(A) $C_p - C_v = 28R$ (B) $C_p - C_v = R/28$ (C) $C_p - C_v = R/14$ (D) $C_p - C_v = R$

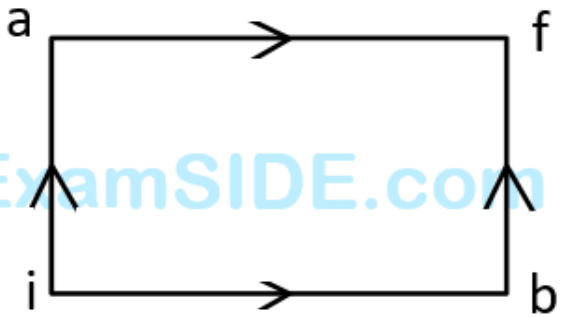
Open solution

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Q382. AIEEE 2007

When a system is taken from state i to state f along the path iaf, it is found that $Q = 50$ cal and $W = 20\ cal$. Along the path $ibf\ Q = 36\ cal$. W along the path ibf is

Open solution

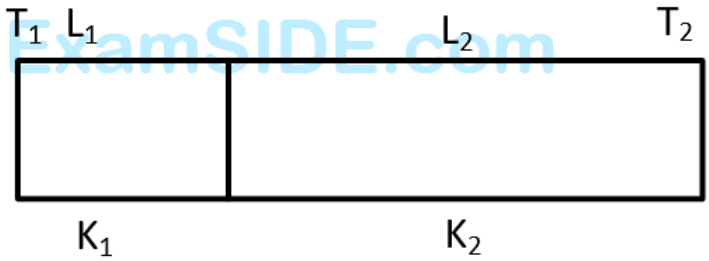


(A) $14\ cal$ (B) $6\ cal$ (C) $16\ cal$ (D) $66\ cal$

Q383. AIEEE 2007

One end of a thermally insulated rod is kept at a temperature T_1 and the other at T_2. The rod is composed of two sections of length L_1 and L_2 and thermal conductivities K_1 and K_2 respectively. The temperature at the interface of the two section is

Open solution



(A) $\frac{(K_1L_1T_1+K_2L_2T_2)}{(K_1L_1+K_2L_2)}$

(B) $\frac{(K_2L_2T_1+K_1L_1T_2)}{(K_1L_1+K_2L_2)}$

(C) $\frac{(K_2L_1T_1+K_1L_2T_2)}{(K_2L_1+K_1L_2)}$

(D) $\frac{(K_1L_2T_1+K_2L_1T_2)}{(K_1L_2+K_2L_1)}$

Q384. AIEEE 2006

Assuming the Sun to be a spherical body of radius R at a temperature of TK, evaluate the total radiant power incident on Earth at a distance r from the Sun
Where r_0 is the radius of the Earth and σ is Stefan's constant.

Open solution

(A) $4\pi r_0^2 R^2 \sigma \frac{T^4}{r^2}$ (B) $\pi r_0^2 R^2 \sigma \frac{T^4}{r^2}$

(C) $r_0^2 R^2 \sigma \frac{T^4}{4\pi r^2}$ (D) $R^2 \sigma \frac{T^4}{r^2}$

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Q385. AIEEE 2006

Two rigid boxes containing different ideal gases are placed on a table. Box A contains one mole of nitrogen at temperature T_0, while Box contains one mole of helium at temperature $\left(\frac{7}{3}\right)T_0$. The boxes are then put into thermal contact with each other, and heat flows between them until the gases reach a common final temperature (ignore the heat capacity of boxes). Then, the final temperature of the gases, T_f in terms of T_0 is

Open solution

(A) $T_f = \frac{3}{7}T_0$ (B) $T_f = \frac{7}{3}T_0$ (C) $T_f = \frac{3}{2}T_0$ (D) $T_f = \frac{5}{2}T_0$

Q386. AIEEE 2006

The work of $146\ kJ$ is performed in order to compress one kilo mole of gas adiabatically and in this process the temperature of the gas increases by $7^{\circ}C$. The gas is $\left(R = 8.3J\ mol^{-1}\ K^{-1}\right)$

Open solution

(A) diatomic (B) triatomic

(C) a mixture of monoatomic and diatomic (D) monoatomic

Q387. AIEEE 2005

A gaseous mixture consists of $16\ g$ of helium and $16\ g$ of oxygen. The ratio $\frac{C_p}{C_v}$ of the mixture is

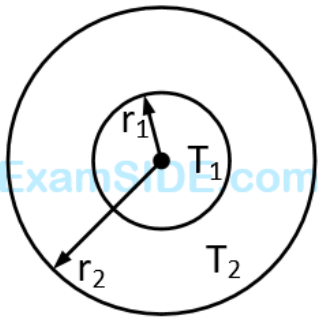
Open solution

(A) 1.62 (B) 1.59 (C) 1.54 (D) 1.4

Q388. AIEEE 2005

The figure shows a system of two concentric spheres of radii r_1 and r_2 are kept at temperatures T_1 and T_2, respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to

Open solution



(A) $In\left(\frac{r_2}{r_1}\right)$ (B) $\frac{(r_2 - r_1)}{(r_1 r_2)}$

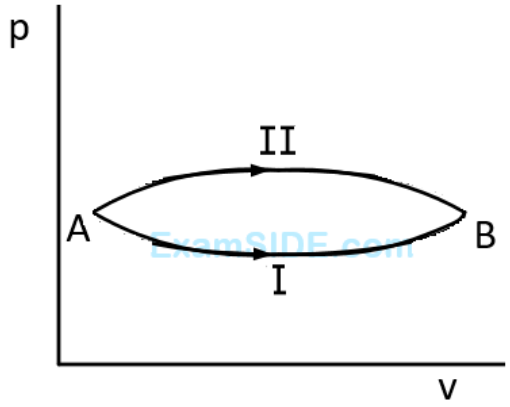
(C) $(r_2 - r_1)$ (D) $\frac{r_1 r_2}{(r_2 - r_1)}$

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Q389. AIEEE 2005

A system goes from A to B via two processes I and II as shown in figure. If ΔU_1 and ΔU_2 are the changes in internal energies in the processes I and II respectively, then

Open solution



(A) relation between ΔU_1 and ΔU_2 can not be determined

(B) $\Delta U_1 = \Delta U_2$

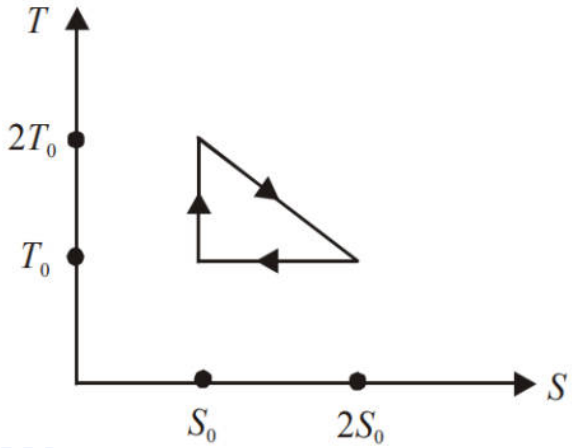
(C) $\Delta U_2 < \Delta U_1$

(D) $\Delta U_2 > \Delta U_1$

Q390. AIEEE 2005

The temperature-entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is

Open solution



(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Q391. AIEEE 2004

One mole of ideal monatomic gas $(\gamma = 5/3)$ is mixed with one mole of diatomic gas $(\gamma = 7/5)$. What is γ for the mixture? γ Denotes the ratio of specific heat at constant pressure, to that at constant volume

Open solution

(A) $35/23$ (B) $23/15$ (C) $3/2$ (D) $4/3$

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Q392. AIEEE 2004

If the temperature of the sun were to increase from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on earth to what it was previously will be

(A) 32 (B) 16 (C) 4 (D) 64

Open solution

Q393. AIEEE 2004

Which of the following statements is **correct** for any thermodynamic system ?

(A) The change in entropy can never be zero

(B) Internal energy and entropy are state functions

(C) The internal energy changes in all processes

(D) The work done in an adiabatic process is always zero

Open solution

Q394. AIEEE 2004

Two thermally insulated vessels 1 and 2 are filled with air at temperatures (T_1, T_2), volume (V_1, V_2) and pressure (P_1, P_2) respectively. If the valve joining the two vessels is opened, the temperature inside the vessel at equilibrium will be

(A) $T_1T_2\,(P_1V_1 + P_2V_2)/\,(P_1V_1T_2 + P_2V_2T_1)$

(B) $(T_1 + T_2)/2$

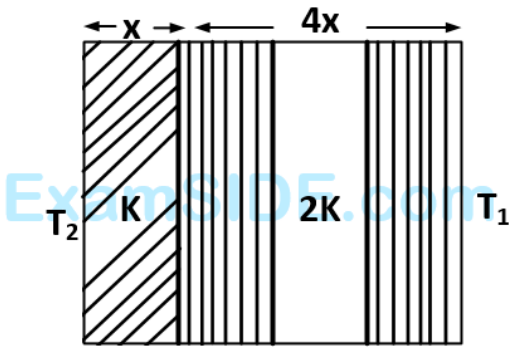
(C) $T_1 + T_2$

(D) $T_1T_2\,(P_1V_1 + P_2V_2)/\,(P_1V_1T_1 + P_2V_2T_2)$

Open solution

Q395. AIEEE 2004

The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$, respectively, are T_2 and $T_1\,(T_2 > T_1)$. The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2-T_1)K}{x}\right)f$, with f equal to



(A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1}{3}$

Open solution

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Q396. AIEEE 2004

Time taken by a $836\ W$ heater to heat one litre of water from $10^{\circ}C$ to $40^{\circ}C$ is

(A) $150\ s$ (B) $100\ s$ (C) $50\ s$ (D) $200\ s$

Open solution

Q397. AIEEE 2003

"Heat cannot by itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of :

(A) second law of thermodynamics (B) conservation of momentum

(C) conservation of mass (D) first law of thermodynamics

Open solution

Q398. AIEEE 2003

During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio C_p/C_V for the gas is

(A) $\frac{4}{3}$ (B) 2 (C) $\frac{5}{3}$ (D) $\frac{3}{2}$

Open solution

Q399. AIEEE 2003

Which of the following parameters does not characterize the thermodynamic state of mattter?

(A) Temperature (B) Pressure (C) Work (D) Volume

Open solution

Q400. AIEEE 2003

The earth radiates in the infra-red region of the spectrum. The spectrum is correctly given by

(A) Rayleigh Jeans law (B) Planck's law of radiation

(C) Stefan's law of radiation (D) Wien's law

Open solution

Q401. AIEEE 2002

Heat given to a body which raises its temperature by $1^{\circ}C$ is

(A) water equivalent (B) thermal capacity

(C) specific heat (D) temperature gradient

Open solution

Q402. AIEEE 2002

Infrared radiation is detected by

(A) spectrometer (B) pyrometer (C) nanometer (D) photometer

Open solution

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Q403. AIEEE 2002

Which of the following is more close to a black body?

(A) black board paint
(B) green leaves
(C) black holes
(D) red roses

Open solution

Q404. AIEEE 2002

Cooking gas containers are kept in a lorry moving with uniform speed. The temperature of the gas molecules inside will

(A) increase
(B) decrease
(C) remain same
(D) decrease for some, while increase for others

Open solution

Q405. AIEEE 2002

At what temperature is the $r.m.s$ velocity of a hydrogen molecule equal to that of an oxygen molecule at $47^{\circ}C$?

(A) $80K$ (B) $-73\ K$ (C) $3\ K$ (D) $20\ K$

Open solution

Q406. AIEEE 2002

1 mole of a gas with $\gamma = 7/5$ is mixed with 1 mole of a gas with $\gamma = 5/3$, then the value of γ for the resulting mixture is

(A) $7/5$ (B) $2/5$ (C) $3/2$ (D) $12/7$

Open solution

Q407. AIEEE 2002

Two spheres of the same material have radii $1\ m$ and $4\ m$ and temperatures $4000\ K$ and $2000\ K$ respectively. The ratio of the energy radiated per second by the first sphere to that by the second is

(A) $1:1$ (B) $16:1$ (C) $4:1$ (D) $1:9$

Open solution

Numerical 59 questions

Q408. JEE Main 2026 (Online) 5th April Evening Shift

The heat extracted out of x gram of water initially at $50^{\circ}\mathrm{C}$ to cool it down to $0^{\circ}\mathrm{C}$ is sufficient to evaporate $(1000 - x)$ gram of water also initially at $50^{\circ}\mathrm{C}$. The value of x (closest integer) is ____.

(Take latent heat of water $2256\ \mathrm{kJ/kg}$, specific heat capacity of water $4200\ \mathrm{J/kg.K}$)

Open solution

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Q409. JEE Main 2026 (Online) 2nd April Evening Shift

5 moles of unknown gas is heated at constant volume from 10°C to 20°C. The molar specific heat of this gas at constant pressure $c_p = 8$ cal/mol.°C and $R = 8.36$ J/mol.°C. The change in the internal energy of the gas is _____ calorie.

Open solution

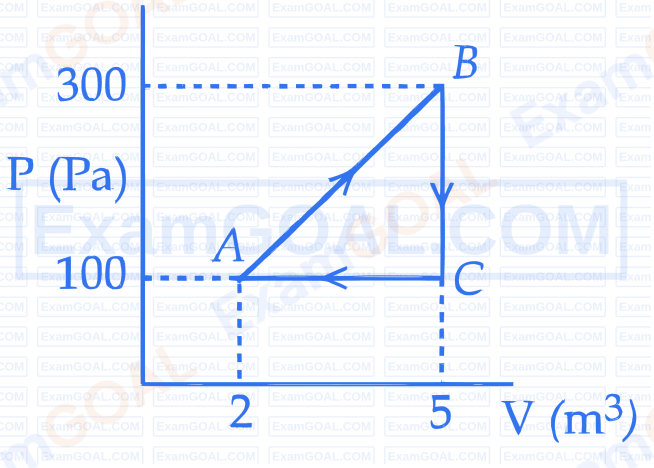
Q410. JEE Main 2026 (Online) 2nd April Morning Shift

A vessel contains 0.15 m^3 of a gas at pressure 8 bar and temperature 140 °C with $c_p = 3R$ and $c_v = 2R$. It is expanded adiabatically till pressure falls to 1 bar. The work done during this process is _____ kJ. (R is gas constant)

Open solution

Q411. JEE Main 2026 (Online) 28th January Evening Shift

A thermodynamic system is taken through the cyclic process *ABC* as shown in the figure. The total work done by the system during the cycle *ABC* is _____ J.



Open solution

Q412. JEE Main 2026 (Online) 24th January Evening Shift

When 300 J of heat given to an ideal gas with $C_p = \frac{7}{2}R$ its temperature raises from $20^\circ C$ to $50^\circ C$ keeping its volume constant. The mass of the gas is (approximately) ____ g. $(R = 8.314\ J/mol.\ K)$

Open solution

Q413. JEE Main 2026 (Online) 24th January Morning Shift

A gas of certain mass filled in a closed cylinder at a pressure of 3.23 kPa has temperature $50^\circ C$. The gas is now heated to double its temperature. The modified pressure is ____ Pa .

Open solution

Q414. JEE Main 2026 (Online) 22nd January Evening Shift

An insulated cylinder of volume $60\ cm^3$ is filled with a gas at $27^\circ C$ and 2 atmospheric pressure. Then the gas is compressed making the final volume as $20\ cm^3$ while allowing the temperature to rise to $77^\circ C$. The final pressure is ____ atmospheric pressure.

Open solution

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Q415. JEE Main 2026 (Online) 21st January Evening Shift

A diatomic gas $(\gamma = 1.4)$ does 100 J of work when it is expanded isobarically. Then the heat given to the gas ____ J.

Open solution

Q416. JEE Main 2026 (Online) 21st January Morning Shift

10 mole of oxygen is heated at constant volume from $30^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$. The change in the internal energy of the gas is ____ cal. (The molecular specific heat of oxygen at constant pressure, $C_P = 7\mathrm{cal/mol.}\,^{\circ}\mathrm{C}$ and $\mathrm{R} = 2\mathrm{cal.}\,/\mathrm{mol.}\,^{\circ}\mathrm{C}.$)

Open solution

Q417. JEE Main 2025 (Online) 7th April Evening Shift

Two cylindrical rods A and B made of different materials, are joined in a straight line. The ratios of lengths, radii and thermal conductivities of these rods are: $\frac{\mathrm{L_A}}{\mathrm{L_B}} = \frac{1}{2}, \frac{\mathrm{r_A}}{\mathrm{r_B}} = 2$ and $\frac{\mathrm{K_A}}{\mathrm{K_B}} = \frac{1}{2}$. The free ends of rods A and B are maintained at 400 K , 200 K , respectively. The temperature of rods interface is _____ K, when equilibrium is established.

Open solution

Q418. JEE Main 2025 (Online) 7th April Morning Shift

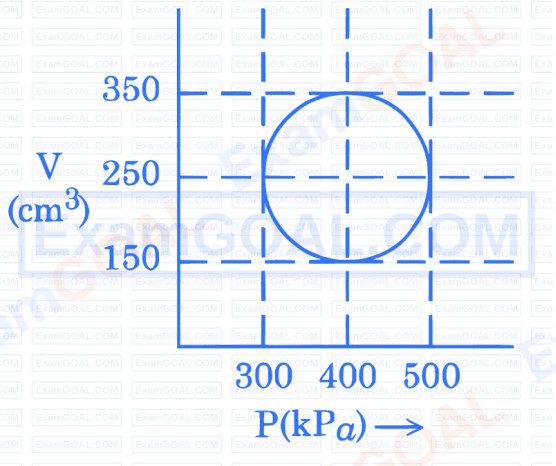
A wire of length 10 cm and diameter 0.5 mm is used in a bulb. The temperature of the wire is $1727^{\circ}\mathrm{C}$ and power radiated by the wire is 94.2 W . Its emissivity is $\frac{x}{8}$ where $x =$ _____.

(Given $\sigma = 6.0 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}, \pi = 3.14$ and assume that the emissivity of wire material is same at all wavelength.)

Open solution

Q419. JEE Main 2025 (Online) 7th April Morning Shift

An ideal gas has undergone through the cyclic process as shown in the figure. Work done by the gas in the entire cycle is _____ $\times 10^{-1}$ J. (Take $\pi = 3.14$)



Open solution

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Q420. JEE Main 2025 (Online) 2nd April Evening Shift

The internal energy of air in $4\,\mathrm{m} \times 4\,\mathrm{m} \times 3\,\mathrm{m}$ sized room at 1 atmospheric pressure will be_____$\times 10^6$ J

(Consider air as diatomic molecule)

Open solution

Q421. JEE Main 2025 (Online) 2nd April Morning Shift

γ_A is the specific heat ratio of monoatomic gas A having 3 translational degrees of freedom. γ_B is the specific heat ratio of polyatomic gas B having 3 translational, 3 rotational degrees of freedom and 1 vibrational mode. If $\frac{\gamma_A}{\gamma_B} = \left(1 + \frac{1}{n}\right)$, then the value of n is _____ .

Open solution

Q422. JEE Main 2025 (Online) 29th January Morning Shift

A container of fixed volume contains a gas at 27°C. To double the pressure of the gas, the temperature of gas should be raised to _____ °C.

Open solution

Q423. JEE Main 2025 (Online) 24th January Morning Shift

The temperature of 1 mole of an ideal monoatomic gas is increased by $50^\circ\mathrm{C}$ at constant pressure. The total heat added and change in internal energy are E_1 and E_2, respectively. If $\frac{E_1}{E_2} = \frac{x}{9}$ then the value of x is _____.

Open solution

Q424. JEE Main 2025 (Online) 23rd January Morning Shift

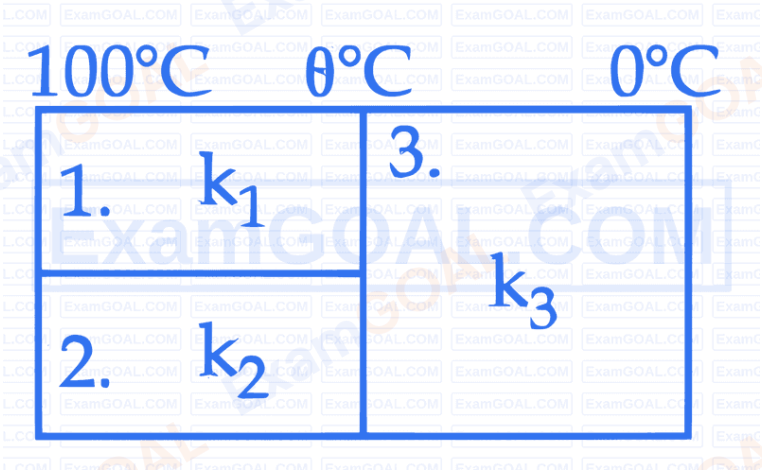
An ideal gas initially at $0^\circ\mathrm{C}$ temperature, is compressed suddenly to one fourth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is $3/2$, the change in temperature due to the thermodynamic process is _____ K.

Open solution

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Q425. JEE Main 2025 (Online) 22nd January Morning Shift

Three conductors of same length having thermal conductivity k_1, k_2 and k_3 are connected as shown in figure.



Area of cross sections of 1st and 2nd conductor are same and for 3rd conductor it is double of the 1st conductor. The temperatures are given in the figure. In steady state condition, the value of θ is _____ °C. (Given : $k_1 = 60 Js^{-1} m^{-1} K^{-1}, k_2 = 120 Js^{-1} m^{-1} K^{-1}, k_3 = 135 Js^{-1} m^{-1} K^{-1}$)

Open solution

Q426. JEE Main 2023 (Online) 13th April Evening Shift

Two plates A and B have thermal conductivities $84\ Wm^{-1} K^{-1}$ and $126\ Wm^{-1} K^{-1}$ respectively. They have same surface area and same thickness. They are placed in contact along their surfaces. If the temperatures of the outer surfaces of A and B are kept at $100°C$ and $0°C$ respectively, then the temperature of the surface of contact in steady state is _____ °C.

Open solution

Q427. JEE Main 2023 (Online) 8th April Evening Shift

A steel rod of length $1\ m$ and cross sectional area $10^{-4}\ m^2$ is heated from $0°C$ to $200°C$ without being allowed to extend or bend. The compressive tension produced in the rod is _____ $\times 10^4$ N. (Given Young's modulus of steel $= 2 \times 10^{11} Nm^{-2}$, coefficient of linear expansion $= 10^{-5} K^{-1}$)

Open solution

Q428. JEE Main 2023 (Online) 31st January Evening Shift

A water heater of power $2000\ W$ is used to heat water. The specific heat capacity of water is $4200\ J\ kg^{-1} K^{-1}$. The efficiency of heater is 70%. Time required to heat $2\ kg$ of water from $10°C$ to $60°C$ is _____ s.

(Assume that the specific heat capacity of water remains constant over the temperature range of the water).

Open solution

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Q429. JEE Main 2023 (Online) 30th January Evening Shift

A faulty thermometer reads $5^{\circ}\mathrm{C}$ in melting ice and $95^{\circ}\mathrm{C}$ in steam. The correct temperature on absolute scale will be _____ K when the faulty thermometer reads $41^{\circ}\mathrm{C}$.

Open solution

Q430. JEE Main 2023 (Online) 24th January Morning Shift

A hole is drilled in a metal sheet. At $27^{\circ}\mathrm{C}$, the diameter of hole is 5 cm. When the sheet is heated to $177^{\circ}\mathrm{C}$, the change in the diameter of hole is $\mathrm{d} \times 10^{-3}$ cm. The value of d will be _____ if coefficient of linear expansion of the metal is $1.6 \times 10^{-5}/^{\circ}\mathrm{C}$.

Open solution

Q431. JEE Main 2022 (Online) 29th July Morning Shift

The pressure P_1 and density d_1 of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes suddenly to $\mathrm{P}_2\,(> \mathrm{P}_1)$ and d_2 respectively during an adiabatic process. The temperature of the gas increases and becomes _____ times of its initial temperature. (given $\frac{\mathrm{d}_2}{\mathrm{d}_1} = 32$)

Open solution

Q432. JEE Main 2022 (Online) 29th July Morning Shift

One mole of a monoatomic gas is mixed with three moles of a diatomic gas. The molecular specific heat of mixture at constant volume is $\frac{\alpha^2}{4}\mathrm{R\,J/mol\,K}$; then the value of α will be _____. (Assume that the given diatomic gas has no vibrational mode).

Open solution

Q433. JEE Main 2022 (Online) 28th July Evening Shift

At a certain temperature, the degrees of freedom per molecule for gas is 8. The gas performs 150 J of work when it expands under constant pressure. The amount of heat absorbed by the gas will be _____ J.

Open solution

Q434. JEE Main 2022 (Online) 25th July Evening Shift

A block of ice of mass 120 g at temperature 0°C is put in 300 g of water at 25°C. The x g of ice melts as the temperature of the water reaches 0°C. The value of x is _____.

[Use specific heat capacity of water = 4200 $\mathrm{Jkg^{-1}K^{-1}}$, Latent heat of ice = 3.5×10^5 $\mathrm{Jkg^{-1}}$]

Open solution

Q435. JEE Main 2022 (Online) 25th July Morning Shift

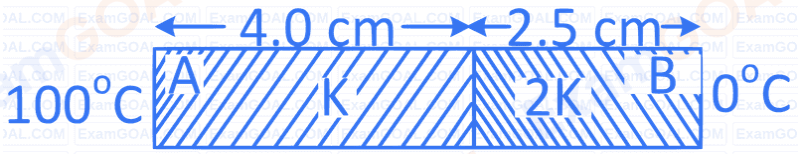
A unit scale is to be prepared whose length does not change with temperature and remains $20\ \mathrm{cm}$, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their lengths remains constant. If length of brass is $40\ \mathrm{cm}$, then length of iron will be _____ cm. $\left(\alpha_{\text{iron}} = 1.2 \times 10^{-5}\,\mathrm{K^{-1}}\right.$ and $\left.\alpha_{\text{brass}} = 1.8 \times 10^{-5}\,\mathrm{K^{-1}}\right)$.

Open solution

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Q436. JEE Main 2022 (Online) 29th June Morning Shift

As per the given figure, two plates A and B of thermal conductivity K and 2 K are joined together to form a compound plate. The thickness of plates are 4.0 cm and 2.5 cm respectively and the area of cross-section is 120 cm^2 for each plate. The equivalent thermal conductivity of the compound plate is $\left(1+\frac{5}{\alpha}\right)$ K, then the value of α will be _____.



Open solution

Q437. JEE Main 2022 (Online) 28th June Morning Shift

The total internal energy of two mole monoatomic ideal gas at temperature T = 300 K will be _____ J. (Given R = 8.31 J/mol.K)

Open solution

Q438. JEE Main 2022 (Online) 27th June Evening Shift

A diatomic gas (γ = 1.4) does 400J of work when it is expanded isobarically. The heat given to the gas in the process is _____ J.

Open solution

Q439. JEE Main 2022 (Online) 26th June Evening Shift

A geyser heats water flowing at a rate of 2.0 kg per minute from 30°C to 70°C. If geyser operates on a gas burner, the rate of combustion of fuel will be _____ g min^{-1}.

[Heat of combustion = 8×10^3 Jg^{-1}, Specific heat of water = 4.2 Jg^{-1} $°C^{-1}$]

Open solution

Q440. JEE Main 2022 (Online) 25th June Evening Shift

When a gas filled in a closed vessel is heated by raising the temperature by 1°C, its pressure increases by 0.4%. The initial temperature of the gas is _____ K.

Open solution

Q441. JEE Main 2022 (Online) 24th June Evening Shift

A monoatomic gas performs a work of $\frac{Q}{4}$ where Q is the heat supplied to it. The molar heat capacity of the gas will be _____ R during this transformation. Where R is the gas constant.

Open solution

Q442. JEE Main 2022 (Online) 24th June Morning Shift

0.056 kg of Nitrogen is enclosed in a vessel at a temperature of 127°C. The amount of heat required to double the speed of its molecules is _____ k cal.

(Take R = 2 cal $mole^{-1}$ K^{-1})

Open solution

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Q443. JEE Main 2021 (Online) 1st September Evening Shift

The temperature of 3.00 mol of an ideal diatomic gas is increased by 40.0°C without changing the pressure of the gas. The molecules in the gas rotate but do not oscillate. If the ratio of change in internal energy of the gas to the amount of workdone by the gas is $\frac{x}{10}$. Then the value of x (round off to the nearest integer) is _____. (Given R = 8.31 J mol^{-1} K^{-1})

Open solution

Q444. JEE Main 2021 (Online) 1st September Evening Shift

The average translational kinetic energy of N_2 gas molecules at .....°C becomes equal to the K.E. of an electron accelerated from rest through a potential difference of 0.1 volt. (Given k_B = 1.38 $\times$ 10^{-23} J/K) (Fill the nearest integer).

Open solution

Q445. JEE Main 2021 (Online) 31st August Evening Shift

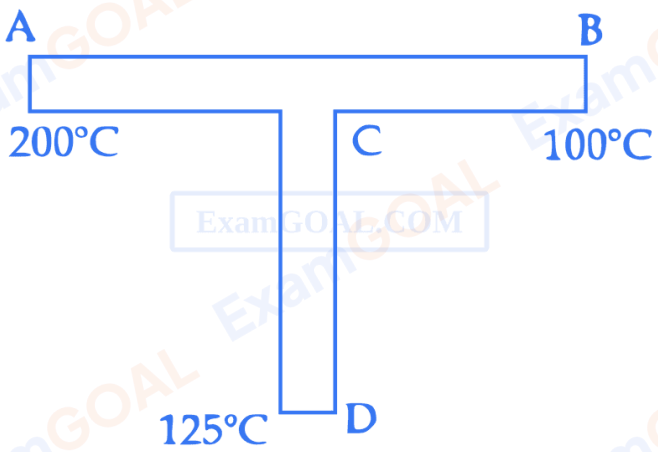
A sample of gas with γ = 1.5 is taken through an adiabatic process in which the volume is compressed from 1200 cm^3 to 300 cm^3. If the initial pressure is 200 kPa. The absolute value of the workdone by the gas in the process = _____ J.

Open solution

Q446. JEE Main 2021 (Online) 27th August Morning Shift

A rod CD of thermal resistance 10.0 KW^{-1} is joined at the middle of an identical rod AB as shown in figure. The end A, B and D are maintained at 200°C, 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is ..... .

Open solution



Q447. JEE Main 2021 (Online) 25th July Evening Shift

A system consists of two types of gas molecules A and B having same number density 2 $\times$ $10^{25}/m^3$. The diameter of A and B are 10 $\mathring{A}$ and 5 $\mathring{A}$ respectively. They suffer collision at room temperature. The ratio of average distance covered by the molecule A to that of B between two successive collision is _____ $\times$ 10^{-2}

Open solution

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Q448. JEE Main 2021 (Online) 22th July Evening Shift

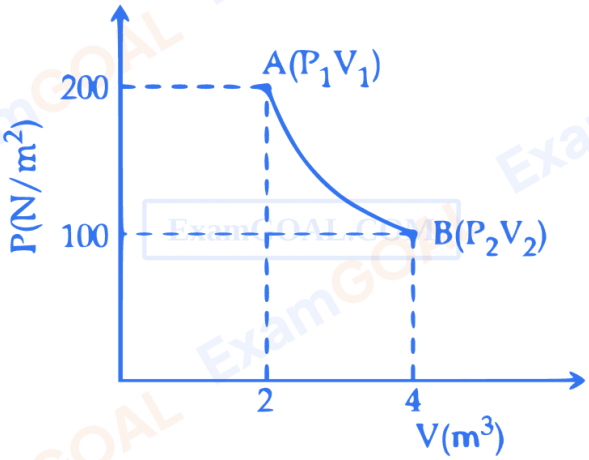
The area of cross-section of a railway track is 0.01 m^2. The temperature variation is 10°C. Coefficient of linear expansion of material of track is 10^{-5}/°C. The energy stored per meter in the track is _____ J/m.

(Young's modulus of material of track is 10^{11} Nm^{-2})

Open solution

Q449. JEE Main 2021 (Online) 20th July Evening Shift

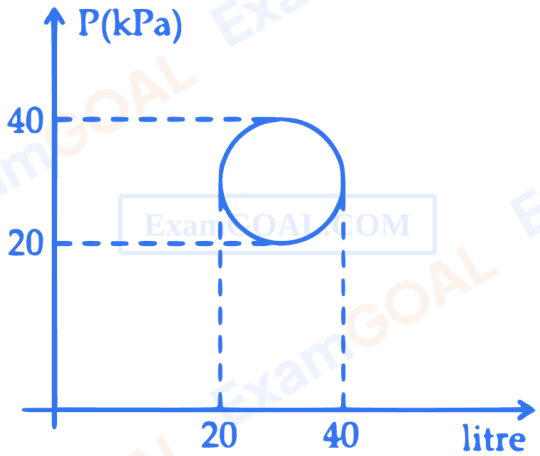
One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be _____ $\times 10^{-1}$ J. [Given : R = 8.3 J/mole K, ln2 = 0.6931] (Round off to the nearest integer)



Open solution

Q450. JEE Main 2021 (Online) 20th July Morning Shift

In the reported figure, heat energy absorbed by a system in going through a cyclic process is _____ πJ.



Open solution

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Q451. JEE Main 2021 (Online) 26th February Evening Shift

1 mole of rigid diatomic gas performs a work of $\frac{Q}{5}$ when heat Q is supplied to it. The molar heat capacity of the gas during this transformation is $\frac{xR}{8}$. The value of x is _____. [R = universal gas constant]

Open solution

Q452. JEE Main 2021 (Online) 26th February Evening Shift

The volume V of a given mass of monoatomic gas changes with temperature T according to the relation $V = KT^{\frac{2}{3}}$. The workdone when temperature changes by 90K will be xR. The value of x is _____. [R = universal gas constant]

Open solution

Q453. JEE Main 2021 (Online) 26th February Morning Shift

A container is divided into two chambers by a partition. The volume of first chamber is 4.5 litre and second chamber is 5.5 litre. The first chamber contain 3.0 moles of gas at pressure 2.0 atm and second chamber contain 4.0 moles of gas at pressure 3.0 atm. After the partition is removed and the mixture attains equilibrium, then, the common equilibrium pressure existing in the mixture is $x \times 10^{-1}$ atm. Value of x is _____.

Open solution

Q454. JEE Main 2021 (Online) 25th February Morning Shift

A monoatomic gas of mass 4.0 u is kept in an insulated container. Container is moving with velocity 30 m/s. If container is suddenly stopped then change in temperature of the gas (R = gas constant) is $\frac{x}{3R}$. Value of x is _____.

Open solution

Q455. JEE Main 2021 (Online) 25th February Morning Shift

In a certain thermodynamical process, the pressure of a gas depends on its volume as kV^3. The work done when the temperature changes from 100°C to 300°C will be _____ nR, where n denotes number of moles of a gas.

Open solution

Q456. JEE Main 2021 (Online) 24th February Evening Shift

The root mean square speed of molecules of a given mass of a gas at 27°C and 1 atmosphere pressure is 200 ms^{-1}. The root mean square speed of molecules of the gas at 127°C and 2 atmosphere pressure is $\frac{x}{\sqrt{3}}$ ms^{-1}. The value of x will be _____.

Open solution

Q457. JEE Main 2020 (Online) 6th September Morning Slot

Initially a gas of diatomic molecules is contained in a cylinder of volume V_1 at a pressure P_1 and temperature 250 K. Assuming that 25% of the molecules get dissociated causing a change in number of moles. The pressure of the resulting gas at temperature 2000 K, when contained in a volume $2V_1$ is given by P_2 . The ratio $\frac{P_2}{P_1}$ is _____.

Open solution

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Q458. JEE Main 2020 (Online) 5th September Evening Slot

Nitrogen gas is at 300°C temperature. The temperature (in K) at which the rms speed of a H_2 molecule would be equal to the rms speed of a nitrogen molecule, is _____.
(Molar mass of N_2 gas 28 g).

Open solution

Q459. JEE Main 2020 (Online) 4th September Evening Slot

The change in the magnitude of the volume of an ideal gas when a small additional pressure ΔP is applied at a constant temperature, is the same as the change when the temperature is reduced by a small quantity ΔT at constant pressure. The initial temperature and pressure of the gas were 300 K and 2 atm. respectively.
If $|\Delta T| = C|\Delta P|$ then value of C in (K/atm.) is _____.

Open solution

Q460. JEE Main 2020 (Online) 4th September Morning Slot

A closed vessel contains 0.1 mole of a monoatomic ideal gas at 200 K. If 0.05 mole of the same gas at 400 K is added to it, the final equilibrium temperature (in K) of the gas in the vessel will be close to _____.

Open solution

Q461. JEE Main 2020 (Online) 3rd September Morning Slot

A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with Vm volume (at 30°C) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(beaker)} = 6 \times 10^{-6}\ ^oC^{-1}$ and $\gamma_{(mercury)} = 1.5 \times 10^{-4}\ ^oC^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to ____.

Open solution

Q462. JEE Main 2020 (Online) 2nd September Morning Slot

An engine takes in 5 moles of air at 20°C and 1 atm, and compresses it adiabatically to $1/10^{th}$ of the original volume. Assuming air to be a diatomic ideal gas made up of rigid molecules, the change in its internal energy during this process comes out to be X kJ. The value of X to the nearest integer is_____.

Open solution

Q463. JEE Main 2020 (Online) 9th January Evening Slot

Starting at temperature 300 K, one mole of an ideal diatomic gas ($\gamma = 1.4$) is first compressed adiabatically from volume V_1 to $V_2 = \frac{V_1}{16}$. It is then allowed to expand isobarically to volume $2V_2$. If all the processes are the quasi-static then the final temperature of the gas (in °K) is (to the nearest integer) _____.

Open solution

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Q464. JEE Main 2020 (Online) 8th January Evening Slot

Three containers C_1, C_2 and C_3 have water at different temperatures. The table below shows the final temperature T when different amounts of water (given in litres) are taken from each containers and mixed (assume no loss of heat during the process)

Open solution

C_1	C_2	C_3	T
$1l$	$2l$	–	60°C
–	$1l$	$2l$	30°C
$2l$	–	$1l$	60°C
$1l$	$1l$	$1l$	θ

The value of θ (in °C to the nearest integer) is .....

Q465. JEE Main 2020 (Online) 7th January Evening Slot

M grams of steam at 100°C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produced liquid water at 40°C [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g] the value of M is ____

Open solution

Q466. JEE Main 2020 (Online) 7th January Morning Slot

A non-isotropic solid metal cube has coefficients of linear expansion as :
$5 \times 10^{-5}/^{o}$C along the x-axis and $5 \times 10^{-6}/^{o}$C along the y and the z-axis. If the coefficient of volume expansion of the solid is C $\times$ $10^{-6}/^{o}$C then the value of C is

Open solution

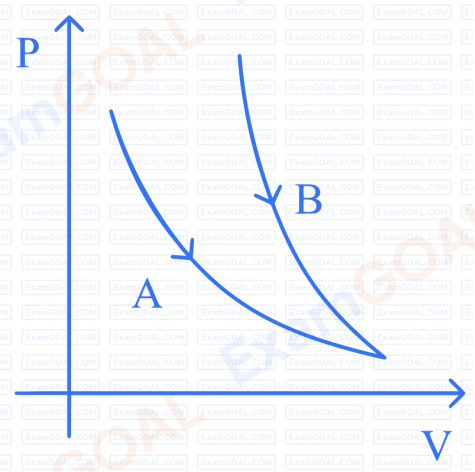
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MCQ (More than One Correct Answer) 2 questions

Q467. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Choose the correct statement for processes A & B shown in figure.



(A) $PV = k$ for process B and A.

(B) $\frac{P^{\gamma-1}}{T^{\gamma}} = k$ for process B and $T = k$ for process A.

(C) $\frac{T^{\gamma}}{P^{\gamma-1}} = k$ for process A and $PV = k$ for process B.

(D) $PV^{\gamma} = k$ for process B and $PV = k$ for process A.

Q468. AIEEE 2005

Open solution

Which of the following is **incorrect** regarding the first law of thermodynamics?

(A) It is a restatement of the principle of conservation of energy

(B) It is not applicable to any cyclic process

(C) It introduces the concept of the entropy

(D) It introduces the concept of the internal energy

Out of syllabus 47 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q469. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Open solution

An engine operating between the boiling and freezing points of water will have

A. efficiency more than 27%.

B. efficiency less than the efficiency of a Carnot engine operating between the same two temperatures.

C. efficiency equal to 27%

D. efficiency less than 27%

Choose the correct answer from the options given below:

(A) B and C only (B) B and D only (C) B, C and D only (D) A and B only

Q470. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Open solution

Work done by a Carnot engine operating between temperatures $127^{\circ}\mathrm{C}$ and $27^{\circ}\mathrm{C}$ is $2\,\mathrm{kJ}$. The amount of heat transferred to the engine by the reservoir is :

(A) 8 kJ (B) 2 kJ (C) 4 kJ (D) 2.67 kJ

Q471. JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

Open solution

A Carnot engine operating between two reservoirs has efficiency $\frac{1}{3}$. When the temperature of cold reservoir is raised by x, its efficiency decreases to $\frac{1}{6}$. The value of x, if the temperature of hot reservoir is 99°C, will be :

(A) 66 K (B) 62 K (C) 16.5 K (D) 33 K

Q472. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A: Efficiency of a reversible heat engine will be highest at $-273^{\circ}\mathrm{C}$ temperature of cold reservoir.

Reason R: The efficiency of Carnot's engine depends not only on the temperature of the cold reservoir but it depends on the temperature of the hot reservoir too and is given as $\eta = \left(1 - \frac{T_2}{T_1}\right)$

In the light of the above statements, choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is false but $\mathbf{R}$ is true

(D) A is true but $\mathbf{R}$ is false

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Q473. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

A Carnot engine with efficiency 50% takes heat from a source at 600 K. In order to increase the efficiency to 70%, keeping the temperature of sink same, the new temperature of the source will be :

(A) 300 K (B) 1000 K (C) 900 K (D) 360 K

Open solution

Q474. JEE Main 2022 (Online) 30th July Evening Shift · MCQ (Single Correct Answer)

A Carnot engine is operating between the two reservoirs of temperature $227\ ^{\circ}\mathrm{C}$ and $127\ ^{\circ}\mathrm{C}$ respectively. If the engine performs $1.04 \times 10^5\ \mathrm{J}$ work per cycle. Then, the heat absorbed from hot reservoir per cycle will be :

(A) $2.08 \times 10^4\ \mathrm{J}$ (B) $1.04 \times 10^5\ \mathrm{J}$ (C) $5.20 \times 10^5\ \mathrm{J}$ (D) $2.36 \times 10^5\ \mathrm{J}$

Open solution

Q475. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

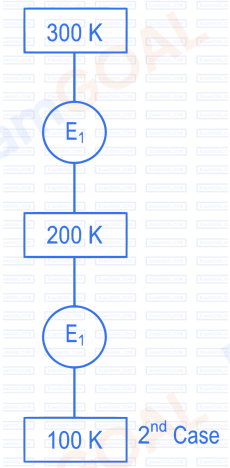
A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by $40^{\circ}\mathrm{C}$, its efficiency increases by 30%. The temperature of the source will be:

(A) 166.7 K (B) 255.1 K (C) 266.7 K (D) 367.7 K

Open solution

Q476. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

In 1st case, Carnot engine operates between temperatures $300\ \mathrm{K}$ and $100\ \mathrm{K}$. In 2nd case, as shown in the figure, a combination of two engines is used. The efficiency of this combination (in 2nd case) will be :



(A) same as the 1st case.

(B) always greater than the 1st case.

(C) always less than the 1st case.

(D) may increase or decrease with respect to the 1st case.

Open solution

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Q477. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Let η_1 is the efficiency of an engine at $T_1 = 447^\circ C$ and $T_2 = 147^\circ C$ while η_2 is the efficiency at $T_1 = 947^\circ C$ and $T_2 = 47^\circ C$. The ratio $\frac{\eta_1}{\eta_2}$ will be :

(A) 0.41 (B) 0.56 (C) 0.73 (D) 0.70

Open solution

Q478. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

The efficiency of a Carnot's engine, working between steam point and ice point, will be :

(A) 26.81% (B) 37.81% (C) 47.81% (D) 57.81%

Open solution

Q479. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

A Carnot engine takes 5000 kcal of heat from a reservoir at 727°C and gives heat to a sink at 127°C. The work done by the engine is

(A) 3×10^6 J (B) Zero (C) 12.6×10^6 J (D) 8.4×10^6 J

Open solution

Q480. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

A Carnot engine whose heat sink is at 27°C, has an efficiency of 25%. By how many degrees should the temperature of the source be changed to increase the efficiency by 100% of the original efficiency?

(A) Increases by 18°C (B) Increases by 200°C
(C) Increases by 120°C (D) Increases by 73°C

Open solution

Q481. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

A reversible engine has an efficiency of $\frac{1}{4}$. If the temperature of the sink is reduced by 58°C, its efficiency becomes double. Calculate the temperature of the sink :

(A) 174°C (B) 280°C (C) 180.4°C (D) 382°C

Open solution

Q482. JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

A refrigerator consumes an average 35W power to operate between temperature $-10^\circ C$ to $25^\circ C$. If there is no loss of energy then how much average heat per second does it transfer?

(A) 263 J/s (B) 298 J/s (C) 350 J/s (D) 35 J/s

Open solution

Q483. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

Two Carnot engines A and B operate in series such that engine A absorbs heat at T_1 and rejects heat to a sink at temperature T. Engine B absorbs half of the heat rejected by Engine A and rejects heat to the sink at T_3. When workdone in both the cases is equal, to value of T is :

(A) $\frac{2}{3}T_1 + \frac{3}{2}T_3$ (B) $\frac{1}{3}T_1 + \frac{2}{3}T_3$
(C) $\frac{3}{2}T_1 + \frac{1}{3}T_3$ (D) $\frac{2}{3}T_1 + \frac{1}{3}T_3$

Open solution

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Q484. JEE Main 2021 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

A heat engine has an efficiency of $\frac{1}{6}$. When the temperature of sink is reduced by 62°C, its efficiency gets doubled. The temperature of the source is :

(A) 124°C (B) 37°C (C) 62°C (D) 99°C

Open solution

Q485. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

A Carnot's engine working between 400 K and 800 K has a work output of 1200 J per cycle. The amount of heat energy supplied to the engine from the source in each cycle is :

(A) 2400 J (B) 1600 J (C) 1800 J (D) 3200 J

Open solution

Q486. JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

A heat engine is involved with exchange of heat of 1915 J, − 40J, + 125 J and −Q J, during one cycle achieving an efficiency of 50.0%. The value of Q is

(A) 980 J (B) 40 J (C) 400 J (D) 640 J

Open solution

Q487. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

A carnot engine having an efficiency of $\frac{1}{10}$ is being used as a refrigerator. If the work done on the refrigerator is 10 J, the amount of heat absorbed from the reservoir at lower temperature is :

(A) 90 J (B) 99 J (C) 1 J (D) 100 J

Open solution

Q488. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

Two ideal Carnot engines operate in cascade (all heat given up by one engine is used by the other engine to produce work) between temperature, T_1 and T_2 . The temperature of the hot reservoir of the first engine is T_1 and the temperature of the cold reservoir of the second engine is T_2 . T is temperature of the sink of first engine which is also the source for the second engine. How is T related to T_1 and T_2 , if both engines perform equal amount of work?

(A) $T = \frac{2T_1T_2}{T_1+T_2}$ (B) $T = \sqrt{T_1T_2}$

(C) $T = \frac{T_1+T_2}{2}$ (D) T = 0

Open solution

Q489. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

A Carnot engine has an efficiency of $\frac{1}{6}$. When the temperature of the sink is reduced by 62°C, its efficiency is doubled. The temperatures of the source and the sink are, respectively

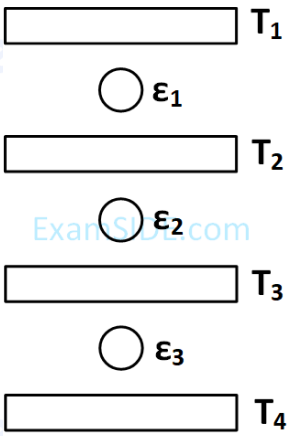
(A) 99°C, 37°C (B) 124°C, 62°C (C) 37°C, 99°C (D) 62°C, 124°C

Open solution

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Q490. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

Three Carnot engines operate in series between a heat source at a temperature T_1 and a heat sink at temperature T_4 (see figure). There are two other reservoirs at temperature T_2 and T_3, as shown, with $T_1 > T_2 > T_3 > T_4$. The three engines are equally efficient if -



(A) $T_2 = (T_1^3T_4)^{1/4}$; $T_3 = (T_1T_4^3)^{1/4}$

(B) $T_2 = (T_1T_4)^{1/2}$; $T_3 = (T_1^2T_4)^{1/3}$

(C) $T_2 = (T_1T_4^2)^{1/3}$; $T_3 = (T_1^2T_4)^{1/3}$

(D) $T_2 = (T_1^2T_4)^{1/3}$; $T_3 = (T_1T_4^2)^{1/3}$

Open solution

Q491. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

Two Carnot engines A and B are operated in series. The first one, A, receives heat at T_1 (= 600 K) and rejects to a reservoir at temperature T_2. The second engine B receives heat rejected by the first engine and, in turn, rejects to a heat reservoir at T_3 (=400 K). Calculate the temperature T_2 if the work outputs of the two engines are equal :

(A) 600 K

(B) 400 K

(C) 300 K

(D) 500 K

Open solution

Q492. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

Two Carnot engines A and B are operated in series. Engine A receives heat from a reservoir at 600 K and rejects heat to a reservoir at temperature T. Engine B receives heat rejected by engine A and in turn rejects it to a reservoir at 100 K. If the efficiencies of the two engines A and B are represented by η_A and η_B, respectively, then what is the value of $\frac{\eta_B}{\eta_A}$?

(A) $\frac{12}{7}$

(B) $\frac{7}{12}$

(C) $\frac{12}{5}$

(D) $\frac{5}{12}$

Open solution

Q493. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

A Carnot's engine works as a refrigerator between 250 K and 300 K. It receives 500 cal heat from the reservoir at the lower temperature. The amount of work done in each cycle to operate the refrigerator is :

(A) $420\ J$

(B) $772\ J$

(C) $2100\ J$

(D) $2520\ J$

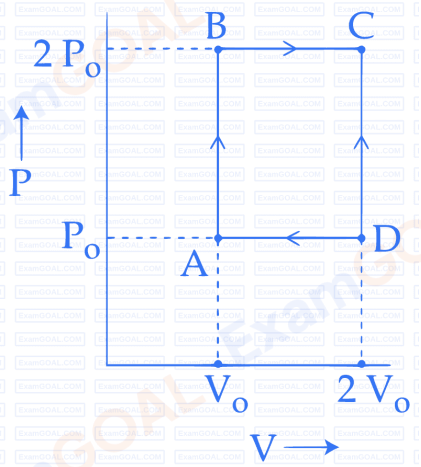
Open solution

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Q494. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

An engine operates by taking n moles of an ideal gas through the cycle ABCDA shown in figure. The thermal efficiency of the engine is : (Take C_v = 1.5 R, where R is gas constant)

Open solution



(A) 0.24 (B) 0.15 (C) 0.32 (D) 0.08

Q495. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

A Carnot freezer takes heat from water at 0^oC inside it and rejects it to the room at a temperature of 27^oC. The latent heat of ice is 336×10^3 J kg^{-1}. If 5 kg of water at 0^oC is converted into ice at 0^oC by the freezer, then the energy consumed by the freezer is close to :

Open solution

(A) 1.67×10^5 J (B) 1.68×10^6 J (C) 1.51×10^5 J (D) 1.71×10^7 J

Q496. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

A Carnot engine absorbs 1000 J of heat energy from a reservoir at 127^oC and rejects 600 J of heat energy during each cycle. The efficiency of engine and temperature of sink will be :

Open solution

(A) 20% and -43^oC (B) 40% and -33^oC (C) 50% and -20^oC (D) 70% and -10^oC

Q497. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

The door of a working refrigerator is left open in a well insulated room. The temperature of air in the room will

Open solution

(A) decrease

(B) increase in winters and decrease in summers

(C) remain the same

(D) increase

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Q498. AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: An inventor claims to have constructed an engine that has an efficiency of 30% when operated between the boiling and freezing points of water. This is not possible.
Statement 2: The efficiency of a real engine is always less than the efficiency of a Carnot engine operating between the same two temperatures.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

Q499. AIEEE 2012 · MCQ (Single Correct Answer)

Open solution

A Carnot engine, whose efficiency is 40%, takes in heat from a source maintained at a temperature of $500\ K$. It is desired to have an engine of efficiency 60%. Then, the intake temperature for the same exhaust (sink) temperature must be :

(A) efficiency of Carnot engine cannot be made larger than 50%

(B) $1200\ K$

(C) $750\ K$

(D) $600\ K$

Q500. AIEEE 2011 · MCQ (Single Correct Answer)

Open solution

A Carnot engine operating between temperatures T_1 and T_2 has efficiency $\frac{1}{6}$. When T_2 is lowered by $62\ K$ its efficiency increases to $\frac{1}{3}$. Then T_1 and T_2 are, respectively:

(A) $372\ K$ and $330\ K$ (B) $330\ K$ and $268\ K$ (C) $310\ K$ and $248\ K$ (D) $372\ K$ and $310\ K$

Q501. AIEEE 2010 · MCQ (Single Correct Answer)

Open solution

A diatomic ideal gas is used in a Carnot engine as the working substance. If during the adiabatic expansion part of the cycle the volume of the gas increases from V to $32\ V$, the efficiency of the engine is

(A) 0.5 (B) 0.75 (C) 0.99 (D) 0.25

Q502. AIEEE 2007 · MCQ (Single Correct Answer)

Open solution

A Carnot engine, having an efficiency of $\eta = 1/10$ as heat engine, is used as a refrigerator. If the work done on the system is $10\ J$, the amount of energy absorbed from the reservoir at lower temperature is

(A) $100\ J$ (B) $99\ J$ (C) $90\ J$ (D) $1\ J$

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Q503. AIEEE 2003 · MCQ (Single Correct Answer)

A carnot engine takes 3×10^6 cal. of heat from a reservoir at $627^\circ C$, and gives it to a sink at $27^\circ C$. The work done by the engine is

(A) $4.2 \times 10^6 J$ (B) $8.4 \times 10^6 J$ (C) $16.8 \times 10^6 J$ (D) zero

Open solution

Q504. AIEEE 2002 · MCQ (Single Correct Answer)

Which statement is incorrect?

(A) all reversible cycles have same efficiency

(B) reversible cycle has more efficiency than an irreversible one

(C) Carnot cycle is a reversible one

(D) Carnot cycle has the maximum efficiency in all cycles.

Open solution

Q505. AIEEE 2002 · MCQ (Single Correct Answer)

Even Carnot engine cannot give 100% efficiency because we cannot

(A) prevent radiation (B) find ideal sources

(C) reach absolute zero temperature (D) eliminate friction.

Open solution

Q506. JEE Main 2022 (Online) 29th June Morning Shift · Numerical

300 cal. of heat is given to a heat engine and it rejects 225 cal. If source temperature is 227°C, then the temperature of sink will be _____ °C.

Open solution

Q507. JEE Main 2022 (Online) 27th June Morning Shift · Numerical

In a carnot engine, the temperature of reservoir is 527°C and that of sink is 200 K. If the work done by the engine when it transfers heat from reservoir to sink is 12000 kJ, the quantity of heat absorbed by the engine from reservoir is _____ $\times 10^6$ J.

Open solution

Q508. JEE Main 2022 (Online) 26th June Evening Shift · Numerical

A heat engine operates with the cold reservoir at temperature 324 K. The minimum temperature of the hot reservoir, if the heat engine takes 300 J heat from the hot reservoir and delivers 180 J heat to the cold reservoir per cycle, is _____ K.

Open solution

Q509. JEE Main 2022 (Online) 25th June Morning Shift · Numerical

A steam engine intakes 50 g of steam at 100°C per minute and cools it down to 20°C. If latent heat of vaporization of steam is 540 cal g^{-1}, then the heat rejected by the steam engine per minute is _____ $\times 10^3$ cal.

(Given : specific heat capacity of water : 1 cal g^{-1} $^\circ C^{-1}$)

Open solution

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Q510. JEE Main 2021 (Online) 27th August Evening Shift · **Numerical**

A heat engine operates between a cold reservoir at temperature T_2 = 400 K and a hot reservoir at temperature T_1. It takes 300 J of heat from the hot reservoir and delivers 240 J of heat to the cold reservoir in a cycle. The minimum temperature of the hot reservoir has to be _____ K.

Open solution

Q511. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

For an ideal heat engine, the temperature of the source is 127°C. In order to have 60% efficiency the temperature of the sink should be _____°C. (Round off to the Nearest Integer)

Open solution

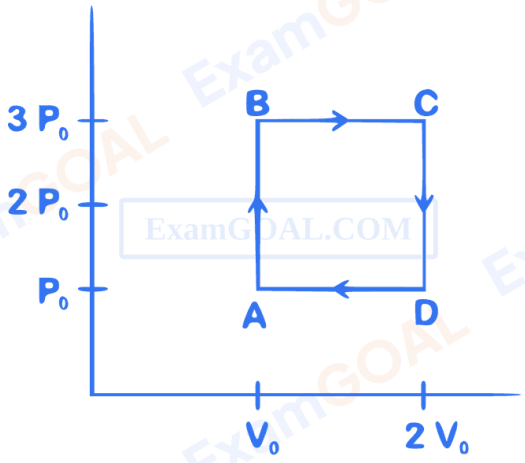
Q512. JEE Main 2021 (Online) 25th February Evening Shift · **Numerical**

A reversible heat engine converts one-fourth of the heat input into work. When the temperature of the sink is reduced by 52K, its efficiency is doubled. The temperature in Kelvin of the source will be _____.

Open solution

Q513. JEE Main 2020 (Online) 6th September Evening Slot · **Numerical**

An engine operates by taking a monoatomic ideal gas through the cycle shown in the figure. The percentage efficiency of the engine is close to _____.



Open solution

Q514. JEE Main 2020 (Online) 3rd September Evening Slot · **Numerical**

If minimum possible work is done by a refrigerator in converting 100 grams of water at 0^oC to ice, how much heat (in calories) is released to the surroundings at temperature 27^oC (Latent heat of ice = 80 Cal/gram) to the nearest integer ?

Open solution

Q515. JEE Main 2020 (Online) 7th January Morning Slot · **Numerical**

A Carnot engine operates between two reservoirs of temperatures 900 K and 300 K. The engine performs 1200 J of work per cycle. The heat energy (in J) delivered by the engine to the low temperature reservoir, in a cycle, is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **B**	4. **D**	5. **A**
6. **C**	7. **B**	8. **C**	9. **C**	10. **D**
11. **B**	12. **B**	13. **C**	14. **D**	15. **C**
16. **C**	17. **B**	18. **D**	19. **A**	20. **A**
21. **A**	22. **C**	23. **D**	24. **D**	25. **C**
26. **D**	27. **B**	28. **C**	29. **D**	30. **B**
31. **C**	32. **D**	33. **C**	34. **A**	35. **A**
36. **D**	37. **B**	38. **C**	39. **D**	40. **B**
41. **D**	42. **A**	43. **C**	44. **B**	45. **D**
46. **B**	47. **B**	48. **A**	49. **D**	50. **A**
51. **B**	52. **B**	53. **A**	54. **D**	55. **C**
56. **D**	57. **B**	58. **C**	59. **C**	60. **D**
61. **B**	62. **D**	63. **B**	64. **D**	65. **D**
66. **B**	67. **A**	68. **C**	69. **B**	70. **B**
71. **D**	72. **B**	73. **B**	74. **C**	75. **B**
76. **B**	77. **C**	78. **D**	79. **A**	80. **C**
81. **C**	82. **B**	83. **A**	84. **A**	85. **B**
86. **B**	87. **B**	88. **A**	89. **B**	90. **C**
91. **C**	92. **D**	93. **D**	94. **D**	95. **B**
96. **B**	97. **A**	98. **A**	99. **B**	100. **A**
101. **A**	102. **A**	103. **C**	104. **A**	105. **D**
106. **B**	107. **A**	108. **C**	109. **B**	110. **C**
111. **D**	112. **D**	113. **C**	114. **C**	115. **D**
116. **D**	117. **D**	118. **A**	119. **A**	120. **B**
121. **D**	122. **C**	123. **C**	124. **A**	125. **C**
126. **B**	127. **B**	128. **D**	129. **B**	130. **A**
131. **D**	132. **C**	133. **C**	134. **B**	135. **B**
136. **D**	137. **A**	138. **D**	139. **D**	140. **D**
141. **B**	142. **B**	143. **D**	144. **C**	145. **D**
146. **D**	147. **D**	148. **B**	149. **B**	150. **B**

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151. **A**	152. **B**	153. **B**	154. **C**	155. **C**
156. **D**	157. **B**	158. **C**	159. **A**	160. **C**
161. **B**	162. **A**	163. **B**	164. **C**	165. **B**
166. **B**	167. *Scan QR*	168. **D**	169. **B**	170. **D**
171. **D**	172. **C**	173. **C**	174. **B**	175. **B**
176. **A**	177. **A**	178. **B**	179. **B**	180. **B**
181. **D**	182. **B**	183. **B**	184. **C**	185. **B**
186. **B**	187. **C**	188. **B**	189. **A**	190. **C**
191. **A**	192. **A**	193. **A**	194. **D**	195. **C**
196. **C**	197. **C**	198. **A**	199. **A**	200. **B**
201. **C**	202. **B**	203. **D**	204. **B**	205. **B**
206. **A**	207. **C**	208. **A**	209. **C**	210. **A**
211. **A**	212. **C**	213. **D**	214. **A**	215. **A**
216. **C**	217. **B**	218. **C**	219. **B**	220. **A**
221. **D**	222. **B**	223. **A**	224. **D**	225. **C**
226. **A**	227. **C**	228. **A**	229. **C**	230. **C**
231. **B**	232. **C**	233. **1**	234. **C**	235. **A**
236. **D**	237. **A**	238. **A**	239. **D**	240. **A**
241. **A**	242. **B**	243. **D**	244. **B**	245. **B**
246. **C**	247. **A**	248. **B**	249. **A**	250. **D**
251. **B**	252. **C**	253. **C**	254. **B**	255. **B**
256. **B**	257. **D**	258. **B**	259. **A**	260. **A**
261. **D**	262. **C**	263. **B**	264. **A**	265. **B**
266. **C**	267. **B**	268. **A**	269. **C**	270. **A**
271. **C**	272. **B**	273. **D**	274. **C**	275. **B**
276. **A**	277. **C**	278. **C**	279. **B**	280. **A**
281. **B**	282. **B**	283. **C**	284. **D**	285. **B**
286. **A**	287. **B**	288. **B**	289. **B**	290. **C**
291. **D**	292. **C**	293. **A**	294. **B**	295. **D**
296. **D**	297. **B**	298. **D**	299. **A**	300. **D**
301. **D**	302. **C**	303. **A**	304. **A**	305. **A**
306. **A**	307. **C**	308. **B**	309. **D**	310. **A**
311. **B**	312. **C**	313. **B**	314. **A**	315. **C**

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316. **D**	317. **C**	318. **D**	319. **C**	320. **A**
321. **D**	322. **D**	323. **D**	324. **C**	325. **D**
326. **A**	327. **C**	328. **D**	329. **C**	330. **C**
331. **C**	332. **A**	333. **B**	334. **D**	335. **A**
336. **A**	337. **C**	338. **A**	339. **D**	340. **A**
341. **C**	342. **B**	343. **A**	344. **A**	345. **B**
346. **C**	347. **D**	348. **B**	349. **C**	350. **B**
351. **B**	352. **B**	353. **C**	354. **C**	355. **B**
356. **C**	357. **C**	358. **C**	359. **B**	360. **B**
361. **D**	362. **A**	363. **D**	364. **C**	365. **D**
366. **A**	367. **A**	368. **B**	369. **D**	370. **A**
371. **C**	372. **A**	373. **A**	374. **A**	375. **A**
376. **B**	377. **A**	378. **A**	379. **A**	380. **A**
381. **B**	382. **B**	383. **D**	384. **B**	385. **C**
386. **A**	387. **A**	388. **D**	389. **B**	390. **D**
391. **C**	392. **D**	393. **B**	394. **A**	395. **D**
396. **A**	397. **A**	398. **D**	399. **C**	400. **D**
401. **B**	402. **B**	403. **A**	404. **C**	405. **D**
406. **C**	407. **A**			

Numerical

408. **922**	409. **300**	410. **120**	411. **300**	412. **481**
413. **3730 or 6460**	414. **7**	415. **350**	416. **500**	417. **360**
418. **5**	419. **314**	420. **12**	421. **3**	422. **327**
423. **15**	424. **273**	425. **40**	426. **40**	427. **4**
428. **300**	429. **313**	430. **12**	431. **4**	432. **3**
433. **750**	434. **90**	435. **60**	436. **21**	437. **7479**
438. **1400**	439. **42**	440. **250**	441. **2**	442. **12**
443. **25**	444. **500**	445. **480**	446. **2**	447. **25**
448. **05**	449. **17258**	450. **100**	451. **25**	452. **60**
453. **25**	454. **3600**	455. **50**	456. **400**	457. **5**
458. **40 to 41**	459. **150**	460. **266**	461. **20**	462. **46**
463. **1818 to 1819**	464. **50**	465. **40**	466. **60**	

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MCQ (More than One Correct Answer)

467. **B, D**	468. **B, C**

Out of syllabus

469. **B**	470. **A**	471. **B**	472. **A**	473. **B**
474. **C**	475. **C**	476. **A**	477. **B**	478. **A**
479. **C**	480. **B**	481. **A**	482. **A**	483. **D**
484. **D**	485. **A**	486. **A**	487. **A**	488. **C**
489. **A**	490. **D**	491. **D**	492. **A**	493. **A**
494. **B**	495. **A**	496. **B**	497. **D**	498. **D**
499. **C**	500. **D**	501. **B**	502. **C**	503. **B**
504. **A**	505. **C**	506. **102**	507. **16**	508. **540**
509. **31**	510. **500**	511. **-113**	512. **208**	513. **19**
514. **8791**	515. **600**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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